

The 2026 assessment for eastern Bering Sea snow crab

Grant Adams and Cody Szuwalski

September 2026

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This report may be cited as: Adams, G. and C. Szuwalski. 2026. The 2026 assessment for eastern Bering Sea snow crab. North Pacific Fishery Management Council, Anchorage, AK. Available from <https://www.npfmc.org/library/safe-reports/>

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Executive Summary

This document is the September 2026 stock assessment for eastern Bering Sea (EBS) snow crab (*Chionoecetes opilio*), prepared for the Crab Plan Team and the Scientific and Statistical Committee, and provides the basis for the 2026/27 harvest specifications.

The stock was declared overfished in October 2021 and is rebuilding under Amendment 53. The directed fishery was closed for the 2022 and 2023 seasons and has since reopened at a reduced scale, with 4.13 kt retained in 2025/26. Mature male biomass is estimated at 113.9 kt, or 77 percent of the B_{MSY} proxy of 148.2 kt, and is above the minimum stock size threshold of 74.1 kt. The stock is therefore not overfished on the current estimate, and the point estimate has not reached the biomass at which it would be considered rebuilt. The 95 percent interval on that biomass runs from 85.4 to 152.1 kt: the stock is above the minimum stock size threshold across the whole of it, but its upper end reaches the B_{MSY} proxy, so the rebuilt threshold is not excluded.

Following the June 2026 SSC recommendation, the overfishing limit is calculated under Tier 4 while a simpler and more stable Tier 3 model is developed. The recommended OFL is 30.76 kt, and the ABC after a 20 percent buffer is 24.61 kt. [[author: confirm the buffer, and state the currency and control rule the recommendation rests on – the choice of currency changes the OFL by more than a factor of four (Table 6), and the September 2025 assessment recommended ≥ 95 mm carapace width rather than morphometric maturity.]]

6 models are presented. Model 25.2e (new data) is the author-recommended Tier 3 model and carries the corrections endorsed by the CPT in May 2026 and the SSC in June 2026; two sensitivities examine the removal of the female population and the truncation of the data to 2019. The Tier 3 models are presented to document progress and are not proposed for specification. Convergence remains the principal difficulty: jittering continues to locate multiple optima whose separation in likelihood is small but whose management quantities differ materially.

[[author: the SAFE guidelines specify ten numbered Executive Summary items – stock, catches, data sources, stock biomass, recruitment, a management performance table, a basis-for-the-OFL table, the OFL probability distribution, the ABC basis, and a rebuilding summary. Items 6 and 7 are tables that have not yet been built.]]

A. Summary of Major Changes

1. **Management:** The eastern Bering Sea snow crab population was declared overfished in October 2021 and the directed fishery was closed by the State of Alaska for the 2022 and 2023 seasons. A small fishery occurred in 2024, with 2.15 kt retained, and a larger one in 2025, with 4.13 kt retained. The stock remains under a rebuilding plan.
2. **Input data:** The 2026 NMFS summer trawl survey and the 2025 fishery data have been added. Three corrections have been made to the inputs themselves. The total-male size composition has been corrected, resolving a discrepancy between the file used in September 2025 and the file produced by the script that translates the ADF&G delivery. The plus group has been enlarged to include all crab greater than 135 mm carapace width, which the previous configuration discarded. The size-composition bin convention has been changed so that a crab falling exactly on a 5 mm cutoff is assigned to the upper bin, and a transcription error in the growth data has been corrected, a 26.3 mm crab's molt increment having been recorded as 73 mm rather than 7.3 mm. Hybrid *Chionoecetes* catch and survey data were evaluated in the May 2026 preliminary assessment and are not included in any model presented here, following CPT and SSC advice; an immature survey index was likewise evaluated and is not carried forward.
3. **Assessment methodology:** Management quantities were derived from maximum likelihood estimates of model parameters in a size-based, integrated assessment method using GMACS, version 2.20.34. Retrospective and jittering analyses were performed for four of the six models presented (Table 9); Model 25 and Model 25.2e were not re-run for either. The probability of terminal molt at size is now taken from the maturity workflow developed by the Kodiak laboratory (Ryznar, in prep) rather than the design-based series used previously. Fits that did not meet the conventional maximum-gradient criterion were polished with Newton steps taken using the inverse Hessian, which reduces the gradient without materially moving the fit. Following the June 2026 SSC recommendation, the Tier 4 calculation is brought forward as the basis for the harvest specification while a simpler and more stable Tier 3 model is developed; the Tier 3 models are presented to document progress and are not proposed for specification.
4. **Assessment results:** [[author: state the recommended OFL and ABC for 2026, the tier and currency they rest on, and the F_{OFL} applied, in the same form as the 2025 document. Two comparisons against 2025 are available and they tell different stories, so state which one is meant. The 2025 *author-preferred* OFL was 3.26 kt at $F_{OFL} = 0.18$, on Tier 3 with ≥ 95 mm carapace width as the currency; the specification the SSC went on to adopt for 2025/26 was Tier 4b with an OFL of 20.11 kt and an ABC of 12.07 kt. The 2025 *Tier 4* calculation, which is the like-for-like comparison now that the SSC has directed Tier 4 for this cycle, gave 28.41, 8.64 and 6.11 kt on the morphometric, ≥ 95 mm and > 101 mm currencies against 30.765, 6.620 and 4.591 kt this cycle – higher on morphometric by 8% and lower on both large-male currencies by about 24%.]]

B. SSC and CPT comments + author responses

2025(10) SSC. Snow crab was declared overfished in 2021, but the SAFE did not include a section on “Rebuilding Analysis and Update.” The SSC recommends that this section be included in future SAFEs as appropriate.

An update on progress toward rebuilding will be included in the September SAFE when new survey data are available.

2025(10) SSC. As with a similar SSC recommendation for Tanner crab, the SSC recommends consideration of different inclusions/exclusions of hybrids into the snow crab assessment (i.e., in survey and catch data) to evaluate sensitivity to these options and ensure an internally consistent approach.

In the May 2026 preliminary assessment we evaluated the inclusion of hybrids into the snowcrab by including them in the survey and fishery data. The inclusion of hybrids had a minor effect on management advice. However, we recommend excluding hybrids from further assessments because they are avoided by the fishery. One challenge is that retained catch currently includes both hybrids and snow crab and there is no data available to separate them. As a consequence we recommend a smaller buffer on OFL to account for retained catch of hybrids that are counted against the fishery.

2025(10) SSC. The SSC recommends that research be conducted to evaluate whether clutch fullness is a reliable indicator of female reproductive output, using histology or lab rearing experiments.

This has been added to the wish list of laboratory studies.

2025(10) SSC. The SSC is willing to participate in a collaborative working group during spring 2026 to facilitate additional progress on the development of an ABC control rule for snow crab.

We appreciate the SSC’s willingness to collaborate and will follow up in the future.

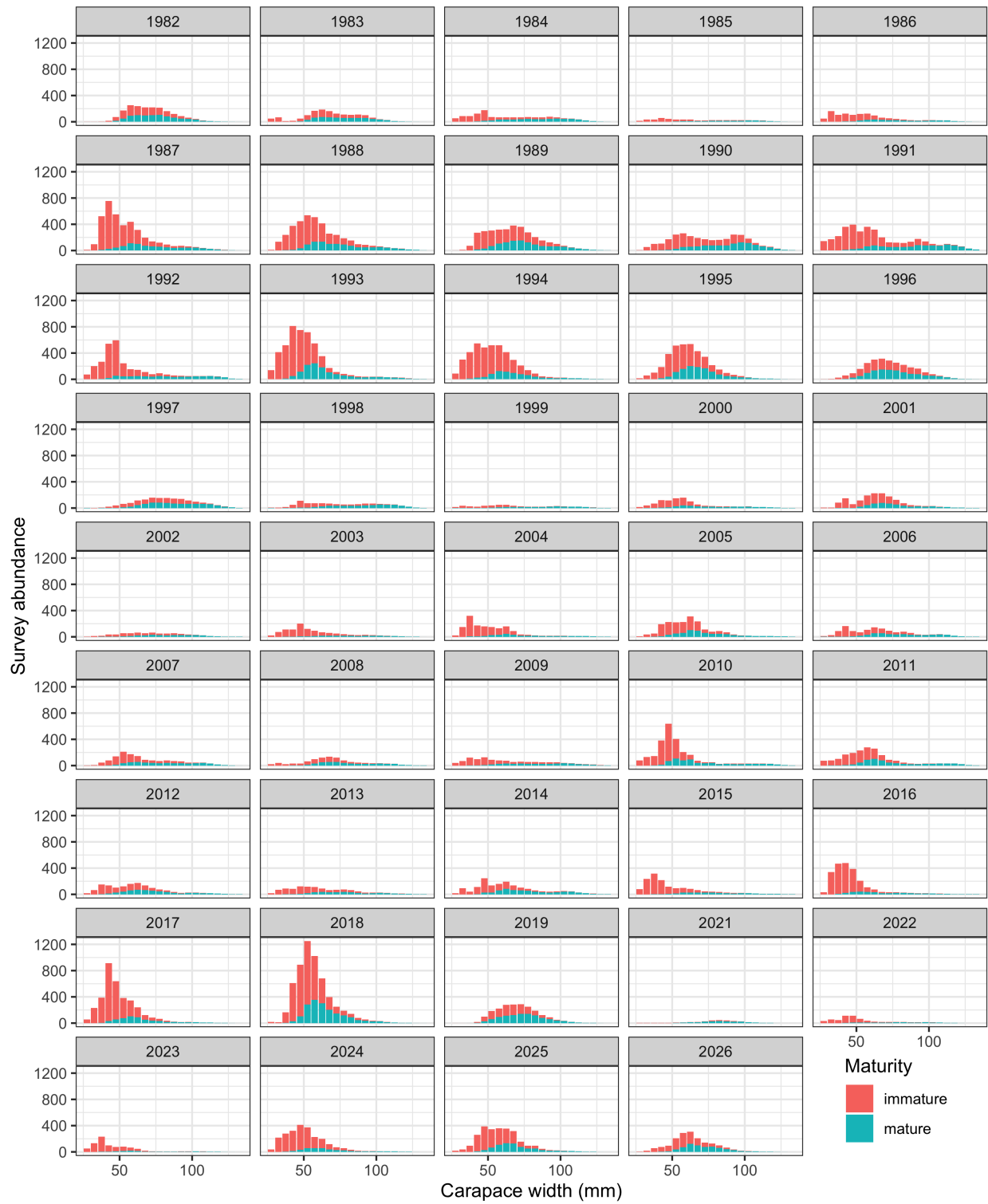
2025(10) SSC. The SSC also reiterates its general comments from December 2024 suggesting that authors use likelihood profiles to describe uncertainty in model parameters and management quantities and use jittering analyses only to ensure that the results provided come from the model with the lowest negative log likelihood (the “MLE”).

The May 2026 preliminary assessment included a jittering analysis across a set of models representing the axes of uncertainty under evaluation, and jittering is reported again here for every model presented (Section E). [[author: likelihood profiles were promised for this document and have not been produced. The June 2026 SSC repeated the request, asking specifically for profiles on terminal MMB, recent recruitment, M and the OFL as part of a focused convergence section. Either produce them or state when they will follow.]]

2025(10) SSC. *The SSC recommends that morphometric maturity status be overlaid on the size distribution plots (Figure 9) to allow visual tracking of growth and the transition to maturity by cohort.*

The figure below plots the abundance of immature and mature crab at size observed in the survey by year. The historical methods for calculating the probability of terminal molt were applied to these data. An updated figure will be provided in September with the new year of survey data and the maturity methodology recommended for adoption by the CPT.

Numbers at size by maturity state by year



2025(6) SSC. *The maximin analysis should be completed assuming a Ricker stock-recruitment relationship, but including the same compensation ratios as the original Clark (1991) analysis.*

The Ricker curves used in Clark were implemented in the maximin analysis presented last cycle (and now published in Fisheries Research: Szuwalski and Punt (2025) “Spawning-biomass-per-recruit proxies for fisheries reference points under multiple axes of uncertainty”). Incorporating these into the analysis shifted the SPBR% from ~36% to ~34%.

2025(6) SSC. *As the figures presented on the updated 1991+ catch data appear to indicate substantial differences in male discards, the SSC requests that the September document more clearly describe changes in the discard estimation and accounting process.*

A presentation from January 2025 from ADFG (available on the council website) noted that the largest changes occurred for total catch through implementing consistent data filtering, expansion using directed pot lifts, and adjustments to biological strata. This process now allows for the production of a data file with all years of available data for each assessment cycle, rather than single years at a time which were appended to an existing data file. An apparent shift upwards in discards of males from the directed fishery results primarily from a change in when discard mortality is applied (before input into the data file [before 2025] or within the assessment code itself [after 2025]).

2025(6) SSC. *Given the findings described in Mullowney and Baker (2021) from Canadian research indicating that the molt to maturity may occur at smaller sizes when lower densities of large males are present, it would be useful to determine if there is evidence for the same process occurring in EBS snow crab, and whether fishing mortality on the large males is consistently high enough to result in a strong effect. Further, it would be useful to evaluate whether clutch fullness may be related to size at maturity or the abundance of large-sized crab.*

Given that natural mortality events seem to switch among years and sexes depending on the model or input data, it would be prudent to investigate whether it is possible to estimate a direct link between natural mortality and a bottom temperature covariate of appropriate spatial and temporal scale.

A manuscript describing both of these relationships for Bering Sea snow crab was presented in May/June 2024 (Figure 3; “Density dependence can modulate climate change impacts on populations”) and was published in 2026 (Szuwalski et al., 2026). In that manuscript, warmer temperatures were associated with elevated mortality and higher densities of large males were associated with lower probabilities of terminally molting at size.

The SSC responded to this presentation with the following text: “The SSC emphasizes that projections of climate change impacts, while useful, are highly uncertain and. . . caution should be used when presenting these results.”

Clutch fullness does not appear to be related to the abundance of large-size crab based on analyses in “Spatial depletion rates in the eastern Bering Sea snow crab fishery” (also presented in May 2024 and now in review for external publication). It is unclear if there is any relationship between clutch fullness and recruitment, though preliminary analyses suggest clutch fullness is not an important covariate for predicting recruitment strength.

These analyses come with the caveat that the Kodiak lab is revising their methods for analyzing the maturity data for snow crab.

2025(6) SSC. *To explore development of an ABC control rule, the SSC requests that a yield per recruit analysis be developed for snow crab.*

Time did not allow for this exercise.

2024(10) SSC. *SSC recommends standardizing the approach to the Tier 4 fallback across BBRKC, Tanner and snow crab assessments so that the same methods are used for each including all mature male biomass, a BMSY proxy based on the time-series of REMA-smoothed survey estimates and an FMSY proxy based on the best estimate of natural mortality (from the Tier 3 model). As the SSC intends the Tier 4 calculation only as a fallback if the Tier 3 analysis fails to converge, no other Tier 4 calculations need to be included in future assessments.*

In the 2025 document, Tier 4 HCRs were applied to rema-smoothed survey biomass estimates of three different size ranges: morphometrically mature males, large males ($\geq 95\text{mm}$), and preferred males ($>101\text{mm}$). The target biomass was set as the average of the time series from 1982-2024 and the target fishing mortality was set as 0.27, the median of the prior on natural mortality used in the integrated assessment. No kink exists in this harvest control rule. The resulting OFLs were 28.41 kt, 8.64 kt, and 6.11 kt for morphometric, large, and preferred males, respectively.

2024(10) SSC. *The SSC highlighted that more research is needed on the basic reproductive biology of snow crab, specifically addressing the question of whether small males can and do mate with large females and produce progeny that will be large males under the right environmental conditions. The SSC strongly supports the proposed collaborative research to explore in situ observations of mating crab using a camera sled.*

We agree and preliminary information from CamSled work done by Weems et al. observed small morphometrically mature males grasping mature females in the eastern Bering Sea ($n=21$). A publication is being prepared to share these results, but improved sampling coverage is needed to draw actionable conclusions.

2024(10) SSC. *The highest priority would be to continue to refine the Maximin analysis as requested by the SSC in June 2024, specially using values of steepness of 0.50, 0.67, and 0.80, and considering both the Beverton-Holt and Ricker stock recruitment relationships. The yield analysis also indicated that fishing mortality rates much lower than $F_{35\%}$ achieved a high percentage of MSY, indicating potential flexibility in specifying reference points. The SSC suggested that some type of collaborative work during the spring, perhaps including SSC members and/or others might facilitate additional progress on this topic. The SSC is interested in developing a wider range of options for reference points for snow crab for consideration in the next assessment cycle.*

The analysis was not extended beyond what was presented in appendix A of the 2024 SAFE and would be of limited use until we know more about reproductive contribution of large male snow crab. The specific values for steepness recommended by the SSC, unfortunately, do not match estimates or available data on the reproductive dynamics of snow crab. We suggest focusing efforts on understanding the dynamics of large males given the reliance of the fishery on large males that are at historical lows.

2024(10) SSC. *The SSC again requests an analysis of the probability of maturing/terminal molt which addresses the observation error in these data and the lack of a monotonically increasing curve. A hierarchical analysis that treats years as random effects might be a starting point. The SSC would also like to better understand the sampling design for the molt data and is concerned about the weighting of the spatial samples in the analysis; weighting should be based on abundance if the sampling rate differs by area (which it would, unless abundance were uniform and/or the targets were in direct proportion to abundance).*

The Kodiak lab gave a presentation describing how these data are analyzed in January 2025 to the CPT and a summary was given to the SSC. Potential changes to this workflow produces different values for the probability of having undergone terminal molt over time. Two models runs are included within the 2025 preliminary assessment and the current document to explore the impact of different assumptions about the probability of terminal molt.

2024(10) SSC. *Investigate whether there is information outside the assessment model (e.g., larval or post-settlement data) or in the model, supporting estimated skewed sex-ratios at recruitment and the mismatch between recent large recruitments for males and females occurring in different years. Explore whether the estimated large differences in male and female recruitment years could be related to the lack of fit to molt-increment data.*

No new information has come to light on why this might be the case. Previous discussion centered around the possibility that different spatial distributions by sex of small crab could expose them to different environments, which could alter what we perceive as recruitment five years after settlement. While this difference is an interesting scientific question to pursue, solving it will not address the central problem in management: commercially preferred males are at historic lows over the last decade. Removing the females from the model (as has been done in the research models shared over the past several years) entirely might be a prudent step towards focusing on the most pressing management problem at hand given conflicting trends.

2024(10) SSC. *Geostatistical (e.g. VAST) modeling of trawl survey data including both the NBS and EBS should be prioritized. This could help understand some of the inconsistent recruitment/growth trends observed in recent years as well as prepare for potential changes in stock distribution or productivity under future warming of the Bering Sea. Geostatistical modeling should evaluate alternative error distributions and other model configurations as appropriate.*

The Kodiak lab presented indices for mature females and large males in the 2025 May CPT. We are hesitant to include both the NBS in the VAST modeling because of the extrapolation of estimates in years which there are no data.

2024(10) SSC. *While the model the author presented produced Hessian matrices with small gradients, a concerning issue arose during the jittering analysis: jittering produced two clouds of model solutions at or near the MLE with OFLs that differed by 5,000 t. The author explained that the difference in OFLs resulted from differences in the terminal year estimates of MMB, driven by differences in recent recruitment estimates. In order to determine whether the absence of the 2020 survey data point is contributing to this issue, the CPT requested seeing the results of the jittering analysis from the accepted model but only including data through 2019.*

Time did not allow for this exercise.

2024(10) SSC. *For future work in which results are shown from the same model configuration using different size-at-maturity, the CPT recommended against running separate jittering analyses for each scenario. Instead, after jittering is complete for the first scenario and the putative MLE has been identified, the parameter estimates from that run should be used for all comparisons that differ only in the size-at-maturity scenario because the MLE would be the same for all such scenarios (i.e., the size-at-maturity scenario should not affect the parameter estimates).*

Management quantities from both models are compared here.

2024(10) SSC. *The CPT further requested that the author investigate whether it is necessary to use the same number of size bins for males and females, noting that the BBRKC model uses different numbers of bins for the sexes.*

Time did not allow for this exercise.

2024(10) CPT. *There is a strong consensus by the CPT that the best available scientific information supports the conclusion that the status quo definition is not appropriate in terms of conserving the reproductive potential for the stock, or in terms of supporting optimal yield of commercial-sized animals for harvest. The CPT therefore again recommended a change to the definition of maturity for this stock, from morphometric maturity to ≥ 95 mm CW.*

We agree, but have used morphometric maturity for all analyses included in this document. A comparison of both definitions can be included in the September document.

2024(6) CPT. *Tier 4 fallback - CPT supports: The author suggested an alternative for snow crab to the basic Tier 4 calculation that is being brought forward for other stocks. Specifically, a fallback Tier 4 option would be calculated as the vulnerable biomass in this year's survey (not smoothed, and vulnerable defined by a cutoff of 95 mm CW), decremented by the proportion of natural mortality that occurs between the time of the survey and the fishery, and M applied as the proxy for FMSY to calculate the Tier 4 OFL. The motivations identified for using the annual survey design-based estimate of vulnerable biomass rather than a REMA smooth through the survey time series were: 1) that the high number of positive stations for snow crab on survey (238 stations in 2023) make the design-based estimate a robust measure of annual biomass; and 2) that the large interannual fluctuations in biomass observed in some recent years would be poorly captured by a smooth. The motivation for decremending the biomass estimate by the proportion of natural mortality occurring between the survey and the fishery is that failing to account for this mortality could lead to unintentionally high values of F if the Tier 4 fallback was used.*

The SSC requested Tier 4 fallback will be provided in September.

2024(6) SSC. *Concerning the GMACS assessment model, the SSC continues to recommend that the assessment author explore ways to incorporate the molt to maturity data in the model in a way that reflects the*

observation error associated with those estimates. An analysis in a GLMM modeling framework, which treats years as random effects, would provide smoother estimates, accommodate differing sample sizes by year and length, and deal appropriately with years in which data are missing. Another possibility that was suggested in the CPT report was to include the annual observed probabilities of terminal molt as data and then fit them, as in the Tanner crab assessment.

Observation errors are not considered when preparing the data that enters the assessment. The Kodiak lab is working on these analyses and will present their workflow in the May 2026 PT meeting. We will revisit this after the discussion of the new maturity workflow. Buck has incorporated the observation error when fitting to maturity data, but the snow crab model assumes those as known inputs. That is an area of improvement for the future.

2024(6) SSC. *The SSC recommends that this model be brought forward in the fall but requests that an additional Tier 4 model be provided for comparison, as recommended in the Simpler Modeling Workshop report and requested in the SSC's June 2023 and October 2023 Reports. This additional model would use the random effects model (REMA) to smooth survey estimates and would not decrement with natural mortality.*

This has been included regularly in 2024 SAFE document and will be included in the September 2026 document.

C. Introduction

Studies and data relevant to key population and fishery processes are discussed below to provide background for the modeling choices made in this assessment. A model description is available on the github repository for GMACS and the files needed to reproduce these assessments also have a github repo, both of which are linked at the end of this document.

Distribution

Snow crab (*Chionoecetes opilio*) are distributed on the continental shelf of the Bering Sea, Chukchi Sea, and in the western Atlantic Ocean as far south as Maine. In the Bering Sea, snow crab are distributed widely over the shelf and are common at depths less than ~200 meters (see tech memo; Zacher et al., 2025). Smaller crab tend to occupy more inshore northern regions and mature crab occupy deeper areas to the south of the juveniles (see Zacher et al., 2025; Zheng et al. 2001). The eastern Bering Sea population within U.S. waters is managed as a single stock; however, the distribution of the population may extend into Russian waters to an unknown degree.

Natural Mortality

Relatively few targeted studies exist to determine natural mortality for snow crab in the Bering Sea. Nevissi, et al. (1995) used radiometric techniques to estimate shell age from last molt (??). The total sample size was 21 male crab (a combination of Tanner and snow crab) from a collection of 105 male crab from various hauls in the 1992 National Marine Fishery Service (NMFS) Bering Sea survey. Representative samples for the 5 shell condition categories were collected from the available crab. Shell condition 5 crab (SC5 = very, very old shell) had a maximum age of 6.85 years (s.d. 0.58, 95% CI approximately 5.69 to 8.01 years; carapace width of 110 mm). The average age of 6 crab with SC4 (very old shell) and SC5, was 4.95 years (range: 2.70 to 6.85 years). Given the small sample size, this maximum age may not represent the 1.5% percentile of the population that is approximately equivalent to Hoenig’s method (1983). Tag recovery evidence from eastern Canada revealed observed maximum ages in exploited populations of 17-19 years (Nevissi, et al. 1995, Sainte-Marie 2002). A maximum time at large of 11 years for tag returns of terminally molted mature male snow crab in the North Atlantic has been recorded since tagging started about 1993 (Fonseca, et al. 2008). Fonseca, et al. (2008) estimated a maximum age of 7.8 years post terminal molt using data on dactal wear.

In recent years, the mean for the prior for natural mortality used in the eastern Bering Sea snow crab assessment was based on the assumption that longevity would be at least 20 years in an unfished population of snow crab, informed by the studies above. Under negative exponential depletion, the 99th percentile corresponding to age 20 of an unexploited population corresponds to a natural mortality rate of 0.23. Using Hoenig’s (1983) method a natural mortality equal to 0.23 corresponds to a maximum age of 18 years.

In contrast to the implied natural mortalities from the methodology used above, Murphy et al. (2018) estimated time-varying natural mortality for eastern Bering Sea snow crab with a mean of 0.49 for females and 0.36 for males based on the output of state-space models fit to NMFS survey data. Further, natural mortality estimates produced from empirical analyses by Then et al. (2015) and Hamel (2015) using similar assumed maximum ages as the methodology above produced natural mortalities larger than 0.23 (Table 1). Then et al. (2015) compared several major empirical estimation methods for M (including Hoenig’s method) with an updated data set and found that maximum age was the best available predictor. A maximum age of 20 years corresponded to an M of ~0.315 in Then et al.’s analysis. Hamel (2015) developed priors in a similar manner to Then et al., but forced the regression of observed natural mortality onto maximum age through the intercept, which resulted in an M of ~0.27 for an assumed maximum age of 20 years.

Table 1: Empirical estimates of natural mortality (per year) for a range of methods over a range of assumed maximum ages (column header). These are empirical estimators, not output of any assessment model.

	23	20	17
Then	0.277	0.315	0.365
Hoenig (1983)	0.19	0.212	0.257
Hoenig (2013)	0.194	0.223	0.261
Hamel	0.235	0.271	0.318

In addition to the results of empirical estimates of M from updated methodologies and state-space modeling by Murphy et al. (2018), inspection of the survey data suggests that natural mortality for mature individuals is higher than assumed. A fraction of the mature population (which are assumed not to grow, given evidence for a terminal molt) are not selected in the fishery (e.g. sizes 50-80 mm; ??). Consequently, all mortality observed is ‘natural’. The collapse in recruitment in the 1990s can be used as an instrument to understand natural mortality for mature individuals. The last large recruitment enters these size classes in the mid- to late-1990s and numbers of crab in these size classes return to low levels in less than 5 years.

The median value of the priors used in this assessment are set equal to values resulting from assuming a maximum age of 20 years and applying Hamel’s methodology (0.271). A standard error of 0.0054 was used for initial priors and was estimated using the 95% CI of ± 1.7 years on maximum age estimates from dactyl wear and tag return analysis in Fonseca, et al. (2008). Mortality events in 2018 and 2019 are estimated as additional mortality parameters applied by sex and maturity state to allow the model to fit recent population trends.

Maturity

Maturity of females collected during the NMFS summer survey was determined by the shape of the abdomen, by the presence of brooded eggs, or egg remnants. Maturity for males was determined by chela height measurements, which were available most years starting from the 1989 survey (Otto 1998; ?? & ??). Mature male biomass referenced throughout this document refers to a morphometrically mature male (i.e. large-clawed). A maturity curve for males was estimated using the average fraction mature based on chela height data and applied to years of survey data to estimate mature survey numbers that do not have chela height data available (see Richar and Foy, 2022 for details).

Bering Sea male snow crab appear to have a terminal molt to maturity based on hormone level data and findings from molt stage analysis via setagenesis (Tamone et al. 2005). The models presented here assume a terminal molt for both males and females, which is supported by research on populations in the Bering Sea and the Atlantic Ocean (e.g. Dawe et al. 1991).

Mature male snow crab that do not molt may be important in reproduction. Paul et al. (1995) found that old shell mature male Tanner crab out-competed new shell crab of the same size in breeding in a laboratory study. Recently molted males did not breed even with no competition and may not breed until after ~ 100 days from molting (Paul et al. 1995). Sainte-Marie et al. (2002) stated that only old shell males take part in mating for North Atlantic snow crab.

Mating ratio and reproductive success

Bering Sea snow crab have historically been managed using morphometrically mature male biomass (MMB) as a proxy for reproductive potential. MMB is used as the currency for management because the fishery only retains large male crab, which are nearly 100% mature. Male snow crab are sperm conservers, using less than 4% of their sperm at each mating and females also can mate with more than one male. The amount of stored sperm and clutch fullness varies with sex ratio (Sainte-Marie 2002). If mating with only one male is inadequate to fertilize a full clutch, then females would need to mate with more than one male, necessitating

a sex ratio closer to 1:1 in the mature population, than if one male is assumed to be able to adequately fertilize multiple females. Although mature male biomass is currently the currency of management, some aspect of female reproduction is likely also an important indicator of reproductive potential of the stock.

Clutch fullness is recorded for the females measured in the survey (see ESP; Fedewa et al., 2025). However, quantifying the reproductive potential of the female population from survey data can be difficult. For example, full clutches of unfertilized eggs may be extruded and appear normal to visual examination, and may be retained for several weeks or months by snow crab. Resorption of eggs may occur if not all eggs are extruded resulting in less than a full clutch. Female snow crab at the time of the survey may have a full clutch of eggs that are unfertilized, resulting in overestimation of reproductive potential. Barren females may be a more obvious indication of low reproductive potential and increased in the early 1990s, decreased in the mid-1990s, then increased again in the late 1990s. The highest levels of barren females coincided with periods of high fishing mortality, but even then the proportion of barren females was low (see ESP; Fedewa et al., 2025). Biennial spawning is another confounding factor in determining the reproductive potential of snow crab. Laboratory analyses showed that female snow crab collected in waters colder than 1.5 degrees C from the Bering Sea spawn only every two years.

Further complicating the process of quantifying reproductive capacity, clutch fullness and fraction of unmated females may not account for the fraction of females that may have unfertilized eggs, since these cannot be detected by eye at the time of the survey. The fraction of barren females observed in the survey may not be an accurate measure of fertilization success because females may retain unfertilized eggs for months after extrusion. To examine this hypothesis, NMFS personnel sampled mature females from the Bering Sea in winter and held them in tanks until their eggs hatched in March of the same year (Rugolo et al. 2005). All females then extruded a new clutch of eggs in the absence of males. All eggs were retained until the crab were euthanized near the end of August. Approximately 20% of the females had full clutches of unfertilized eggs. The unfertilized eggs could not be distinguished from fertilized eggs by visual inspection at the time they were euthanized. Indices of fertilized females based on the visual inspection method of assessing clutch fullness and percent unmated females may overestimate fertilized females.

Growth

Several studies are available to estimate the growth per molt of male and female snow crab in the Bering Sea. These studies include:

1. Transit study (2003); 14 crab
2. Cooperative seasonality study; 6 crab
3. Dutch harbor holding study; 9 crab
4. NMFS Kodiak holding study held less than 30 days; 6 crab
5. NMFS Kodiak holding study 2016; 5 crab
6. NMFS Kodiak holding study 2017; 70 crab.
7. BSFRF/NMFS holding study 2018; 4 crab.
8. BSFRF/NMFS holding study 2021; 177 crab.

In the “Transit study”, pre- and post-molt measurements of 14 male crab that molted soon after being captured were collected. The crab were measured when shells were still soft because all died after molting, so measurements may be underestimates of post-molt width (L. Rugolo, pers. com.). The holding studies include only data for crab held less than 30 days because growth of crab held until the next spring’s molting was much lower. Crab missing more than two limbs were excluded due to other studies showing lower growth. Crab from the seasonal study were excluded that were measured less than 3 days after molting due to difficulty in measuring soft crab accurately (L. Rugolo, pers. comm.). In general, growth of snow crab in the Bering Sea appears to be greater than growth of some North Atlantic snow crab stocks (Sainte-Marie 1995). Crab in their first few years of life may molt more than once per year, however, the smallest crab included in the model are approximately 4 years old and would be expected to molt annually.

Management history

ADFG harvest strategy

Before the year 2000, the Guideline Harvest Level (GHL) for retained crab only was a 58% harvest rate of the number of male crab over 101 mm CW estimated from the survey. The minimum legal size limit for snow crab is 78 mm, however, the snow crab market generally only accepts crab greater than 101 mm. In 2000, due to the decline in abundance and the declaration of the stock as overfished, the harvest rate for calculation of the GHL was reduced to 20% of male crab over 101 mm. After 2000, a rebuilding strategy was developed based on simulations by Zheng et al. (2002) using survey biomass estimates. The realized retained catch often exceeded the GHL historically.

The Alaska Department of Fish and Game (ADFG) harvest strategy since 2000 sets harvest rate based on estimated mature biomass. The harvest rate scales with the status of the population relative to a proxy for B_{MSY} , which is calculated as the average total mature biomass at the time of the survey from 1983 to present and MSST is one half the B_{MSY} proxy. The harvest rate begins at 0.10 when total mature biomass exceeds 50% MSST (230 million lbs) and increases linearly to 0.225 when biomass is equal to or greater than the B_{MSY} proxy (Zheng et al. 2002).

$$u = \begin{cases} \text{Bycatch} & \text{if } \frac{TMB}{TMB_{MSY}} \leq 0.25 \\ \frac{0.225(\frac{TMB}{TMB_{MSY}} - \alpha)}{1 - \alpha} & \text{if } 0.25 < \frac{TMB}{TMB_{MSY}} < 1 \\ 0.225 & \text{if } TMB > TMB_{MSY} \end{cases} \quad (1)$$

Where TMB is the total mature biomass and TMB_{BMSY} is the TMB associated with maximum sustainable yield. The maximum retained catch is set as the product of the exploitation rate, u , calculated from the above control rule and survey mature male biomass. If the retained catch in numbers is greater than 58% of the estimated number of new shell crab greater than 101 mm plus 25% of the old shell crab greater than 101 mm, the catch is capped at 58%.

History of BMSY

Prior to adoption of Amendment 24, B_{MSY} was defined as the average total mature biomass (males and females) estimated from the survey for the years 1983 to 1997 (921.6 million lbs; NPFMC 1998) and MSST was defined as 50% of B_{MSY} . Currently, the biological reference point for biomass is calculated using a spawning biomass per recruit proxy, $B_{35\%}$ (Clark, 1993). $B_{35\%}$ is the biomass at which spawning biomass per recruit is 35% of unfished levels and has been shown to provide close to maximum sustainable yield for a range of stock productivities (Clark, 1993). Consequently, it is an often used target when a stock recruit relationship is unknown or unreliable. The range of years of recruitment used to calculate biomass reference points is from 1982 to the present assessment year, minus 1. However, recent analyses suggest SPR-based reference points do not provide a meaningful constraint on the snow crab fishery when the probability of having undergone terminal molt is specified to reflect observations in the survey. This is because a large fraction of the population matures (and ceases growing) at a size smaller than is harvested by the fishery.

Fishery history

Snow crab were harvested in the Bering Sea by the Japanese from the 1960s until 1980 when the Magnuson Act prohibited foreign fishing. After the closure to foreign fleets, retained catches increased from relatively low levels in the early 1980s (e.g. 11.85 kt during 1982) to historical highs in 1990s (retained catches during 1991, 1992, and 1998 were 143.02, 104.67, and 88.15 kt, respectively). The stock was declared overfished in

1999 at which time retained catches dropped to levels similar to the early 1980s (e.g. 11.46 kt). Retained catches slowly increased after 1999 before dropping again in 2016. Total allowable catches were reduced with the collapse of the population in 2021, the fishery was closed for the first time in 2022, and the closure continued through 2023. A small fishery occurred in 2024.

Discard mortality from the directed fishery is the next largest source of mortality after retained catch and approximately tracks the retained catch. The highest estimated discard mortality occurred during 1990 at 35.55 kt which was 23.85% of the retained catch during that year. Discard mortality in 2024 was 0.66 kt; the directed fishery was closed in 2022 and 2023, so no directed discards were recorded in those years. Non-directed mortality continues to be very small at 0.09 kt in 2024.

Discard from the directed pot fishery has been estimated from observer data since 1992 and has ranged from 11-100% of the magnitude of retained catch by numbers. Discards reached 50-100% of the magnitude of retained catch in the lead up to the collapse after the 2018-2019 heatwave because of the large year class entering the population. Female discard catch has been very low compared to male discard catch and has not been a significant source of mortality. Discard mortality rates for the directed fishery are assumed to be 30%. Discard of snow crab in groundfish fisheries has been highest in the yellowfin sole trawl fishery, and decreases down through the flathead sole trawl fishery, Pacific cod bottom trawl fishery, rock sole trawl fishery, and the Pacific cod hook-and-line and pot fisheries, respectively (??). Bycatch in fisheries other than the groundfish trawl fishery has historically been relatively low. Discard mortality rates from non-directed fisheries are assumed to be 80%. Size frequency data and catch per pot have been collected by observers on snow crab fishery vessels since 1992. Observer coverage has been 10% on catcher vessels larger than 125 ft (since 2001), and 100% coverage on catcher processors (since 1992).

Several modifications to pot gear have been introduced to reduce bycatch mortality. In the 1978/79 season, escape panels were required on pots used in the snow crab fishery to prevent ghost fishing. Escape panels consist of an opening with one-half the perimeter of the tunnel eye laced with untreated cotton twine. The size of the cotton laced panel was increased in 1991 to at least 18 inches in length. No escape mechanisms for undersized crab were required until the 1997 season when at least one-third of one vertical surface of pots had to contain not less than 5 inches stretched mesh webbing or have no less than four circular rings of no less than 3 3/4 inches inside diameter. In the 2001 season the escapement provisions for undersized crab was increased to at least eight escape rings of no less than 4 inches placed within one mesh measurement from the bottom of the pot, with four escape rings on each side of the two sides of a four-sided pot, or one-half of one side of the pot must have a side panel composed of not less than 5 1/4 inch stretched mesh webbing.

D. Data

Updated time series of survey indices and size compositions were calculated from data downloaded from the AKFIN database. Bycatch data (biomass and size composition) were updated for the most recent year from the AKFIN database. Retained, total, and discarded catch (in numbers and biomass) and size composition data for each of these data sources were updated for the most recent year based on files provided by the State of Alaska.

Catch data

Catch data and size composition of retained crab from the directed snow crab pot fishery from survey year 1982 to 2025 were used in this analysis (??). Time series of discarded catch from the directed fishery are fit, but size composition data are for total catch. The discard male catch was estimated for survey years 1982 to 1991 in the model using the estimated fishery selectivities based on the observer data for the period of survey year 1992 to 2025. The discards prior to 1992 may be underestimated due to the lack of escape mechanisms for undersized crab in the pots before 1997. See Table 2 for a summary of catch data range.

Table 2: Data included in the assessment, for all models presented. Dates indicate survey year. The 2020 survey was cancelled due to the pandemic.

Data component	Years
Retained male crab pot fishery size frequency by shell condition	1982 - 2025
Discarded Males and female crab pot fishery size frequency	1992 - 2025
Trawl fishery bycatch size frequencies by sex	1991 - 2025
Survey size frequencies by, maturity, sex and shell condition	1982 - 2019, 2021 - 2026
Retained catch estimates	1982 - 2025
Discard catch estimates from crab pot fishery	1992 - 2025
Trawl bycatch estimates	1993 - 2025
Total survey abundance estimates and coefficients of variation	1982 - 2019, 2021 - 2026
Survey selectivity priors (BSFRF)	2009, 2010, 2016-2018

Survey biomass and size composition data

Estimates of the numbers of crab by sex and size from the annual eastern Bering Sea (EBS) bottom trawl survey conducted by NMFS (e.g. ?? & ??; see Lang et al., 2018 for materials and methods; see the most recent tech memo for detailed description) are used to calculate the primary indices of abundance used in this assessment. Additional survey stations were added in 1989, which could alter the interpretation of catchability coefficient for the survey. Consequently, survey selectivity has been historically modeled in two ‘eras’ in the assessment (1982-1988, 1989-present). All survey data in this assessment used measured net widths instead of the fixed 50 ft net width based on Chilton et al.’s (2009) survey estimates. Carapace width and shell conditions were measured and reported for snow crab caught in the survey.

Mature male size composition data were calculated by multiplying the total numbers at length for new shell male crab by a vector of observed proportion of mature males at length. All old shell crab of both sexes were assumed to be mature. New shell crab were demarcated as any crab with shell condition index ≤ 2 . The biomass of new and old shell mature individuals was calculated by multiplying the vector of numbers at length by weight at length. These vectors were then summed by sex to provide the input for assessment (??). Input sample sizes are specified as 100 for both sexes and maturity states for all years of survey size composition data.

Spatial distribution of survey abundance and catch

Snow crab are distributed widely over the eastern Bering Sea shelf, but their density and the extent of their distribution has changed over time (see tech memo). Spatial gradients exist in the survey data by maturity and size for both sexes. For example, larger males have been more prevalent on the southwest portion of the shelf while smaller males have been more prevalent on the northern portion of the shelf. Centroids of mature female abundance early in the history of the survey were farther south, but moved north during the 1990s. Since the late 1990s and early 2000s, the centroids moved south again, but not to the extent seen in the early 1980s. This phenomenon was mirrored in centroids of abundance for large males.

Fishing effort has generally been south of 58.5 N, even when ice cover did not restrict the fishery moving farther north (?? & ??). This is possibly due to the proximity to port and practical constraints of meeting delivery schedules. Unstandardized CPUE (hereafter just ‘CPUE’) in the fishery has varied over time and an increase in average CPUE occurred after rationalization (??). The change in CPUE in a given spatial area within a season can reflect the impact of the fishery on the population in that area. Declines in CPUE can be seen by spatial area over time within a season (??).

The observed distribution of large males during the summer survey and the fishery catch have historically differed, and the origin of this difference is unknown. It is possible that crab move between the fishery and the survey, but it is also possible that fishers do not target all portions of the distribution of large male crab

equally. The underlying explanation of this phenomenon could hold implications for relative exploitation rates spatially and it has been suggested that high exploitation rates in the southern portion of the snow crab range may have resulted in a northward shift in snow crab distribution (Orensanz, 2004). Snow crab larvae likely drift north and east after hatching in spring. Snow crab appear to move south and west as they grow (Parada et al., 2010); however, little tagging data exists to fully characterize the ontogenetic or annual migration patterns of this stock (Murphy et al. 2010).

Experimental study of survey selectivity

The Bering Sea Fisheries Research Foundation (BSFRF) has conducted supplementary surveys in the Bering Sea in which snow crab were caught during 2009, 2010, 2016, 2017, and 2018. The location and extent of these surveys varied over the years as the survey goals changed. In 2009, the survey consisted of 108 tows around 27 survey stations and the goal was to improve understanding snow crab densities and the selectivity of NMFS survey gear (??). Abundances estimated by the industry surveys were generally higher than the NMFS estimates, which suggests that the catchability of the NMFS survey gear is less than 1.

In 2016, 2017, and 2018, snow crab were not the focus of the BSFRF surveys, but were still caught in the BSFRF gear. Comparing the ratio of the number of crab caught at length in the BSFRF gear (which is assumed to have a catchability/selectivity of 1 over all size classes) to the number of crab caught at length within the same area in the NMFS survey gear (which is assumed to have a catchability/selectivity ≤ 1 for at least some of the size classes) can provide an empirical estimate of catchability/selectivity (??). Empirical estimates of catchability/selectivity vary by year and size class across the different BSFRF data sets (?? & ??). The number of snow crab used to develop estimates of numbers at length likely contribute to these differences among years (??), but other factors may also influence catchability/selectivity at size of the NMFS survey gear (e.g. Somerton et al. 2013 show substrate type can influence selectivity). GAMs fit to the empirical estimates of selectivity are used as priors on estimated selectivity in the currently used assessment model. The standard errors from the GAMs are input as weighting factors for each prior of selectivity at size and sex derived from the BSFRF data.

E. Analytic approach

Model description

The Generalized Model for Assessing Crustacean Stocks (GMACS) was adopted as the assessment platform for snow crab in 2022 after a demonstration that GMACS could effectively reproduce the dynamics of the status quo model and offered structural improvements. GMACS is an integrated, size-structured model developed using automatic differentiation software developed as a set of libraries under C++ (ADModel Builder). ADModel Builder can estimate a large number of parameters in a non-linear model using automatic differentiation software extended from Greiwank and Corliss (1991) and developed into C++ class libraries. GMACS version 2.20.34 was used for the author-preferred software after the January 2026 CPT.

The snow crab population dynamics model tracks the number of crab of sex s , maturity state m , during year y at width w , $N_{s,m,y,w}$. A terminal molt was modeled in which crab move from an immature to a mature state, after which no further molting occurred. The mid-points of the size bins tracked in the model spanned from 27.5 to 132.5 mm carapace width, with 5 mm size classes. The number of estimated parameters differs among the models: 407 in Model 25, Model 25.2c and Model 25.2e, 412 in Model 25.2e (new data), 386 in Model 25.2e (2019 data), and 234 in Model 26.1. Parameters estimated within the assessment included those associated with the population processes recruitment, growth, natural mortality (subject to an informative prior and two years of additional ‘mortality events’ estimated in 2018 and 2019), fishing mortality, selectivity (fisheries and survey), and catchability. The yearly probability of undergoing terminal molt, weight at size, discard mortality, bycatch mortality, variance in growth increment, and parameters associated with proportion of recruitment allocated to size bin were estimated outside of the model or

specified. See the GMACS repo linked at the end of this document to peruse the control files that specify the populations dynamics.

A lognormal distribution was assumed for all catch and index time-series, while a robust multinomial was assumed for all size composition data. No likelihood weights were used in the assessment.

A ‘jittering’ approach has been historically used to explore the impact of different starting values on the assessment output (Turnock, 2016). Parameter values were jittered based on a normal distribution centered on one with a standard deviation of 0.10 that was then translated to the parameter space defined by the initial value and its bounds. 100 jitters were run for the author-recommended model (Table 9); each run was started from an independently perturbed parameter vector and re-fit to convergence. Each jitter run used an explicitly recorded random seed, so the analysis is exactly reproducible; the seed used by every run is stored alongside its results. [[VERIFY: the 2025 assessment jittered six candidate models. The September 2026 analysis jitters the single author-recommended model – confirm this is the intended scope before the document goes to the CPT.]]

Model selection and evaluation

Models were evaluated based on their fit to the data, evidence of non-convergence, the credibility of the estimated population processes, and the strength of the influence of the assumptions of the model on the outcomes of the assessment. Note that for some of the model scenarios, data inputs are different and, therefore, model fits are not directly comparable.

Model summaries

The key assumptions of these models include:

- the probability of terminally molting at size varies over time and size and is specified as fixed inputs to the model based on observations from the survey data
- survey selectivity is estimated by era (1982-88; 1989-present) and sex as a non-parametric curve subject to priors based on the BSFRF survey efficiency experiment data
- growth is a linear function of pre-molt carapace width with a specified variability around post-molt size
- all immature crab molt
- natural mortality is estimated by sex and maturity state with additional mortality events estimated in 2018 and 2019 and subject to a prior based on an assumed longevity of 20 years
- total and retained fishery selectivity are estimated logistic curves
- all non-directed bycatch (e.g. snow crab caught in the Tanner crab fishery or crab caught in the non-pelagic trawl fisheries) is lumped into a single ‘fishery’ for which a single selectivity is estimated; these removals are recorded for both sexes combined, so in the male-only sensitivity they are fit by the male population alone and the bycatch fishing mortality estimated for males is correspondingly inflated (the affected removals average 0.48 kt against a mean retained catch of 37.1 kt, and the fit to the series is essentially unchanged)
- recruitment is estimated separately for females and males and is allocated in the first 3 size bins; the male-only sensitivity estimates male recruitment only

Model scenarios

A total of 6 models were explored (Table 9). All models share the same structural assumptions described above and differ only in the version of GMACS, their data inputs, specification of the maturity ogive, and the definition of the plus group. The naming convention reflects an additive series of changes layered onto the September 2025 model (**Model 25**).

The base ('rolled forward') model:

- **Model 25** – the September 2025 author-recommended model, included for reference only. No data are updated and no structural changes are made. It was fitted under an earlier version of GMACS and is reported as originally fitted.

Models carried forward from the May 2026 assessment:

- **Model 25.2c** – the model accepted by the CPT in May 2026. **Model 25** updated to GMACS 2.20.34, with the total-male size composition corrected and the plus group enlarged to include all crab greater than 135 mm carapace width, and with the molting probability controls replaced by the array produced by the new maturity workflow developed by the Kodiak lab (E. Ryznar, in prep). That workflow fits an **sdmTMB** spatiotemporal model to chela-based maturity classifications to produce a smoothed probability of being mature at size and location, integrated to a population-level annual ogive; the updated series shows a weaker temporal trend in the probability of terminal molt at intermediate sizes. Data end in 2024, as in May.
- **Model 25.2e** – as **Model 25.2c**, but with the size-composition bin convention changed so that a crab falling exactly on a 5 mm cutoff is assigned to the upper bin, and with a transcription error in the growth data corrected (a 26.3 mm crab's molt increment recorded as 73 mm where the specimen master gives 7.3 mm). Data end in 2024. Presented to isolate the effect of each correction against the May baseline.

The September 2026 models:

- **Model 25.2e (new data)** – the author-recommended model. The **Model 25.2e** configuration advanced to end year 2025, so that the fishery data run through the 2025/26 season and the survey data through the 2026 summer survey. The data change; the model does not.
- **Model 25.2e (2019 data)** – the same configuration with all data truncated to 2019, so that the fishery data end with the 2019/20 season and the survey with the 2019 summer survey. Requested by the CPT in May 2026 to test whether the convergence problems originate in estimating recruitment following the 2018-19 collapse and across the 2020 survey gap. The estimated parameter count is 386.

Sensitivity to the female data and dynamics:

- **Model 26.1** – **Model 25.2e (new data)** with the female population removed entirely rather than merely down-weighted. The number of sexes is set to one, which removes the female size compositions, the female survey indices, the female discard catch, and the female growth observations from the data, and removes the female recruitment, natural mortality, growth, and selectivity parameters from the model. The estimated parameter count falls from 412 to 234. Fixing the female estimation phases alone would not accomplish this: a combined-sex catch observation is predicted by summing over both sexes, and the recruitment scale and deviations are not sex-specific, so the female population would continue to influence fishing mortality and recruitment. The non-directed bycatch is recorded for both sexes combined and is therefore fit by the male population alone in this run, as noted among the assumptions above. This is the only model presented here that differs structurally from **Model 25.2e**, which is why it takes a new number rather than a parenthetical. Requested by the CPT in May 2026 and presented as a sensitivity only; it is not proposed for specification unless the CPT selects it in September.

The May 2026 preliminary assessment presented a bracket of fourteen candidate models exploring three axes of uncertainty: the treatment of *Chionoecetes* hybrids, the new maturity workflow, and a set of attempts to improve convergence by correcting the total-male composition, expanding the plus group, and adding an immature survey index. That bracket is not repeated here. The CPT and SSC concluded that hybrid catch and survey data should not be included in the assessment model at this time, and agreed that the immature survey index should not be treated as a convergence fix, so neither the hybrid nor the immature-index candidates are carried forward. The composition correction and the expanded plus group, which the SSC supported as clear technical improvements, are incorporated in every model presented above. The full May bracket and its results remain available in the May 2026 preliminary assessment document.

Time series of data used in the assessment are visualized in (Figures 2, 3, 4). The sentence that stood here reported that the maturity and hybrid treatments led to small differences in male survey biomass, and cited a figure comparing the May hybrid and immature-index candidates; both that figure and those models are out of the September set, so the claim and its cross-reference have been removed rather than left pointing at nothing. [[author: if a survey-biomass comparison across the September models is wanted, add it here.]]

Results

Model convergence and comparison

The 2026 groundfish stock assessment guidelines set the convergence standard applied here: Newton steps taken with the inverse Hessian to reduce the maximum gradient below $1e-05$, which the guidelines describe as stronger evidence of convergence than a maximum gradient or an invertible Hessian alone, while noting that it does not diagnose a local minimum. All 6 models presented here inverted their Hessians (Table 13). Where a fit did not initially meet the conventional convergence threshold of a maximum absolute gradient < 0.001 , the solution was polished with Newton steps taken using the inverse Hessian, which reduces the gradient without moving the fit: across the models it changed the objective function by no more than 4×10^{-7} and the reference points by less than 1 part in 10^7 . **Model 25**, which was fitted under an earlier version of GMACS and cannot be re-run with the current executable, is reported as originally fitted. Likelihood components by category for each candidate model are summarized in Table 10, 11, and 12.

A zero gradient should not be read as evidence that the parameters are well determined. It states that the fit sits at a stationary point of the likelihood, not that the likelihood is sharply peaked there, and it says nothing about how many other such points exist. The jitter analysis below speaks to the second question, and the near-zero Hessian eigenvalues discussed there to the first.

The practical cost of the multiple optima is visible in **Model 25.2c**. The jitter analysis located a solution 1.27 negative log-likelihood units better than the fit obtained from the model's own starting values, a difference that would ordinarily be considered small. Adopting it moved B_{MSY} by less than one percent, but moved the total OFL from 50.27 to 43.83 kt, a change of approximately 13 percent. The choice among solutions that the data separate only weakly therefore carries more consequence for the management quantity than several of the structural choices examined in this assessment.

As in past cycles, jittering was used to evaluate sensitivity to starting values. Of the 100 runs, 100 produced a complete set of estimates and 79 attained a maximum absolute gradient below 0.01. 10 of the runs met the conventional threshold of 0.001. For comparison, the accepted fit attained 0.000931 before Newton polishing and reports a maximum gradient of zero after it (Table 13); the jitter runs above are reported on their unpolished gradients, which is what makes them comparable with one another.

The runs settled into 6 distinct optima (Figure 5), confirming that the multimodality reported in the 2025 preliminary and final assessments persists in the 2026 model. The best-fitting mode accounted for 5.1% of the runs that reached an optimum. Across all runs, total OFL ranged from 81.81 to 90.59 kt and mature male biomass in 2025 ranged from 128.7 to 151.7 kt, so the choice of starting values, not the data, moves the management quantities across that range. The same separation is visible in male recruitment (Figure 6) and in MMB (Figure 7).

No jitter run improved materially on the reported fit, which supports the reported solution being the global optimum rather than a local one.

The parameters that separate the optima are summarized in Table 4 and Figure 9. Identifying which parameters differ between the clouds, rather than only observing that the clouds exist, was requested by the SSC and is intended to direct the structural work needed to resolve the multiple optima. [[author: interpret the attribution result – which processes the separating parameters belong to, and what that implies for the structural changes to evaluate before the next cycle.]]

Removing the female population improves the numerical behavior of the model. **Model 26.1** estimates 234 parameters against 412 for **Model 25.2e (new data)**. Its accepted fit attains a maximum absolute gradient of zero to reported precision after Newton polishing, and a Hessian markedly better conditioned than that of **Model 25.2e (new data)** (condition number 3.3e05 against 1.4e06, smallest eigenvalue 120 against 35), with no non-positive eigenvalues. Jittering the male-only model over 100 runs reached the best-fitting mode in 25.3% of the runs that reached an optimum, against 5.1% for **Model 25.2e (new data)**; the multimodality that has characterized this assessment is therefore reduced but not removed by dropping the females. The retrospective pattern is also smaller: Mohn’s rho for mature male biomass is -0.09, or -0.08 when the terminal survey year is withheld, against -0.11 for **Model 25.2e (new data)**, and both remain inside the Hurtado-Ferro et al. (2015) bounds. The terminal-year recruitment artifact is substantially reduced, with Mohn’s rho for male recruitment falling from approximately 8 to 0.7. This is consistent with the expectation that removing the females reduces the confounding among growth, natural mortality, and selectivity. Dropping the females also lowers the reference points: B_{MSY} falls from 144.97 to 131.34 kt and the total-catch OFL from 85.65 to 77.00 kt, with terminal mature male biomass falling from 141.53 to 115.90 kt (Table 16).

Table 3: Summary of the jitter analysis of the author-recommended model, Model 25.2e (new data). Objective function values are negative log-likelihood and gradients are dimensionless; OFL and mature male biomass are in 1,000 t.

Quantity	Value
Jitter runs	100
Runs returning a complete fit	100
Runs with max gradient < 0.01	79
Runs meeting the conventional 0.001	10
Distinct optima identified	6
Runs at the best-fitting optimum (%)	5.1
Negative log likelihood, base fit	-23546.346
Negative log likelihood, best jitter run	-23546.346
Total OFL, range across runs (1,000 t)	81.81 - 90.59
MMB in 2025, range across runs (1,000 t)	128.7 - 151.7

Table 4: Parameters that most separate the jitter clouds of Model 25.2e (new data), ranked by the standardized difference in fitted value between the best-fitting optimum and each alternative optimum.

Parameter	Comparison	Mean in A	Mean in B	Difference	Std. difference
logit_rec_prop_est:	A vs B	3.4830	2.6643	0.8188	3.26
S_pars_est[13]	A vs C	-1.1547	-1.1512	-0.0036	-3.08
T_pars_est[81]	A vs E	-13.7017	-13.7276	0.0259	2.98

Table 4: Parameters that most separate the jitter clouds of Model 25.2e (new data), ranked by the standardized difference in fitted value between the best-fitting optimum and each alternative optimum.

Parameter	Comparison	Mean in A	Mean in B	Difference	Std. difference
S_pars_est[75]	A vs F	-3.1262	-3.0096	-0.1166	-2.93
M_pars_est[9]	A vs F	0.2661	0.2655	0.0005	2.93
S_pars_est[78]	A vs B	-0.1956	-0.1860	-0.0096	-2.93
G_pars_est[1]	A vs B	0.0384	0.0403	-0.0019	-2.91
S_pars_est[55]	A vs E	-1.3558	-1.3170	-0.0388	-2.90
T_pars_est[64]	A vs E	10.1030	10.0986	0.0044	2.89
S_pars_est[49]	A vs B	4.2047	4.2036	0.0011	2.87
T_pars_est[59]	A vs E	13.3566	13.3193	0.0373	2.83
S_pars_est[51]	A vs E	-5.9495	-4.9462	-1.0033	-2.82
M_pars_est[14]	A vs B	1.7518	1.8087	-0.0569	-2.82
S_pars_est[4]	A vs B	2.3815	2.3886	-0.0071	-2.79
S_pars_est[3]	A vs B	4.8480	4.8528	-0.0049	-2.76

The guidelines also ask whether any parameters were inestimable or reached their bounds. In the accepted fit no parameter reached a bound and the Hessian is well conditioned: its smallest eigenvalue is 34.6 and none is non-positive. Twenty-four of the 545 estimated quantities have a standard error larger than the absolute value of the estimate. Sixteen of those are estimated within 0.1 of zero – recruitment and fishing mortality deviations and selectivity offsets, where a large ratio is arithmetic rather than diagnostic. The remainder are three recruitment deviations, three recruitment proportion parameters, a fishing mortality deviation, and `M_pars_est[2]`, estimated at 0.33 with a standard error of 0.59.

That is a property of this fit and not of the model. In cold starts the model can settle in an optimum where the second natural-mortality block for immature females, `M_pars_est[15]`, is unidentified: it drifts toward its upper bound of 10 with a standard error of order 5×10^3 , the smallest Hessian eigenvalue falls to 1.1×10^{-6} against a next-smallest of 19.8, and the likelihood acquires a single flat direction along which a cold start lands wherever floating-point noise takes it. In the accepted fit that same parameter is estimated at 1.83 with a standard error of 0.58, consistent with its block-one counterpart at 1.80, and the flat direction is absent. Whether an exponential time-varying natural mortality block on immature females is identifiable from these data at all is a question for the Plan Team; the alternative is to fix it, as the corresponding rows for the other sex and maturity combinations already are.

Retrospective analysis

Retrospective analyses were performed by sequentially removing the terminal year of data from the model, re-fitting, and comparing the resulting time series of estimated mature male biomass with the estimates from the model fit to the full data set. A retrospective pattern is a consistent directional change in an assessment estimate for a given year as additional years of data are added. Peels of one through 10 years were run, giving terminal years from 2024 back to 2015.

Mohn’s ρ is reported here as the mean across peels of the relative difference between each peel’s terminal-year estimate and the estimate for that same year from the model fit to all of the data,

$$\rho = \frac{1}{P} \sum_{p=1}^P \frac{X_{T_p, p} - X_{T_p, 0}}{X_{T_p, 0}},$$

where T_p is the terminal year of peel p and $X_{\cdot, 0}$ is the unpeeled fit. A positive ρ therefore means the peels sit *above* the full model, i.e. that terminal biomass is revised downward as data accumulate. Hurtado-Ferro et al. (2015) suggest that ρ outside $[-0.22, 0.30]$ for a short-lived stock indicates a retrospective pattern worth reporting.

Two retrospectives are presented. The **standard** retrospective removes the last p years of all data streams. The **dropped-terminal-survey** retrospective additionally withholds the most recent survey year, which isolates the leverage of the single most recent survey observation on the terminal estimate. This distinction matters for this stock because, under the crab-year convention, an assessment with end year N is fit to the summer survey from year $N + 1$, so the terminal MMB estimate rests heavily on one very recent survey point.

For mature male biomass, the standard retrospective gave Mohn’s $\rho = -0.109$ and the dropped-terminal-survey retrospective gave $\rho = -0.110$ (Table 14, Figures 10 and 11). Male recruitment gave $\rho = 7.893$.

Management quantities are not recomputed for each peel. GMACS calculates reference points only when the retrospective peel count is zero (`gmacsbase.TPL 2.20.34` gates both call sites on `nyrRetroNo == 0`), so every peeled run returns B_{MSY} , B/B_{MSY} and the OFL as exactly zero rather than as estimates. This is a property of GMACS and not of the runs presented here – the peels were fit with the Hessian – and it applies equally to the 2025 assessment. The retrospective therefore characterizes mature male biomass and recruitment only, and the movement in the management advice itself under each peel cannot be read from it. [[author: if the CPT wants the advice-level retrospective for a scientific-uncertainty buffer, it needs peel-by-peel reference points computed outside GMACS.]]

Each peel was re-fit with the Hessian rather than being seeded from the full model’s variance estimates. The projection file was left unmodified; GMACS shifts its reference-point averaging windows back by the number of peeled years internally.

Historical analysis

The retrospective analysis above withholds data from a single model. A historical analysis asks a different question: how the perception of the stock has changed across successive assessments accepted for management. Figure 1 shows estimated mature male biomass at mating as each assessment from 2016 to 2025 saw it.

The revisions are large, and they are much larger than the within-model retrospective pattern. The clearest case is the collapse itself. The 2020 assessment put mature male biomass at the time of the survey in 2020 at 486.5 kt; the 2021 assessment, with the 2021 survey in hand, revised that same year to 40.95 kt, a downward revision of 92 percent. The revision then reversed: the 2022 assessment estimated 2020 biomass at mating at 19.1 kt, and the 2023 assessment revised the same year to 205.4 kt, an order-of-magnitude increase in the opposite direction. Terminal-year estimates across the ten vintages range from 26.7 kt (2021 assessment) to 207.2 kt (2020 assessment).

[[author: the guidelines ask for the cause of any obvious differences, and that cause cannot be established from the material in this repository – `data/historical/` holds the published time series but not the configurations that produced them. The candidates worth checking against the 2021–2023 assessment documents are the treatment of the 2018–19 mortality event, changes in the maturity input, and any change in the initialisation. Note that the time-varying probability of terminal molt is not itself new this cycle; what changed for 2026 is the workflow producing it.]]

The practical consequence for management advice is that terminal-year mature male biomass in this assessment has historically been revised by considerably more than the within-model retrospective analysis implies, and in both directions. [[author: the guidelines ask for uncertainty intervals on the vintages. The

published series in `data/historical/` carry point estimates only, so the figure shows no intervals; either recover the intervals from the archived assessments or state here that they were not available.]]

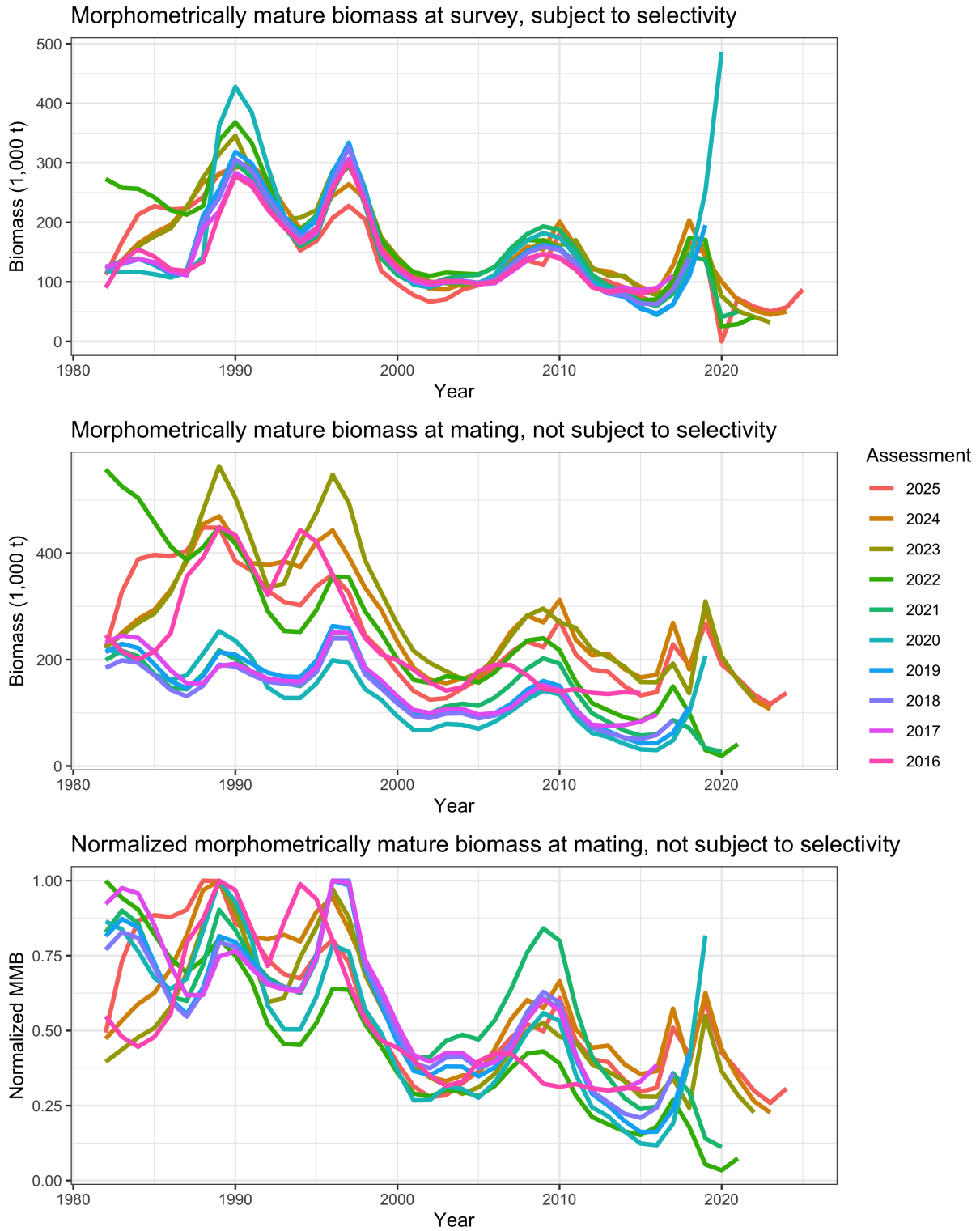


Figure 1: Estimated mature male biomass at mating (1,000 t) as seen by each assessment accepted for management from 2016 to 2025, against the current assessment. Point estimates only, without uncertainty intervals.

Model fits

Fits to the survey indices of mature biomass were broadly similar across the models (Figure 13). The two models that incorporate the 2026 NMFS summer survey, **Model 25.2e (new data)** and **Model 26.1**, continue to have difficulty fitting the terminal year of female biomass in the two-sex case. Inputting a yearly varying probability of terminal molt from the new maturity workflow (**Models 25.2**) shifts the estimated population dynamics modestly but does not resolve the female terminal-year fit (Figure 13).

[[author: this paragraph described the hybrid-survey and `imm_surv` candidates, which are not in the September set – the CPT and SSC advised against both in May and June 2026. Replace with a comparison across the six models actually presented, or drop it.]]

Fits to the non-directed fishery catches were good for all models (Figure 14). Fits to the directed-fishery retained catch were also good across the model set. [[author: a sentence here compared discard fits in the hybrid-fishery models, which are not in the September set. Removed rather than left describing models the document does not present. The underlying observation – that ADF&G reported roughly 18 kt of legal-male hybrid catch in 1997, with hybrid identification quality decreasing further back in the time series – is still recorded in Table 15 and may be worth keeping somewhere.]] The models accommodate the additional discards primarily through changes in estimated discard fishing mortality rather than through major changes in selectivity.

Differences among models in the fits to size composition data for retained catch in the directed fishery are small (size composition figures, below). Fits to the total size composition data from the directed fishery differ noticeably between the standard and the hybrid-fishery models, particularly for the larger size bins where hybrids are most prevalent. No visually discernible differences among models existed for female discards from the directed fishery. Fits to the non-directed fishery had some of the largest misfits of all data sources (particularly for male data), but this is unsurprising given a single estimated selectivity curve and changes in fishery behavior over time. Sample sizes are assumed to be 100 for all composition data except for female discards in 2021, 2024 and 2025 (sample size = 10).

Fits to the survey size composition data were broadly similar across models in most years (size composition figures, below). The plus-group change (extending the plus group to all crab > 135 mm) makes a small visual difference for retained males but improves the runtime of the model substantially. All models had difficulty fitting the size composition data for immature females in 2009 and 2019, which could suggest some unmodeled size-based mortality or time-varying probability of terminal molt. Estimates of mean growth are shown in Figure 15.

Estimated population processes

The estimated abundance of commercially preferred male crab (≥ 101 mm carapace width) varied among models (Figure 16). The early years during which differences in survey selectivity occurred were particularly different. All models agreed that the abundance in recent years has been the lowest in the time series (Figure 17). The estimates of the most recent era of survey selectivity were similar across models for both sexes (Figure 18).

Estimated fisheries selectivities were similar across models (Figure 19). Estimated fully-selected fishing mortality for the directed fishery was highly variable and relatively high for most of the fishery history (Figure 20). The total fishery selectivity curve is shifted far to the right, so only the largest crab experience the fully-selected fishing mortality.

The probability of undergoing terminal molt is input as data (Figure 21); the status quo workflow produces a strong increasing trend through time at intermediate sizes (e.g., at 77 mm the probability rose from ~0.2 in the early 1990s to ~0.5 after 2010), while [[VERIFY: this comparison of the two maturity workflows – the quoted probabilities at 77 mm and the characterisation of one trend as strong and the other as weak – could not be reproduced from the terminal-molt arrays in the control files. Check it against the workflow output directly before this goes out.]] the new maturity workflow produces a much weaker temporal trend (~0.35–0.45 throughout the time series). The new workflow uses a spatiotemporal `sdmTMB` model fit to chela-based

maturity classifications and borrows information across years and locations for size bins with sparse chela data; it does not extrapolate the predicted spatial probability surface beyond stations with measured crab.

Trends in recruitment were very similar across models for the early time series, with some variability in scale and differences in timing between the sexes (Figure 22). Differences in estimated recruitment in the most recent years were observed among models for males and contribute to differences in OFLs across the model set.

Base estimates of natural mortality for mature animals of both sexes were broadly similar across the models presented (Figure 23, Table 16). All models estimated additional mortality events that removed a large fraction of crab by sex and maturity state, except for mature females, at some point during the marine heatwave of 2018–19. In every model the immature-female natural mortality deviation for the second event block is poorly determined, with a standard error far larger than the estimate itself; whether that parameter should be fixed rather than estimated is discussed under model convergence. [[author: the May document also reported an extreme immature-female M in one of the hybrid candidates; those models are not carried forward, so that observation has been removed rather than restated.]]

MMB and management quantities

Estimated mature male biomass time series had similar trends across models but differed in scale (Figure 24). Estimates of B_{MSY} , current MMB relative to B_{MSY} , F_{MSY} , F_{OFL} and the Tier 3 OFL are summarized in Table 16. Across the models presented, the B_{MSY} proxy ranged from 131.3 to 180.1 kt and terminal mature male biomass from 86.9 to 141.5 kt. The three models whose data end in 2024 place the stock below its B_{MSY} proxy (status 0.81 to 0.96), while the two carrying the 2025 fishery and 2026 survey place it above (status 1.23 and 1.32); the additional year of data, rather than any structural change, accounts for that difference.

[[author: a paragraph comparing the hybrid candidates' B_{MSY} proxies stood here. Those models are not carried forward and the paragraph has been removed; the audit-trail note in Section E records that hybrids were evaluated in May and excluded on CPT and SSC advice.]]

F. Calculation of the OFL

Methodology for OFL

Tier 3

Historically, the tier 3 OFL was calculated using proxies for biomass and fishing mortality reference points and a sloped control rule. Proxies for biomass and fishing mortality reference points were calculated using spawning-biomass-per-recruit methods (e.g. Clark, 1991). After fitting the assessment model to the data and estimating population parameters, the model was projected forward 100 years using the estimated parameters under no exploitation and constant recruitment to determine ‘unfished’ mature male biomass-per-recruit. Projections were repeated in which the bisection method was used to identify a fishing mortality that reduced the mature male biomass-per-recruit to 35% of the unfished level (i.e. $F_{35\%}$ and $B_{35\%}$). Calculations of $F_{35\%}$ were made under the assumption that bycatch fishing mortality was equal to the estimated average value over the last 10 years.

Calculated values of $F_{35\%}$ and $B_{35\%}$ were used in conjunction with a Tier 3 control rule to adjust the proportion of $F_{35\%}$ that is applied to the stock based on the status of the population relative to $B_{35\%}$ (Amendment 24, NMFS). To determine the F_{OFL} , the population is projected to the time of fishing for the upcoming fishery under no fishing. If the MMB at that time exceeds 25% of $B_{35\%}$, a fishery can occur and the F_{OFL} is calculated as:

$$F_{OFL} = \begin{cases} Bycatch & \text{if } \frac{MMB}{B_{35}} \leq 0.25 \\ \frac{F_{35}(\frac{MMB}{B_{35}} - \alpha)}{1 - \alpha} & \text{if } 0.25 < \frac{MMB}{B_{35}} < 1 \\ F_{35} & \text{if } MMB > B_{35} \end{cases} \quad (2)$$

Where MMB is the projected morphometrically mature male biomass in the current survey year after fishing at the F_{OFL} , $B_{35\%}$ is the mature male biomass at the time of mating resulting from fishing at $F_{35\%}$, $F_{35\%}$ is the fishing mortality that reduces the morphometrically mature male biomass per recruit to 35% of unfished levels, and α determines the slope of the descending limb of the harvest control rule (set to 0.1 here).

Calculated total Tier 3 OFLs across the models presented ranged from 43.8 to 147.3 kt (Table 16). For reference, the observed survey biomass of morphometrically mature males in 2025 was 117.7 kt and of preferred males (> 101 mm) 23.3 kt. Taking **Model 25.2c**, the model accepted in May, as the comparison baseline at 43.8 kt: correcting the size-composition bin convention and the growth transcription error (**Model 25.2e**) lowered the B_{MSY} proxy from 172.7 to 158.7 kt and raised the OFL to 53.7 kt; adding the 2025 fishery and 2026 survey (**Model 25.2e (new data)**) lowered the proxy further to 145.0 kt, raised status above one, and raised the OFL to 85.6 kt. The male-only sensitivity gives a lower OFL than the model it is derived from, 77.0 kt against 85.6 kt. The run truncated to 2019 is not comparable with the others, since its reference points are computed from a different span of data. [[author: interpret the direction of these changes – in particular why the corrected bin convention and the additional year of data both lower the B_{MSY} proxy.]]

Tier 4

The Tier 4 calculation is the basis for the 2026 harvest specification. The SSC assigned this stock to **Tier 4b** when it set the 2025/26 specification (October 2025 SSC report, Table 2), and that subtier is carried here. [[author: confirm 4b still applies on this cycle’s status.]] A random-effects model (REMA) is fitted to the observed survey biomass series, the B_{MSY} proxy is taken as the mean of the smoothed series over years before 2025, and stock status is the terminal smoothed biomass relative to that proxy. Target fishing mortality is $F = M = 0.27 \text{ yr}^{-1}$, the median of the prior on natural mortality.

Three currencies are carried, as in the September 2025 assessment: morphometrically mature males, large males (≥ 95 mm carapace width), and preferred males (> 101 mm). The observed survey biomass for each, in the terminal survey year and the year preceding it, is:

Table 5: Observed NMFS summer survey male biomass (1,000 t) by currency. These are survey observations, not output of any assessment model. 2025 and 2026. These are the survey observations; the Tier 4 calculation below is applied to the REMA-smoothed series.

Currency	2025	2026
Morphometrically mature males	117.73	114.58
Large males (≥ 95 mm)	33.17	23.98
Preferred males (> 101 mm)	23.28	16.55

The observed mature male biomass from the 2025 survey was 117.73 kt, against 114.58 kt in 2026.

Applying the Tier 4 rule to each currency gives:

Table 6: Tier 4 management quantities by currency (biomass and OFL in 1,000 t; F per year). Computed from the REMA-smoothed survey series, not from a GMACS model. F_{OFL} and OFL are the recommended basis: flat $F_{OFL} = M$ above the closure threshold, with $OFL = F_{OFL} \times B$. The final two columns show two alternatives, each changing one choice - the ramped control rule, and the Baranov catch equation.

Currency	B_{MSY}	B current	Status	M	F_{OFL}	OFL	OFL (ramped F)
morphometric	148.181	113.944	0.769	0.270	0.270	30.765	2
>101 mm	66.577	17.002	0.255	0.270	0.270	4.591	
>95 mm	87.953	24.520	0.279	0.270	0.270	6.620	

The recommended currency is morphometrically mature male biomass, giving an OFL of 30.765 kt and, after the 20% buffer, an ABC of 24.612 kt. [[author: the September 2025 assessment recommended the ≥ 95 mm currency instead, on the grounds that morphometrically mature male biomass “results in reference points that allow for the complete removal of large males”. The choice of currency moves the OFL by more than a factor of four (Table 6) – far more than either the control-rule or catch-equation choice – so the reversal needs an explicit justification here.]]

The biomass the control rule acts on carries real uncertainty, and the guidelines require it to be reported. The random-effects fit puts terminal morphometrically mature male biomass at 113.94 kt with a 95 percent interval of 85.36 to 152.09 kt, a coefficient of variation of 14.8 percent. This is the prediction interval REMA returns for the smoothed series, which reflects the process error the random-effects model estimates. It is not the sampling uncertainty of the survey itself: the survey CVs are an input to the fit, not its output, and the two should not be quoted interchangeably. The interval is lognormal, so the log-scale standard deviation is recovered as (log of the upper bound minus log of the lower bound) divided by 2×1.96 , and the coefficient of variation follows from it.

Propagating that interval through the control rule, with F_{OFL} flat at M across its whole range, gives an OFL of 30.76 kt with a range of 23.05 to 41.07 kt, and a stock status of 0.769 with a range of 0.576 to 1.026. Two features of that status range matter. The lower bound remains above the minimum stock size threshold of 74.09 kt, so the stock is not overfished at any point in the interval. But the upper bound reaches 1.026, so on this measure the biomass at which the stock would be considered rebuilt lies inside the interval, even though the point estimate sits below it. The rebuilding determination in Section H should be read with that in mind.

[[author: this interval covers the random-effects smoothing only. It does not include uncertainty in the B_{MSY} proxy, which is the mean of the same smoothed series and is therefore correlated with the terminal estimate, nor uncertainty in the choice of currency or averaging window, both of which move the OFL by more than this interval does. Say so, or widen the treatment.]]

Two alternatives are reported alongside the recommended basis. The first applies the sloped control rule between the closure threshold and B_{MSY} rather than a flat $F = M$; at the current status this lowers F_{OFL} from 0.2700 to 0.2007. The second computes the OFL from the Baranov catch equation, treating F_{OFL} as an instantaneous rate acting alongside natural mortality over the year, rather than as an annual exploitation fraction. The flat rule is the recommended basis. It follows the groundfish Tier 5 FMP, under which the target rate is the natural mortality rate itself with no ramp, and it follows the Tier 4 calculation presented in the September 2025 assessment, which states explicitly that no kink exists in that harvest control rule. This differs from the basis the SSC used when it set the 2025/26 specification: at a stock status of 0.74 the SSC reported $F_{OFL} = 0.19$ (October 2025 SSC report, Table 2), where flat $F = M$ would have given 0.27. The sloped rule at that status returns $M \times (0.74 - 0.10) / (1 - 0.10) = 0.192$, and the specified OFL of 20.11 kt divided by the MMB of 105.213 kt implies 0.1911, so the specification was set on the sloped rule. The SSC did use the linear catch equation, as this assessment does; the Baranov form would have returned 16.02 kt rather than 20.11 kt. At this cycle’s status of 0.769 the difference between the two rules is an OFL of 30.76 kt against 22.87 kt, and the sloped result is reported above so that the comparison is available. [[author: the divergence from the 2025/26 specification will be asked about. State the argument for the flat rule here

in your own terms – the FMP basis is recorded above, but the reason for departing from the SSC’s most recent application is not.]]

Parameters estimated independently of the assessment

The Tier 4 calculation does not estimate biological parameters. The quantities it takes as given, and their sources, are:

- **Natural mortality**, $M = 0.27 \text{ yr}^{-1}$ for mature males. This is the median of the prior on natural mortality used in the integrated assessment, and it serves as both the target fishing mortality and, under Tiers 4 and 5, the F_{limit} .
- **The probability of terminal molt at size**, supplied as a fixed input from the Kodiak laboratory’s spatiotemporal sdmTMB maturity workflow rather than estimated within the model.
- **Growth per molt and the variance in growth increment**, estimated outside the model from the growth data.
- **Weight at length**, specified from the published relationship.
- **Discard and bycatch mortality rates**, specified rather than estimated.

F_{limit}

Under Tiers 4 and 5 the fishing mortality rate above which the stock is considered to be subject to overfishing is the natural mortality rate itself. For this stock $F_{\text{limit}} = M = 0.27 \text{ yr}^{-1}$. This is a different quantity from the reverse-engineered F reported for Tier 1–3 stocks, which is the rate that would have produced a catch equal to the previous year’s OFL, and that quantity is not computed here.

G. Calculation of the ABC

The recommended acceptable biological catch (ABC) is calculated by subtracting a buffer from the OFL to account for scientific uncertainty. The buffer applied here is 20 percent, giving an ABC of 24.61 kt against an OFL of 30.76 kt. This buffer accounts for scientific uncertainty not directly captured in the assessment, including retrospective patterns and model misspecification.

[[author: the maximum permissible ABC must be reported regardless of the author’s recommendation, with its calculation described (groundfish guidelines 4.11.2; the crab checklist says the same). It is not stated here because the maximum permissible buffer for a Tier 4 crab stock could not be established from the documents to hand — the June 2026 SSC notes only that the default crab control rule gives a minimal buffer between maxABC and the OFL at $P^* = 0.49$. Supply that basis and report maxABC alongside the recommended ABC.]] The June 2026 SSC supported reporting a stock-specific history of the buffers applied to the OFL, and suggested that a historical buffer could initially be set at the average buffer from the previous five assessments (Table 7).

Table 7: History of the ABC buffer applied by the SSC, as a percentage of the OFL. The mean over the 5 recoverable years is 41%. The 2022 value could not be sourced, so this is the average of five assessments but not of the five most recent ones.

Year	SSC buffer (% of OFL)	Source
2,020	25%	2025 SAFE, Section G
2,021	25%	2025 SAFE, Section G
2,022	NA	October 2022 SSC reports are not available online
2,023	50%	2025 SAFE, Section G
2,024	65%	2025 SAFE, Section G
2,025	40%	October 2025 SSC report, Table 2

The 20 percent buffer recommended here is the figure the September 2025 assessment also recommended, and is the buffer historically applied to crab stocks whose data and model complexity are sufficient for Tier 3. It is lower than the buffer the SSC has applied to this stock in recent years. For the 2025/26 specification the SSC applied a 40 percent buffer, setting an ABC of 12.07 kt against an OFL of 20.11 kt (October 2025 SSC report, Table 2), and against the recoverable history in Table 7 a 20 percent buffer would be the smallest applied to this stock in any of the six years shown and less than half the 41 percent average over the five recoverable years. Three of the four risk table categories score Level 1, Normal (Table 8). [[author: the reduction from 40 percent to 20 percent is the single largest departure in this document from the last specification, and it moves the ABC in the same direction as the control-rule choice rather than offsetting it. Compounded, the two put the recommended ABC at 24.61 kt where the basis the SSC used for 2025/26 – sloped rule, 40 percent buffer – would give 13.72 kt on this cycle’s biomass, a factor of 1.79. That is the number the CPT will arrive at, so the document should reach it first. It needs your argument, and the assessment-related risk category – still unscored – is where the supporting evidence for it belongs. The June 2026 SSC cautioned against a predetermined direction in modifying any buffer.]]

Risk table

The risk table follows the four categories used for groundfish, in the tiered form the June 2026 SSC described: a long-term row capturing persistent uncertainty, which is what justifies a base or historical buffer, and a current-year row for year-specific risk. The SSC scopes the long-term row to the assessment and population-dynamics categories only, which is why the ecosystem and fishery cells in that row are not scored.

The population-dynamics, ecosystem and fishery-informed cells below are taken from the 2026 risk table summary prepared from the ESP (Fedewa and Shotwell 2026), the AFSC Spring Preview of Ecosystem and Economic Conditions Workshop, and the eastern Bering Sea Ecosystem Status Report (Siddon 2026, in press). The ESP is Appendix A to this assessment.

Table 8: Risk table for eastern Bering Sea snow crab. Levels for the population dynamics, ecosystem and fishery-informed categories are from the 2026 ESP/ESR risk table summary; the assessment-related category is for the author to score. The long-term row is scoped to the first two categories, following the June 2026 SSC.

Category	Long-term / persistent	Current year
Assessment-related	[[author: assign a level. The evidence: multimodality has persisted across cycles and is the reason the stock moved from Tier 3 to Tier 4. Jittering the accepted model returns to the best mode in 5.1 percent of runs. A likelihood improvement of 1.27 units moved the OFL of Model 25.2c by 12.8 percent. Successive assessments have revised the same historical years by up to an order of magnitude (Section E, Historical analysis), which is far larger than the within-model retrospective pattern. The base model estimates on the order of four hundred parameters, many highly correlated, which the June 2026 SSC identified as consistent with overparameterisation.]]	[[author: assign a level. The evidence: the OFL rests on a Tier 4 calculation whose terminal biomass carries a CV of 14.8 percent, a 95 percent interval of 85.4 to 152.1 kt, and a status interval of 0.58 to 1.03 that spans the rebuilt threshold. Two specification choices move the OFL by more than that interval does: the currency, by more than a factor of four, and the control rule, from 30.8 to 22.9 kt. Against that, every model that could be polished attains a zero maximum gradient, all Hessians inverted, and Mohn’s rho stays inside the Hurtado-Ferro bounds.]]
Population dynamics	[[author: assign a level and supporting evidence. Not scored in the ESP summary.]]	Level 1: Normal. Stock-specific indicators related to natural mortality, growth and recruitment suggest no apparent population dynamics concerns, although a highly female-skewed sex ratio combined with directional downward shifts in male maturity and depressed abundances of large males should continue to be monitored.
Ecosystem	Not scored (long-term row is scoped to the first two categories)	Level 1: Normal. Thermal conditions from winter 2025 to summer 2026 have been favorable, with an extensive cold pool and high sea ice extent. Pelagic feeding conditions have remained favorable since 2023; benthic prey indicators were mixed in 2025. Competition for space remained high, competition for prey low. Predation pressure from Pacific cod was reduced in 2025. Spring 2026 offshore transport moved crab away from suitable benthic habitat. Overall, ecosystem concerns are minor with uncertain impacts on the stock.

Table 8: Risk table for eastern Bering Sea snow crab. Levels for the population dynamics, ecosystem and fishery-informed categories are from the 2026 ESP/ESR risk table summary; the assessment-related category is for the author to score. The long-term row is scoped to the first two categories, following the June 2026 SSC.

Category	Long-term / persistent	Current year
Fishery-informed stock	Not scored (long-term row is scoped to the first two categories)	Level 1: Normal. Fishery-informed indicators generally support stable stock condition. Shifts in fleet behavior appear motivated primarily by weather, sorting, fuel costs and TAC considerations rather than stock health or distribution. Many skippers noted increased hybrid abundance during the 2025/26 fishery, consistent with the increase in hybrid abundance observed on the 2025 EBS bottom trawl survey.

[[author: the risk table’s purpose is to say whether the evidence supports reducing the ABC below the maximum permissible, and if so by how much – either a specific reduction or a recommendation that the SSC determine it. Three of the four categories score Level 1 (Normal). The assessment-related category is the one carrying the weight for this stock, and it is unscored. Complete it, then state the conclusion the table supports for the buffer.]]

Status determination

Is the stock subject to overfishing? No. Overfishing is determined by comparing the catch from the most recent complete crab year with the OFL specified for that year. Total catch in 2025/26 was 7.10 kt — 4.13 kt retained, 2.52 kt of male discards, 0.03 kt of female discards and 0.42 kt of trawl bycatch — against the OFL of 20.11 kt that the SSC specified for 2025/26 in October 2025. Catch was 35 percent of the specified OFL.

Is the stock overfished? The stock was determined to be overfished for 2024/25, with mature male biomass of 60.08 kt against a minimum stock size threshold of 71.40 kt (October 2025 SSC report, Table 1), and it remains in the rebuilding plan adopted under Amendment 53. On this cycle’s estimate mature male biomass is 113.94 kt against a minimum stock size threshold of 74.09 kt, so the stock is above the threshold on the current estimate and across the whole of its 95 percent interval (Section F).

Is the stock approaching an overfished condition? [[author: this determination is required of the author and is not made here. Note that the groundfish guidelines scope both the overfished and the approaching-overfished statements to Tiers 1–3; the crab checklist does not, and the stock is in a rebuilding plan, so both are stated.]]

[[author: the catch convention needs one check. The figure above is observed catch. The SSC’s Table 1 reports total catch of 2.81 kt for 2024/25 where the same summation on the assessment’s own inputs gives 2.90 kt, so the SSC applies a handling-mortality weighting this does not. The determination is unaffected — catch is far below the OFL on either convention — but match the convention before the table is published.]]

Author recommendations

The June 2026 SSC recommended that only the Tier 4 model be brought forward in October, to allow time for the development of a simpler and more stable Tier 3 model. The overfishing limit in this document is calculated on that basis. The Tier 3 models are presented to document progress and are not proposed for specification.

Three questions the May 2026 preliminary assessment put to the CPT and SSC have since been settled and are not reopened here. The composition correction and the expanded plus group were endorsed as technical corrections and are carried in every September model. The immature survey index was rejected as a convergence fix, on the grounds that it shifts the model toward higher natural mortality among mature males and raises a question of duplicative use of the same survey data. Hybrid catch data were judged premature for inclusion in the snow and Tanner crab assessment models, with hybrid-related uncertainty to be handled instead through the risk table, the ABC buffer, or the TAC-setting process.

[[author: state the recommended Tier 4 configuration – currency, control rule, and buffer – and the reasoning. Three decisions are open and each moves the OFL materially: (a) the currency. Morphometric maturity gives 30.76 kt against 6.62 kt on ≥ 95 mm and 4.59 kt on > 101 mm (Table 6). The 2025 assessment recommended ≥ 95 mm, arguing that morphometric reference points permit the removal of essentially all large males while the trend in large males is firmly negative. That argument has not been answered here; (b) the control rule. This document applies a flat $F = M$ above the closure threshold, which follows the 2025 Tier 4 calculation and the groundfish Tier 5 FMP. At a status of 0.769 the stock is below B_{MSY} and under a rebuilding plan, where the crab FMP subtier would ordinarily ramp; the sloped alternative is reported alongside; (c) the buffer. See the ABC section.]]

[[author: the September 2026 meeting notes call for a discussion of author priorities against SSC priorities and where they diverge. The SSC's stated priority is a simplified Tier 3 model – fixing the recruitment sex ratio at 50:50, reducing confounding among growth, natural mortality and selectivity, estimating growth externally, and considering an earlier equilibrium start year – in preference to further iterations of the current base model. The male-only model (Model 26.1) is a first step in that direction and is reported here. State where the author's priorities differ and why.]]

A yield-per-recruit analysis, requested by the SSC in June 2025 and again in June 2026 to inform development of an alternative harvest control rule, has not been completed for this cycle. [[author: give the reason and a date.]]

H. Rebuilding analysis and update

[[author: this section is required both by the SAFE guidelines and by the rebuilding plan itself – Amendment 53 states that the BSAI Crab Plan Team would report stock status and progress towards the rebuilt level in this document. The September 2025 assessment did not include it, which the SSC noted in October 2025. Placed here un-lettered pending the section-lettering decision; the guidelines number it H.]]

Status of the rebuilding plan

On 19 October 2021 NMFS determined, and notified the Council, that the eastern Bering Sea snow crab stock was overfished, the estimated mature male biomass having fallen below the minimum stock size threshold specified in the fishery management plan. The Council developed a rebuilding plan under section 304(e) of the Magnuson-Stevens Act, adopted as **Amendment 53** to the Fishery Management Plan for Bering Sea/Aleutian Islands King and Tanner Crabs and implemented before the start of the 2023/24 fishing season.

The rebuilding analysis set the minimum time to rebuild in the absence of fishing ($T_{\min}$) at 6 years, which under the National Standard 1 guidelines sets a maximum time to rebuild ($T_{\max}$) of 10 years. **The stock is**

considered rebuilt when mature male biomass reaches B_{MSY} . Median projections placed rebuilding in 2029 under no fishing, under bycatch only, and under the State harvest strategy, and in 2030 under fishing at the ABC. The 5th inter-simulation percentile is 2027 across all scenarios. The 95th is 2034 under no fishing and under bycatch only; under the State harvest strategy and under fishing at the ABC the upper tail does not rebuild within the projection horizon. If mortality events of the kind observed in 2018–19 recur in roughly one year in seven, the projected rebuilding time extends to 2034. The analysis concluded that the main driver of the rate of rebuilding is recruitment and the ecosystem conditions that permit it, rather than fishing mortality.

Relationship to previous rebuilding plans

This is the third rebuilding plan for the stock. Snow crab was first declared overfished on 24 September 1999, and a rebuilding plan was adopted as Amendment 14 in 2000. The Amendment 53 environmental assessment’s reference list cites Amendment 15 for that plan, but the 2000 environmental assessment itself is titled for Amendment 14 (NPFMC, ADF&G and NMFS staff, December 2000), and Amendment 14 is the citation carried here. In 2009 the NMFS Alaska Region notified the Council that the stock would not be rebuilt by the end of that rebuilding period, 2009/10, and that a revised plan was required; Amendment 39 followed. Amendment 53 replaces those earlier provisions, which were removed from the fishery management plan when it was adopted.

Progress toward rebuilding

[[author: state progress on the basis actually adopted for the specification. The quantities below are from the Tier 4 calculation on the recommended currency and will need to be restated if the currency or the control rule changes.]]

Mature male biomass is estimated at 113.94 kt against a B_{MSY} proxy of 148.18 kt, or 77 percent of the proxy, with a minimum stock size threshold of 74.09 kt. The stock is therefore above its minimum stock size threshold but has not, on the point estimate, reached the biomass at which it would be considered rebuilt. Accounting for the uncertainty in the random-effects fit (Section F), status is 0.769 with a 95 percent range of 0.576 to 1.026. The lower end remains above the threshold for an overfished determination; the upper end reaches the rebuilt threshold. On this basis the stock cannot be declared rebuilt, but neither can rebuilding be excluded on the terminal-year estimate alone.

Recruitment

Because the rebuilding analysis identifies recruitment as the main driver of the rebuilding rate, recruitment is reported here directly. Estimated male recruitment collapsed to 0.088 and 0.065 billion crab in 2018 and 2019, among the lowest in the time series, before rising to 1.65, 2.36 and 4.28 billion in 2020, 2021 and 2022 and settling near 1.0 billion in 2024 and 2025. The mean over the last five years, 2.01 billion, is above the 1982–2019 mean of 1.22 billion.

That comparison should be treated with caution. Terminal-year recruitment is the least well determined quantity in the assessment: the retrospective analysis gives a Mohn’s rho for male recruitment of 7.89, meaning that recent recruitment estimates revise substantially as data accumulate. The apparent post-collapse recruitment pulse is consistent with the survey observing small crab, but its magnitude in the terminal years should not be relied on. [[author: compare against the survey-derived recruit index produced by `04_plot_recruitment_comparison.R` (`plots/recruitment_survey_vs_model.png`), which is not yet included as a numbered figure; the correlation between the model’s estimated recruitment and the survey recruit index over the overlapping period is negative, which deserves comment.]]

I. Data gaps and research priorities

Several data and structural issues identified warrant follow-up:

- **Hybrid identification consistency between ADF&G and NMFS.** ADF&G classifies hybrid crab into three categories (opilio-type, bairdi-type, and unspecified) using a hybrid identification key based on eye color, epistome shape, and secondary morphological characteristics. ADF&G and NMFS do not currently agree on the number of secondary characteristics required for a positive hybrid identification, and the unspecified-hybrid category are present in a non-trivial portion of the time series. Resolving identification criteria across agencies and propagating the resulting categorization back through the historical observer time series would improve the hybrid-data products available to the assessment. However, the value add to the snow crab assessment may be minimal given the low percentage of opilio-type hybrids in the data.
- **Apportionment of retained catch between snow crab and hybrids.** Hybrids are currently retained under either the snow-crab or the Tanner-crab IFQ depending on regulatory identification but are not separately tracked in the retained-catch data. ADF&G has indicated it does not yet have a method for partitioning historical retained catch into hybrid versus parent species. Developing such a partitioning would allow the hybrid sensitivity to be explored more fully (presently the retained time-series is identical between the hybrid and non-hybrid models because retained catch already contains hybrids).
- **Quality control on early hybrid catch estimates.** The ADF&G hybrid total-catch estimates contain unexpectedly large values in 1997 (~18 kt of legal-male hybrid catch) and 1998 (~15 kt), with high-CPUE pots peppered around the fishery footprint rather than concentrated in obvious hot spots. ADF&G has flagged these years as requiring follow-up before the data should be considered reliable for regular assessment use.
- **Maturity workflow comparison.** The new spatiotemporal maturity workflow produces a much weaker temporal trend in the probability of terminal molt at size than the legacy workflow. Side-by-side documentation of the two workflows, an evaluation of the spatial sample sizes of chela measurements (which are sparse at many stations and many sizes), uncertainty, and quantification of the degrees of freedom consumed by the spatial process would help the CPT and SSC evaluate the two approaches.
- **Source of bimodality in the snow crab assessment.** Although the immature survey index appears to remove the bimodal jitter signal in the May 2026 candidate models, the underlying source of multiple optima is not understood. A focused investigation of which parameter combinations produce the two clouds (e.g., recruitment in the most recent years, M events, or selectivity parameters) would help clarify whether the immature index is solving a real problem or is masking it.
- **Reproductive biology of large male snow crab.** As noted in 2024 and 2025 SSC comments, more research is needed on whether small males can and do mate with large females and produce progeny that will be large males under favorable environmental conditions. This question remains the main axis of uncertainty because the fishery cannot exist without large males and large males are at historical lows.

J. Ecosystem considerations

See the ESP for snow crab specific indices of environmental variation that may be relevant to stock dynamics (Appendix A). See the risk table (Table 8) for factors that may influence the choice of buffer. [[author: the crab checklist lists the buffer history and the risk table as an appendix; both currently sit in Section G, where the ABC reasoning uses them. Move them if the appendix placement is preferred.]] More broadly, scientific frameworks that quantify tradeoffs among fisheries would be useful in decision making and scientific communication.

Supplemental information

Input and output for the models described here can be found at https://github.com/szuwalski/snow_sept.

The new maturity workflow can be found at <https://github.com/eryznar/Chionoecetes.maturity.workflow>.

GMACS code and documentation can be found at: <https://github.com/GMACS-project>.

Acknowledgements

[[author: the guidelines ask for reviewers and their affiliations, together with the names and affiliations of everyone who contributed data, advice or information without being part of the assessment team. For this cycle that plausibly includes the Kodiak laboratory for the maturity workflow, ADF&G for the fishery removals and the hybrid identification caveats, the NMFS eastern Bering Sea bottom-trawl survey team, and the maintainers of GMACS and crabpack. Names and affiliations are yours to supply.]]

Auxiliary files

The following files accompany this assessment and are archived with it. All are in the native format of the assessment program.

- `26_snow_update_newmat_plus_group.dat` — the model data file.
- `snow.ct1` — the control file, naming the estimated parameters, their bounds and phases.
- `snow.prj` — the projection file.
- `gmacs.dat` and `gmacs_files_in.dat` — the run configuration and the file manifest GMACS reads.
- `gmacs.pin` — the starting parameter vector. For the accepted fit this is the jitter winner’s parameter vector rather than a cold start; its provenance is recorded in `jitter/PROMOTION.md`.
- `gmacs` — the model executable (GMACS 2.20.34, macOS arm64). `gmacs.exe` is the corresponding Windows build.
- The six derived input files that form the interface between the data preparation scripts and the model: `directed_catch.csv`, `bycatch_catch.csv`, `fishery_size_comps.csv`, `survey_size_comps.csv`, `survey_indices.csv` and `male_maturity_ogive.csv`.

[[author: confirm this list against what is actually submitted for archiving, and add the equivalent files for any model other than the accepted one that is to be archived.]]

Appendices

Appendix A. Ecosystem and Socioeconomic Profile. The ESP is provided as a separate document: Fedewa, E. and Shotwell, K. 2026. *Ecosystem and Socioeconomic Profile of the eastern Bering Sea snow crab stock — Update*. It is the source of the ecosystem and fishery-informed indicators summarised in the risk table (Table 8) and of the indicator series referenced in Section J.

Appendix B. Removals from sources other than commercial catch. [[author: an appendix of non-commercial removals is required (groundfish guidelines 4.23; the internal review checklist asks for an appendix table of removals from sources other than official catch estimates). No such data exists anywhere in this repository — `data/adfg_removals/` holds directed retained catch, directed discards, trawl bycatch and other-crab bycatch, all of which are commercial removals already carried in the model, and the September 2025 assessment does not report non-commercial removals either. Survey removals, and any research, subsistence or personal-use take, would need to be obtained before this table can be built. State the amounts, or state that they are negligible and on what basis.]]

K. Literature cited

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Tables

Table 9: Overview of the models presented in the September 2026 assessment, w

Model	Updates	Type	Purpose
Model 25	25.1 gmacs	Reference	The 2025 accepted model (G
Model 25.2c	25.1 gmacs (update + compfix + plus group + new_mat)	Research	Carries the composition corr
Model 25.2e	25.1 gmacs (update + compfix + plus group + new_mat + data-fix)	Research	As above with the data corr
Model 25.2e (new data)	26.1 gmacs (update + compfix + plus group + new_mat)	Research	The same configuration adva
Model 26.1	26.1 gmacs (update + compfix + plus group + new_mat + male_only)	Research	Simplification sensitivity: re
Model 25.2e (2019 data)	26.1 gmacs (update + compfix + plus group + new_mat + data2019)	Diagnostic	Convergence diagnostic: data

Table 10: Catch and index likelihood components (negative log-lik

Model	Catch: Pot retained M	Catch: Pot discarded M	Catch: Pot discarded F	Catch: Trawl bycatch	Index
Model 25	120.98	377.21	-71.40	-54.11	
Model 25.2c	124.87	379.02	-71.40	-54.11	
Model 25.2e	129.18	310.35	-71.40	-54.09	
Model 25.2e (new data)	127.02	279.00	-73.14	-55.47	
Model 26.1	133.42	290.59	-55.47	0.00	
Model 25.2e (2019 data)	69.27	107.30	-66.18	-47.15	

Model	Size: Pot M retained	Size: Pot M total	Size: Pot F discarded	Size: Trawl F discarded	Size: Trawl
Model 25	-3,645	-2,639	-2,529	-2,606.0	
Model 25.2c	-3,680	-2,670	-2,528	-2,606.0	
Model 25.2e	-3,810	-2,879	-2,684	-2,550.0	
Model 25.2e (new data)	-3,900	-2,958	-2,757	-2,623.0	
Model 26.1	-3,892	-2,943	-2,673	-560.8	
Model 25.2e (2019 data)	-3,425	-2,645	-2,502	-2,151.0	

Table 12: Other likelihood components (negative log-likelihood, dimensionless) by category and model.

Model	SRR	Tag	Growth	Penalties	Priors	Positive_growth_penalty	Total
Model 25	185.4	7,965	0	760.4	387.0	0	-19,170
Model 25.2c	192.2	0	7,970	761.0	395.0	0	-19,230
Model 25.2e	191.1	0	4,882	754.1	382.7	0	-22,840
Model 25.2e (new data)	197.8	0	4,884	757.6	385.7	0	-23,550
Model 26.1	123.6	0	1,532	406.5	183.1	0	-14,010
Model 25.2e (2019 data)	170.3	0	4,877	752.2	377.6	0	-19,740

Table 13: Maximum absolute value gradient (dimensionless) and corresponding parameter for each candidate model.

Model	N parameters	Parameter	Max gradient
Model 25	407	theta[3]	0.00134
Model 25.2c	407	T_pars_est[3]	0.00000
Model 25.2e	407	T_pars_est[3]	0.00000
Model 25.2e (new data)	412	T_pars_est[3]	0.00000
Model 26.1	234	T_pars_est[3]	0.00000
Model 25.2e (2019 data)	386	T_pars_est[3]	0.00000

Table 14: Retrospective statistics by configuration and model. Mohn’s rho and Ralston’s sigma are dimensionless. Mohn’s rho is the mean over peels of (peel - full model) / full model at each peel’s terminal year, so a positive value means terminal estimates are revised downward as data accumulate. Ralston’s sigma is the log-scale RMSE of the same terminal-year pairs. The dropped-terminal-survey peels are compared against the full model fit to all data.

Quantity	Retrospective	Peels	Mohn’s rho	Ralston’s sigma	Min rel. diff.	Max rel. diff.
Mature male biomass	Standard	10	-0.109	0.301	-0.493	0.139
Mature male biomass	Terminal survey dropped	10	-0.110	0.281	-0.493	0.118
Male recruitment	Standard	10	7.893	2.171	-0.922	32.027
Male recruitment	Terminal survey dropped	10	9.508	2.118	-0.567	33.148

Table 15: Comparison of observed retained catches, discards, and bycatch between ba (1,000 t). [[author: the hybrid cases are not in the September model set – the CPT and catch data premature in May and June 2026. Retain this table only as a record of what say so, or drop it.]]

Year	Retained (kt)	Non-directed Bycatch	Discard Male Pot	Discard Male Pot + hybrids
1,982	11.85	0.37	1.27	1.27
1,983	12.16	0.47	1.24	1.24
1,984	29.94	0.50	2.76	2.76
1,985	44.45	0.43	4.01	4.01
1,986	46.22	0.00	4.25	4.25
1,987	61.40	0.00	5.52	5.52
1,988	67.79	0.00	5.82	5.82
1,989	73.33	0.10	6.68	6.68
1,990	149.07	0.33	35.55	35.55
1,991	143.02	4.45	9.03	9.03
1,992	104.67	2.05	21.18	21.18
1,993	67.94	1.13	22.27	22.27
1,994	34.16	0.70	7.78	7.78
1,995	29.80	1.04	14.73	14.73
1,996	54.24	1.22	23.23	23.23
1,997	110.44	0.73	7.10	7.10
1,998	88.15	0.58	19.50	19.91
1,999	15.10	0.24	4.13	4.13
2,000	11.46	0.25	3.25	3.25
2,001	14.80	0.18	3.98	3.98
2,002	12.84	0.10	4.50	4.50
2,003	10.86	0.14	2.40	2.45
2,004	11.29	0.18	3.58	3.58
2,005	16.77	0.19	0.62	0.62
2,006	16.49	0.36	4.17	4.17
2,007	28.59	0.31	5.77	5.77
2,008	26.56	0.19	5.11	5.11
2,009	21.78	0.39	4.28	4.28
2,010	24.61	0.14	4.47	4.47
2,011	40.29	0.14	3.73	3.73
2,012	30.05	0.15	5.53	5.53
2,013	24.49	0.16	10.61	10.61

Table 15: Comparison of observed retained catches, discards, and bycatch between ba (1,000 t). [[author: the hybrid cases are not in the September model set – the CPT and catch data premature in May and June 2026. Retain this table only as a record of what say so, or drop it.]]

Year	Retained (kt)	Non-directed Bycatch	Discard Male Pot	Discard Male Pot + hybrids
2,014	30.82	0.62	11.62	11.62
2,015	18.42	1.58	10.90	10.90
2,016	9.78	0.04	4.51	4.51
2,017	8.60	0.10	5.88	5.88
2,018	12.51	0.40	8.63	8.62
2,019	15.43	0.21	15.61	15.61
2,020	20.41	0.19	6.12	6.12
2,021	2.52	0.13	1.68	1.68
2,022	0.00	0.06	0.00	0.00
2,023	0.00	0.11	0.00	0.00
2,024	2.15	0.09	0.66	0.66
2,025	0.00	0.00	0.00	2.52

Table 16: Management quantities by model, derived from maximum likelihood estimates using Tier 3 reference points. Biomass and OFL in 1,000 t; fishing and natural mortality rates in per year. Reported natural mortality is for immature and mature males in the terminal year of the assessment, average recruitment is for males, and status and MMB were estimates for February 15 of the completed crab year.

Model	BMSY	Bcurr/BMSY	Fmsy	Fofl	OFL	M (imm males)	M (mat males)
Model 25	180.06	0.88	39.52	34.37	44.29	0.29	0.28
Model 25.2c	172.67	0.96	43.35	41.23	43.83	0.30	0.29
Model 25.2e	158.70	0.81	39.10	30.76	53.70	0.21	0.29
Model 25.2e (new data)	144.97	1.32	55.16	55.16	85.65	0.23	0.29
Model 26.1	131.34	1.23	20.54	20.54	77.00		0.29
Model 25.2e (2019 data)	161.29	1.05	135.66	135.66	147.28	0.08	2.26

Figures

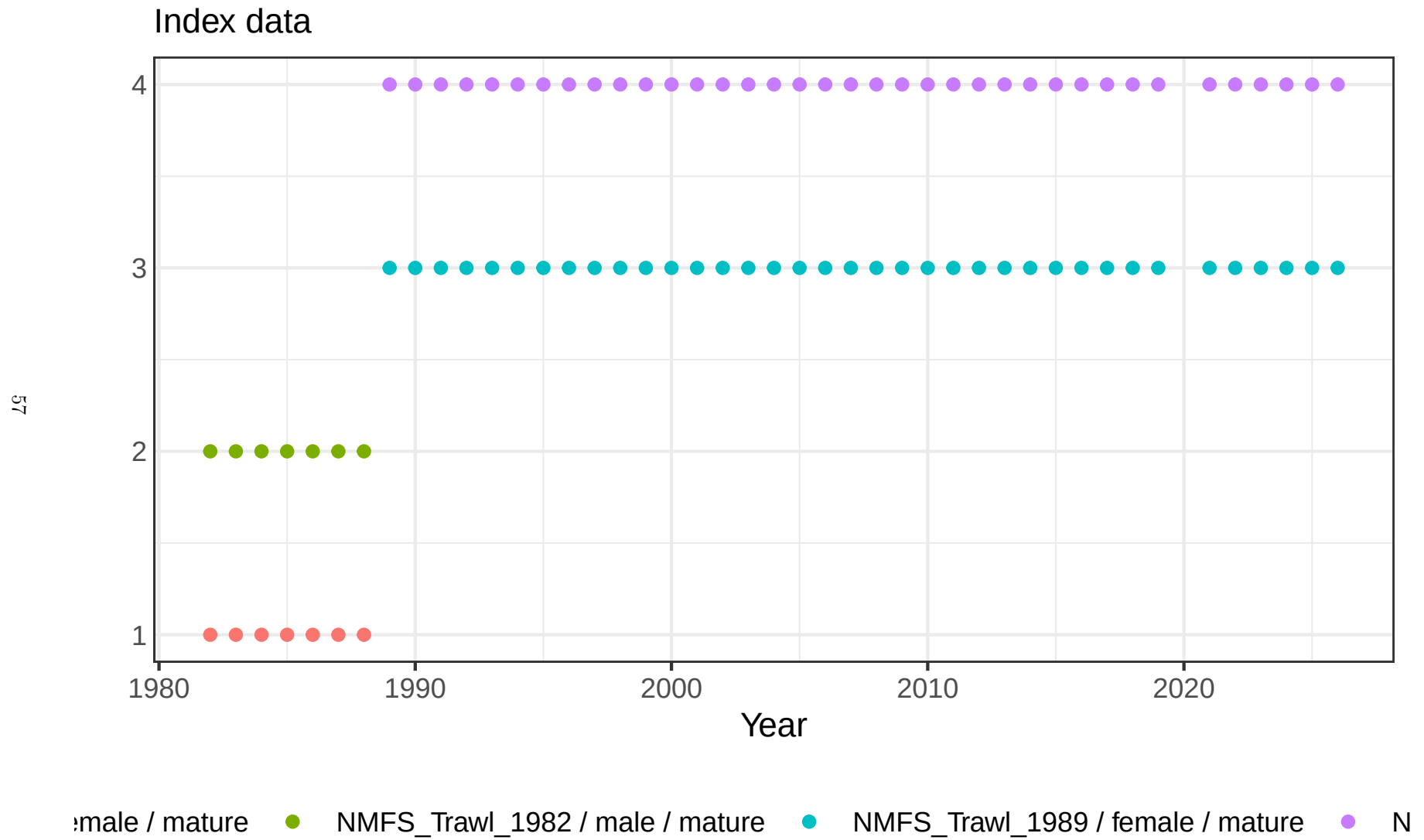


Figure 2: Time series of index data used in the assessment.

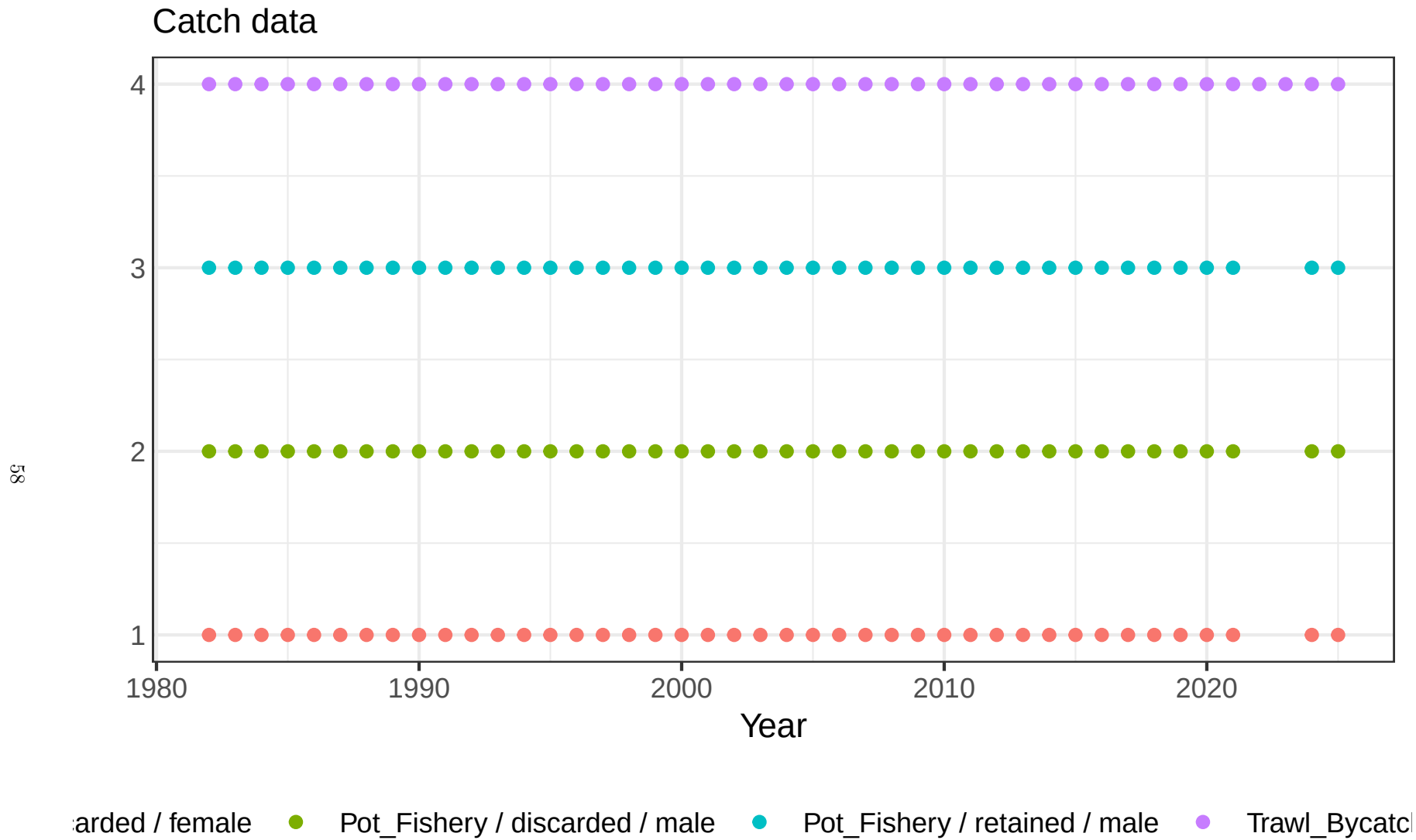
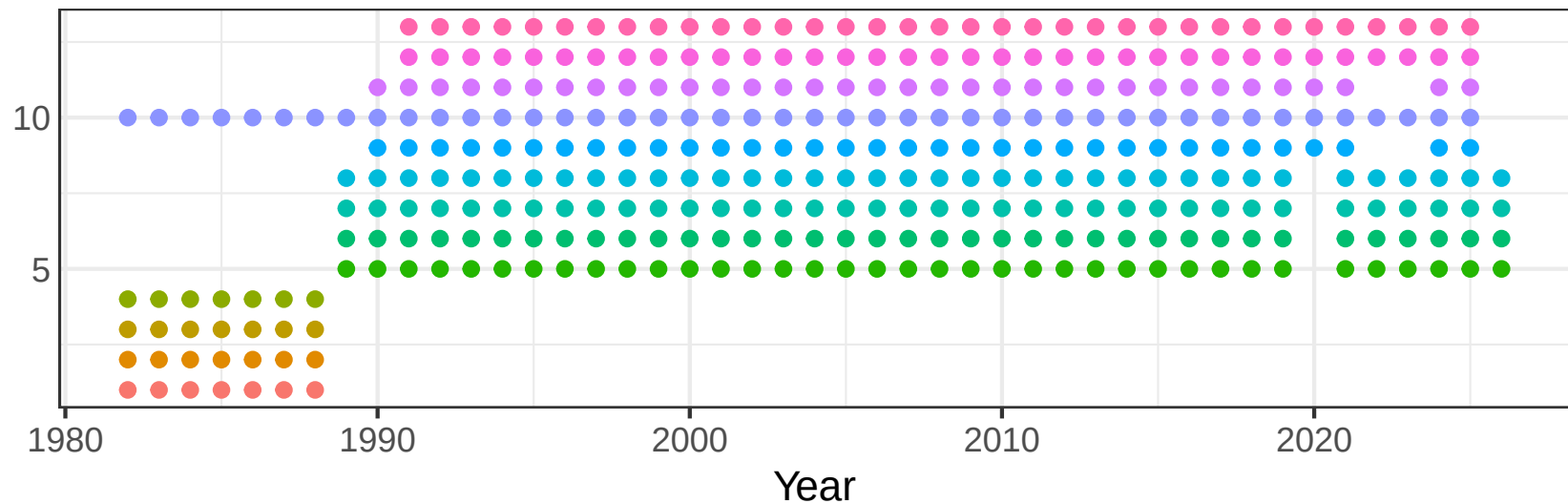


Figure 3: Time series of catch data used in the assessment.

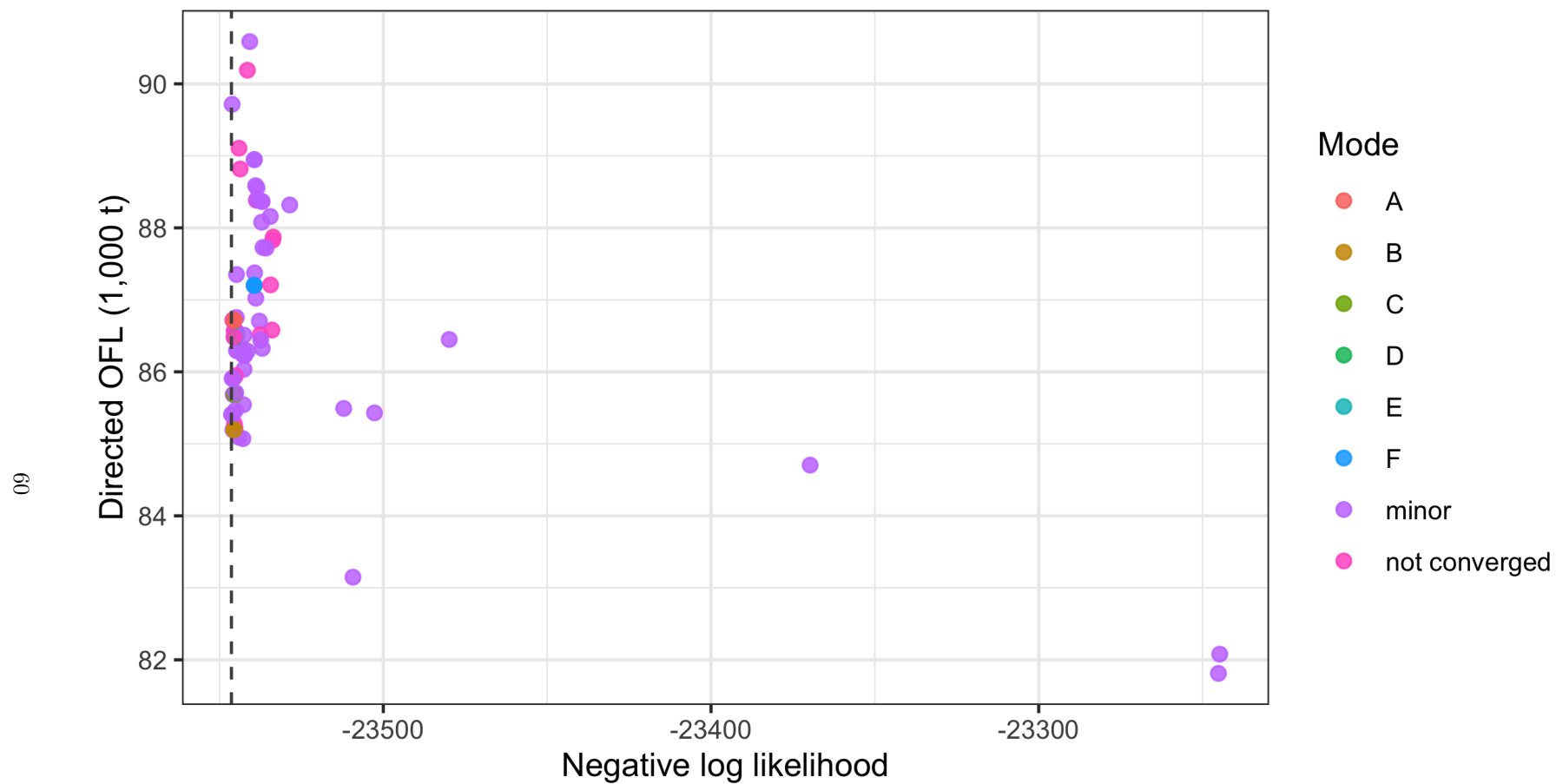
Composition data



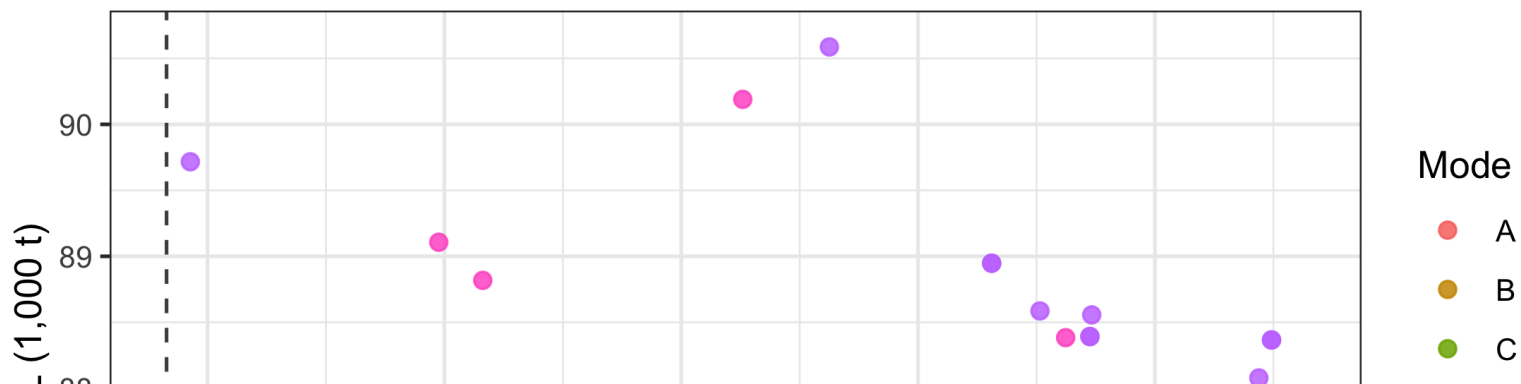
- | | | |
|-------------------------------|---|---------------------|
| discarded / female / immature | ● NMFS_Trawl_1989 / discarded / female / mature | ● Pot_Fishery / tot |
| discarded / female / mature | ● NMFS_Trawl_1989 / discarded / male / immature | ● Trawl_Bycatch / |
| discarded / male / immature | ● NMFS_Trawl_1989 / discarded / male / mature | ● Trawl_Bycatch / |
| discarded / male / mature | ● Pot_Fishery / discarded / female / undetermined | |
| discarded / female / immature | ● Pot_Fishery / retained / male / undetermined | |

Figure 4: Time series of composition data used in the assessment.

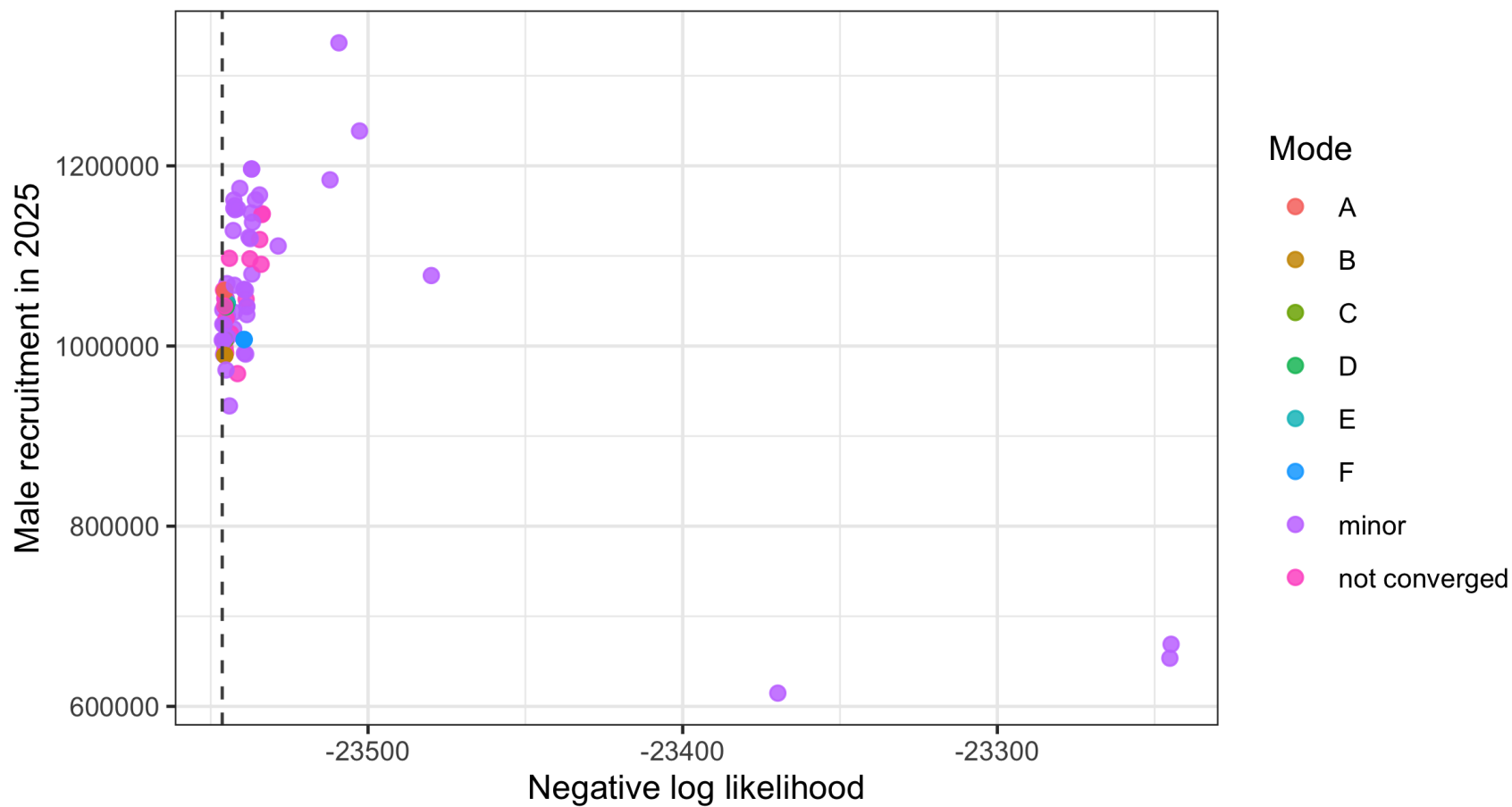
100 jitter runs, sd 0.1 (dashed line = base fit)



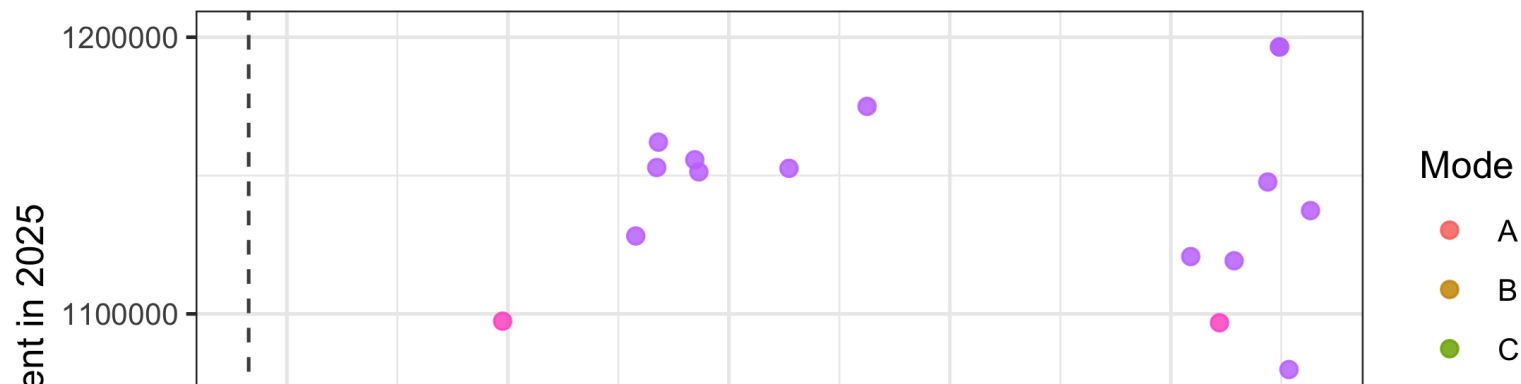
Runs within 10 nll units of the best



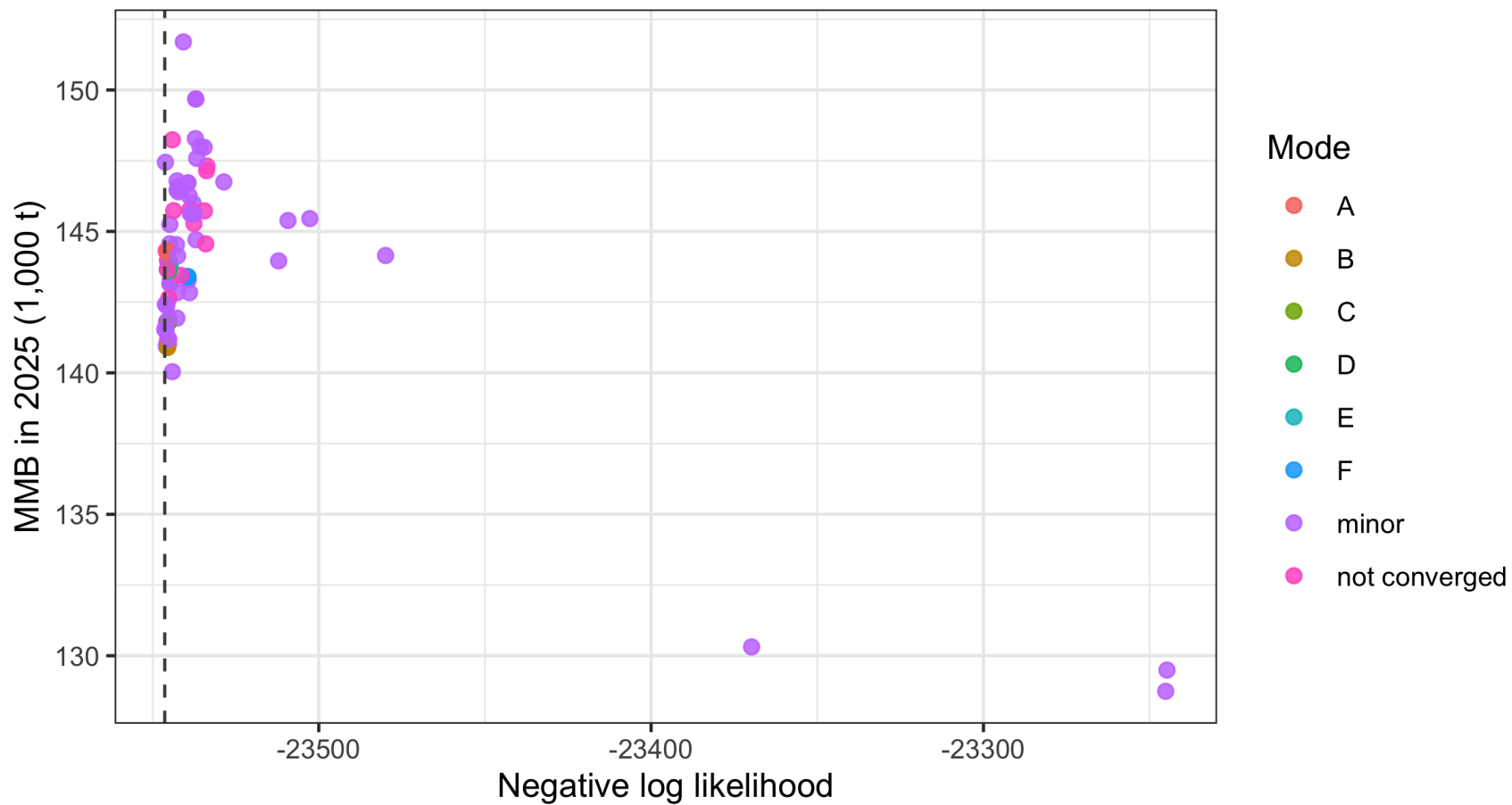
100 jitter runs, sd 0.1 (dashed line = base fit)



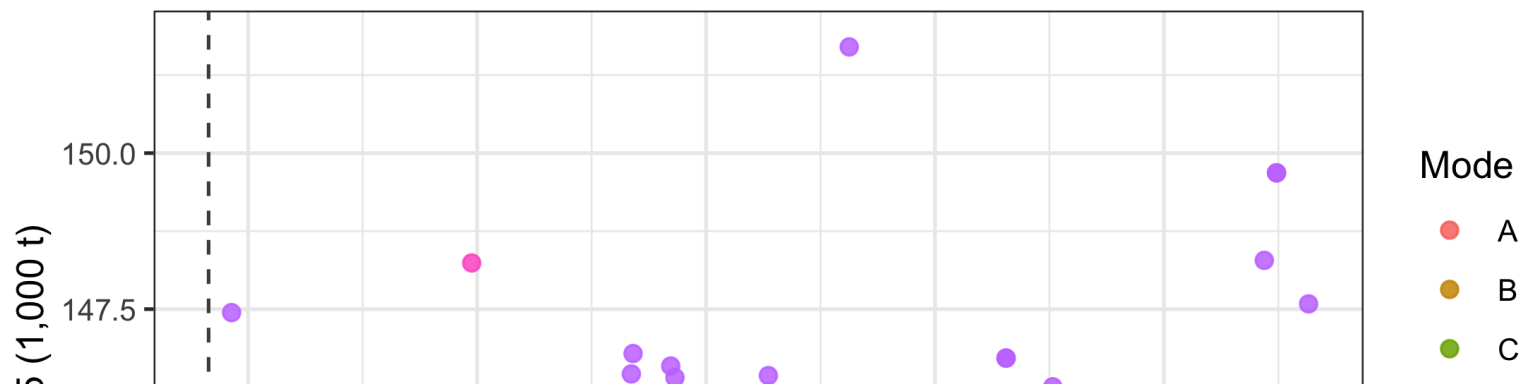
Runs within 10 nll units of the best



100 jitter runs, sd 0.1 (dashed line = base fit)



Runs within 10 nll units of the best



Convergence of 100 jitter runs

dashed = conventional threshold $1e-03$; dotted = base fit (0.000931)

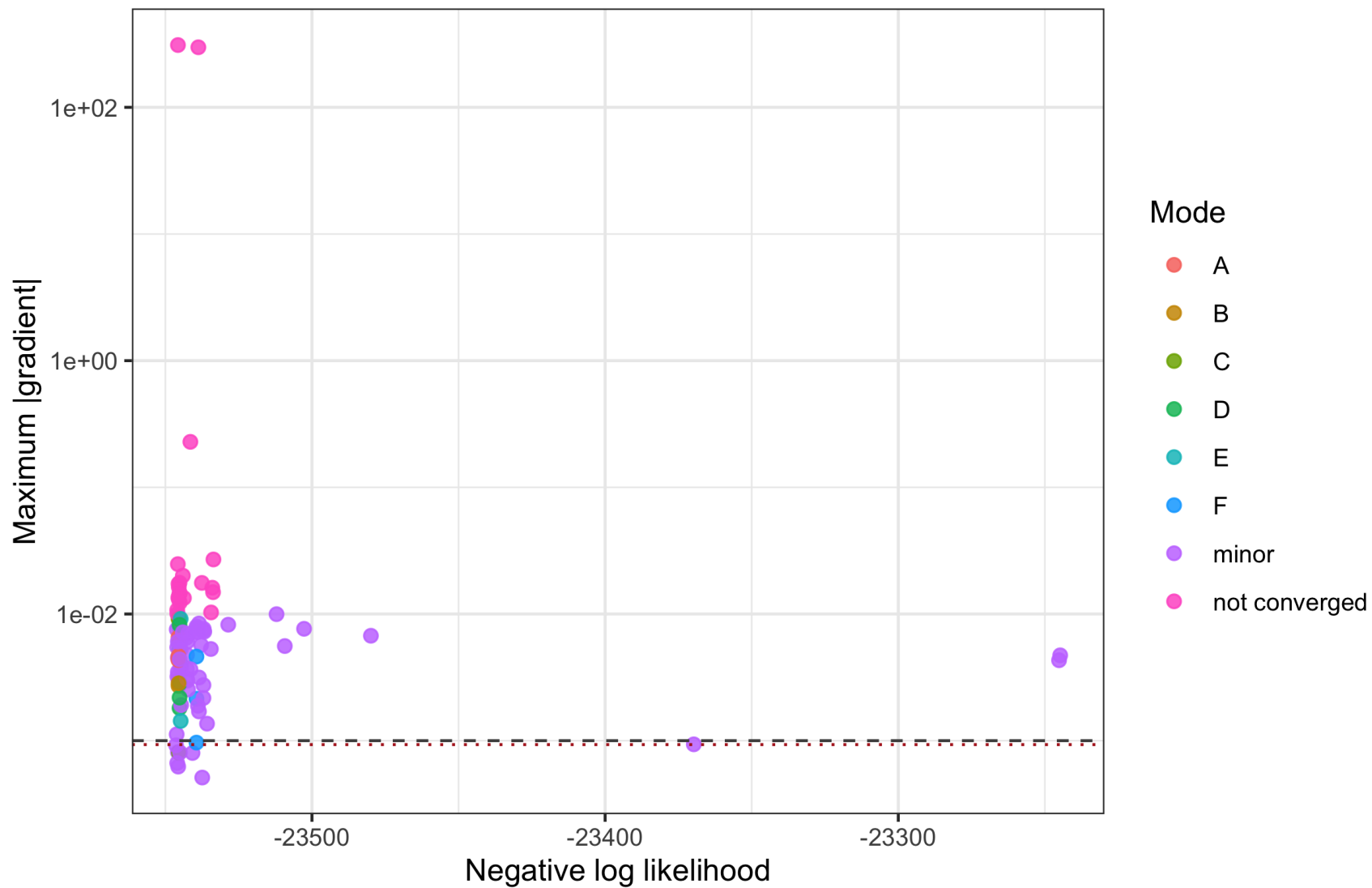


Figure 8: Maximum absolute gradient against negative log likelihood for 100 jitter runs. The dashed line is the conventional convergence threshold of 0.001; the dotted line is the base fit's maximum gradient.

Parameters separating the jitter clouds

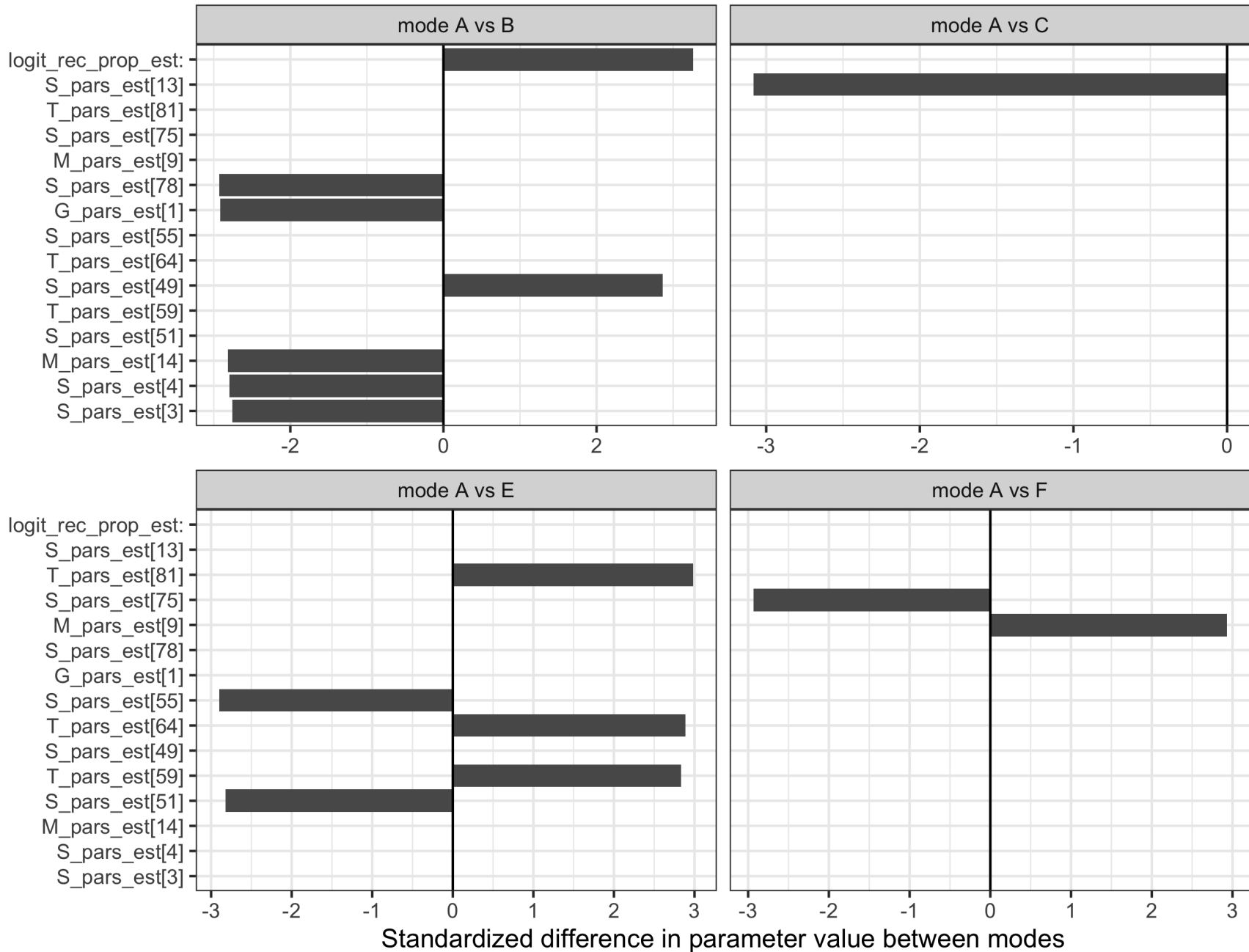


Figure 9: Parameters that most separate the jitter clouds, ranked by the standardized difference in fitted value between the best-fitting mode and

Retrospective analysis, 26_gmacs_update_newmat_plus_group

Peels 0-10. Mohn's rho = -0.109

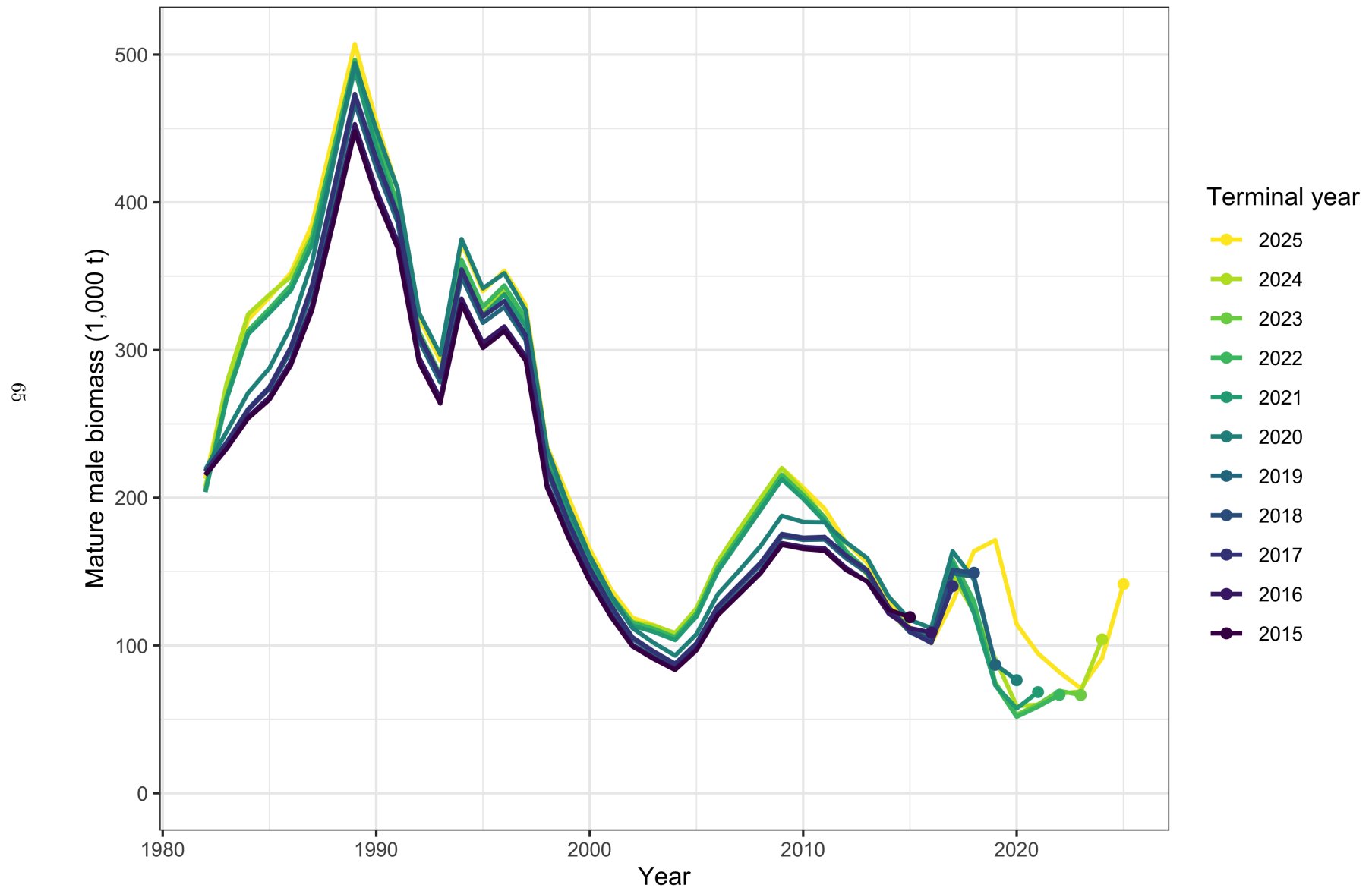


Figure 10: Standard retrospective analysis of Model 25.2e (new data), the accepted 2026 model. Mature male biomass in 1,000 t. Each line is a fit with the terminal years of data sequentially removed; color denotes the terminal year of the peel and the point marks each peel's terminal estimate.

Retrospective with the terminal survey year withheld

Peels 0-10, compared with the full base fit. Mohn's rho = -0.110

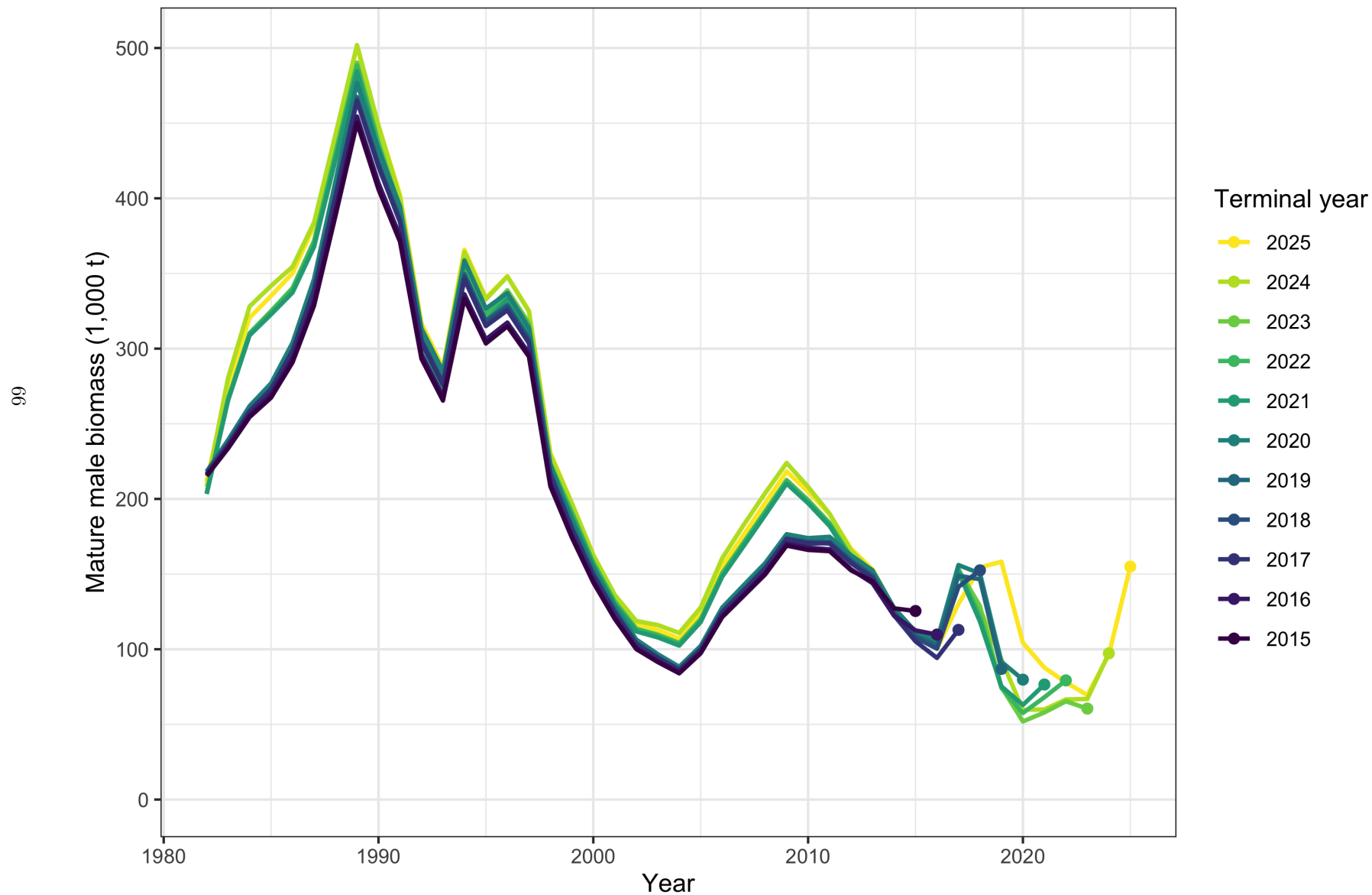


Figure 11: Retrospective analysis with the terminal survey year withheld in addition to the peeled years, isolating the leverage of the most recent survey observation on the terminal biomass estimate. Peels are compared against the model fit to all data.

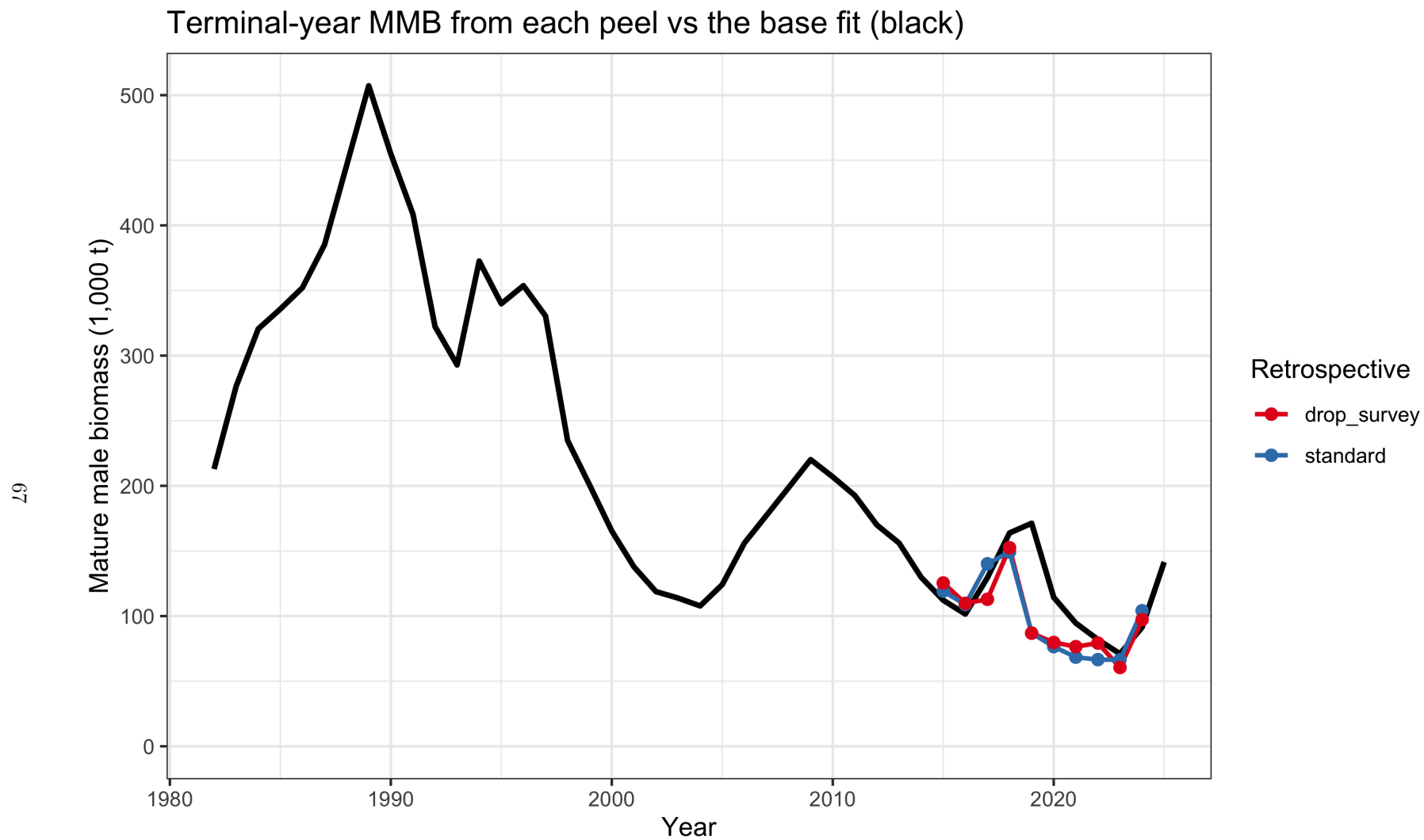
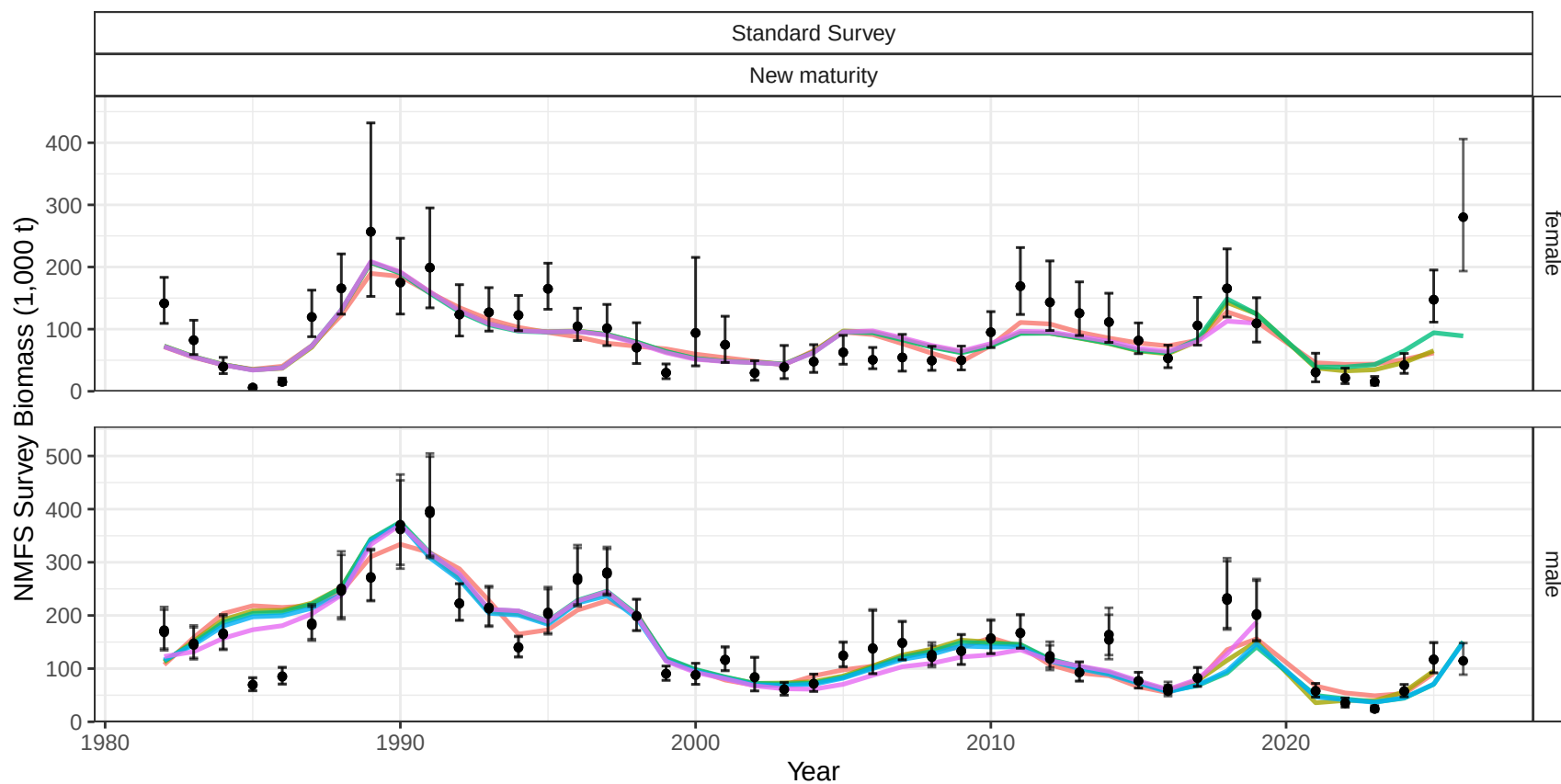


Figure 12: Terminal-year mature male biomass from each retrospective peel (colored) against the mature male biomass time series estimated by the model fit to all data (black). Systematic separation between the colored and black series is the retrospective pattern.

Points/Bars: Shared Observations | Lines: Model Fits



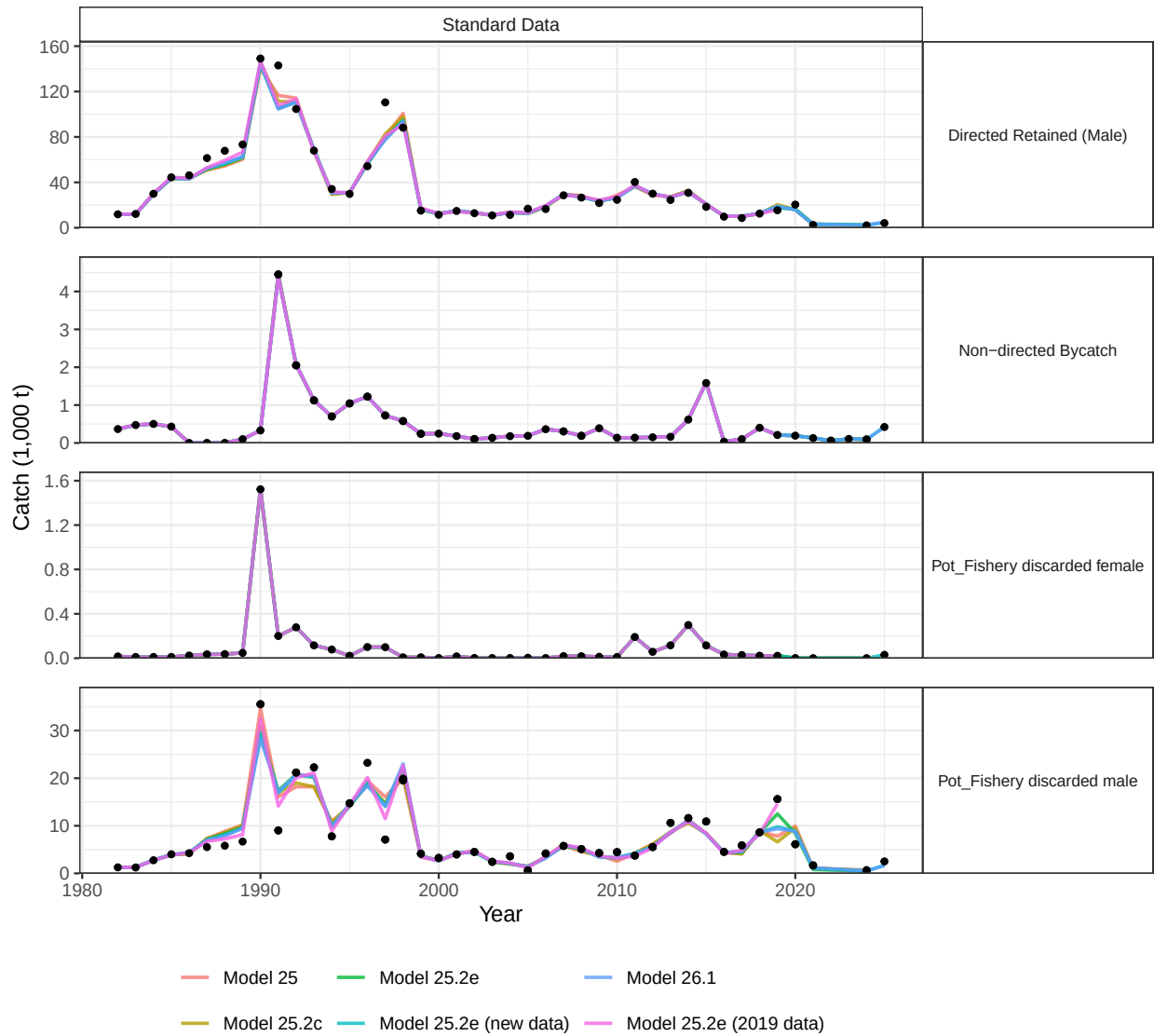
Model Prediction Model 25.2c Model 25.2e Model 25.2e (new data) Model 26.1 Model 25.2e (2019 data)

Error bars represent 95% log-normal confidence intervals on observations.

Figure 13: Model fits to the observed mature biomass at survey (1,000 t), for all models presented.

Fishery Catch and Bycatch Fits

Black points: Observations | Colored lines: Model predictions



). [[author: the hybrid cases are not in the September model set; drop this panel or label it as a May-cycle record.]]

Figure 14: Model fits to catch data (1,000 t), for all models presented.

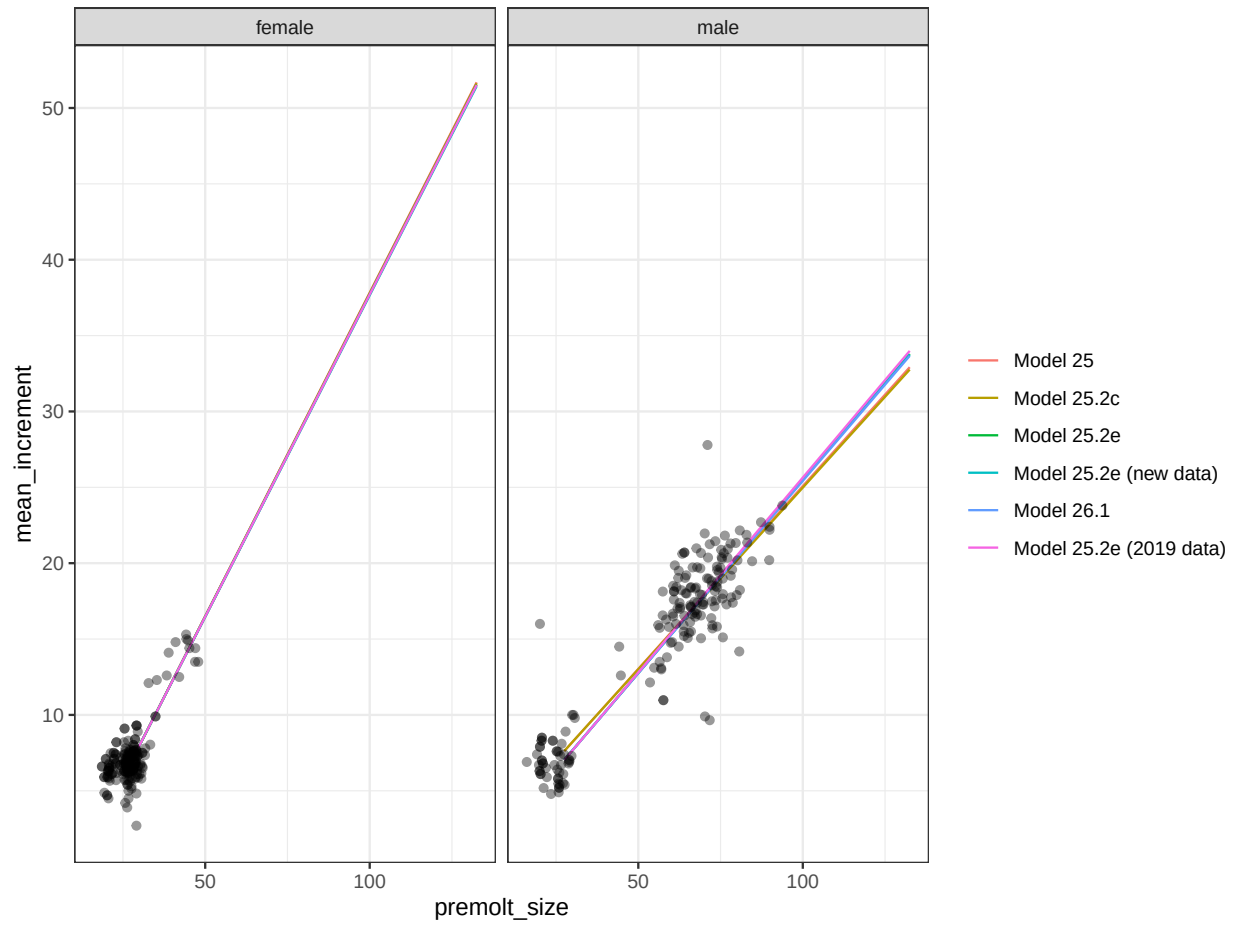


Figure 15: Model fits (colored lines) to the growth data (dots). Black dots are historical observations; red dots are new data for 2025.

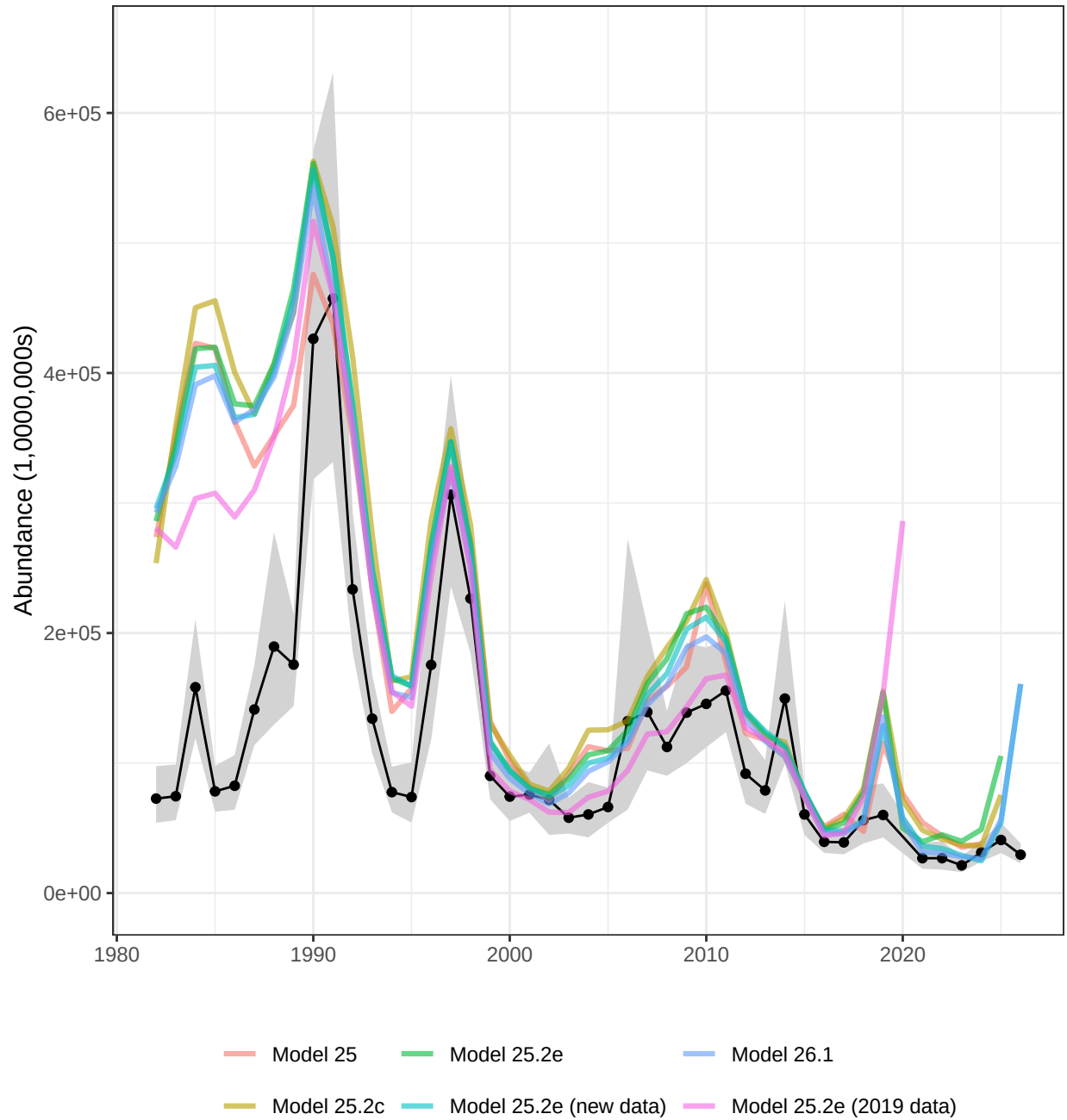


Figure 16: Estimated abundance of male crab >101mm carapace width from the survey (black line and dots with gray 95th CI) and from each model in the assessment (colored lines). [[author: the plotted quantity is the crabpack ABUNDANCE field divided by 1,000; state the resulting unit (thousands or millions of crab) – it could not be established from the source.]]

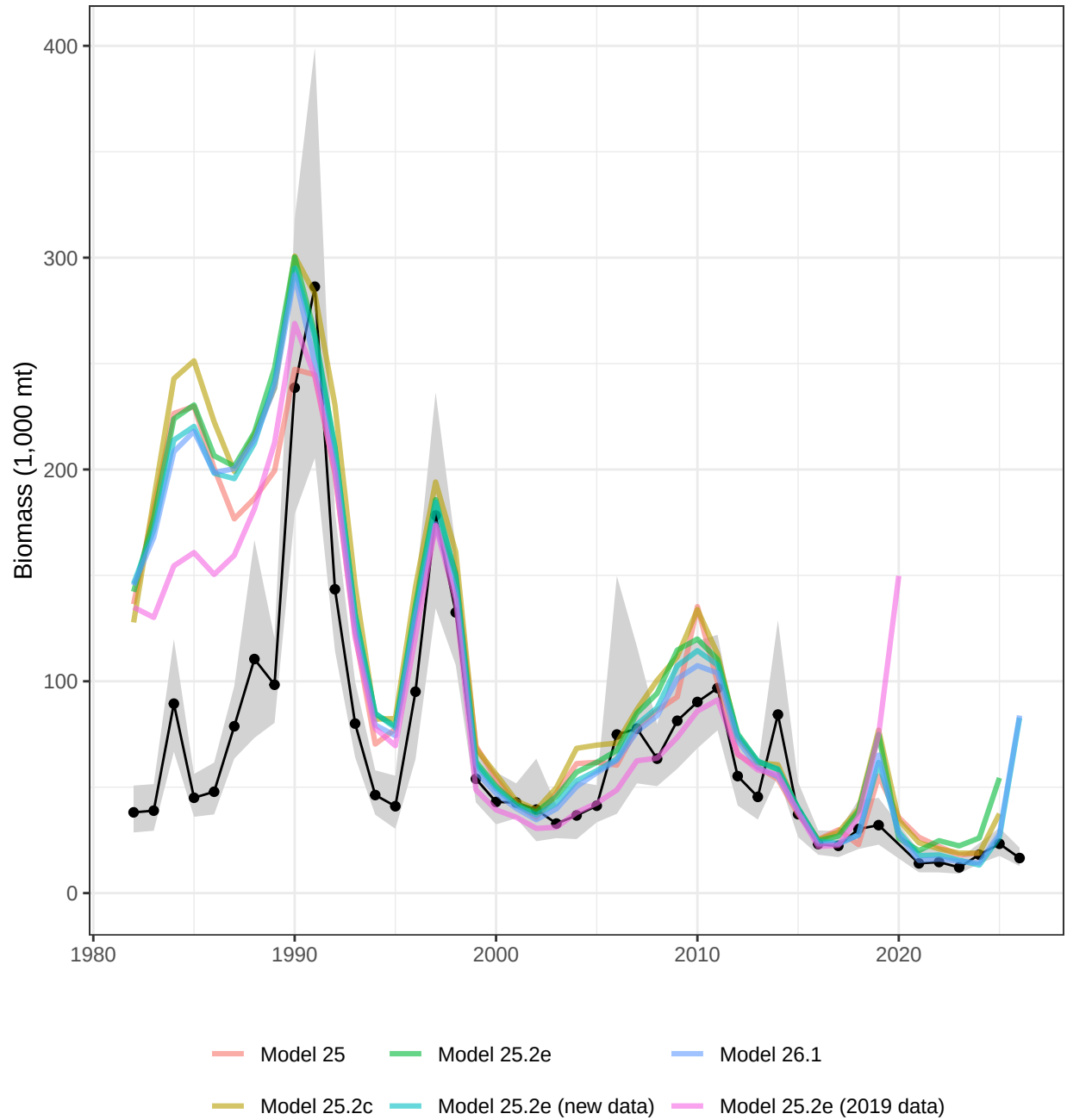


Figure 17: Estimated biomass (1,000 t) of male crab >101mm carapace width from the survey (black line and dots with gray 95th CI) and from each model in the assessment (colored lines).

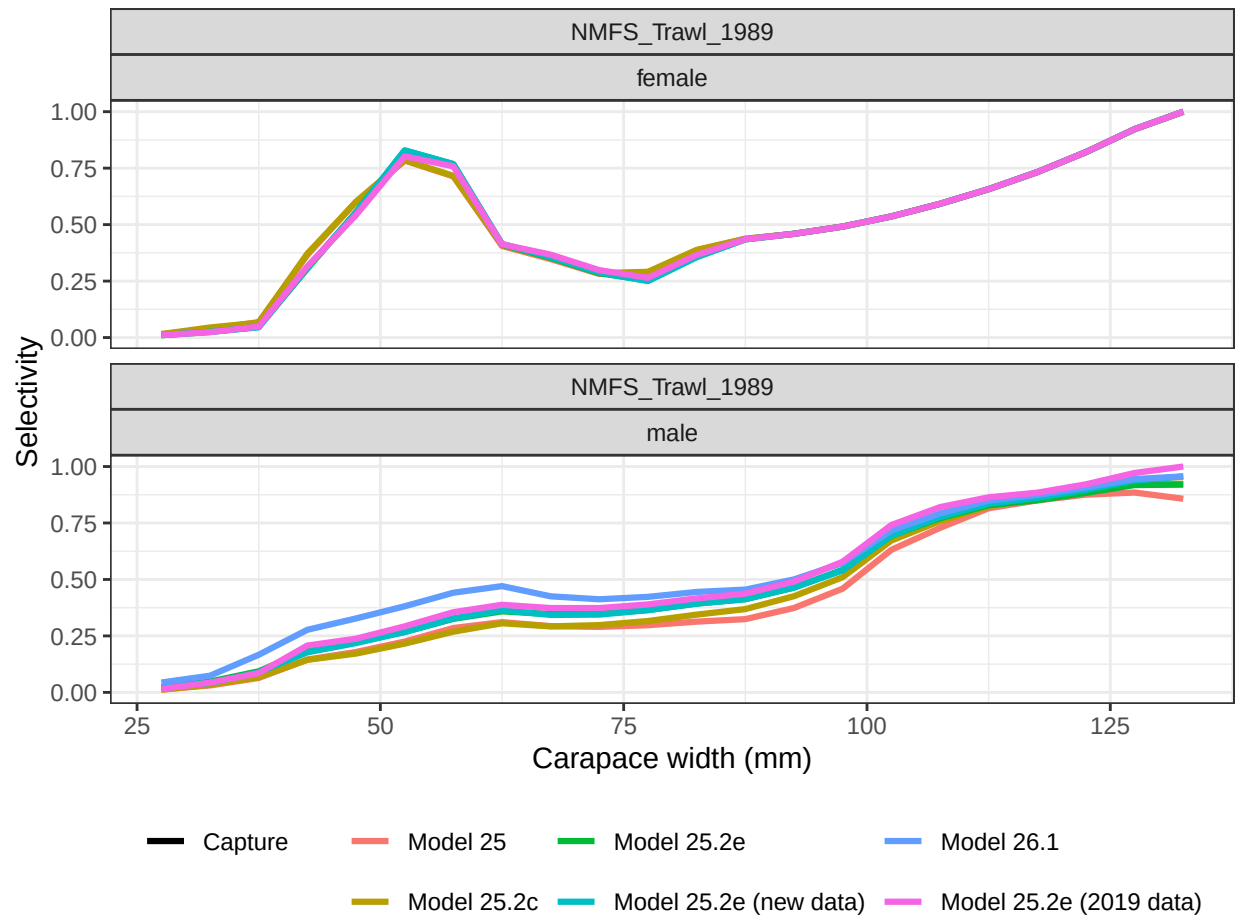


Figure 18: Estimated survey selectivity (lines) with normal priors derived from BSFRF selectivity experiment data. Points are the mean of the prior at a given size; intervals are 95th quantiles based on input CVs.

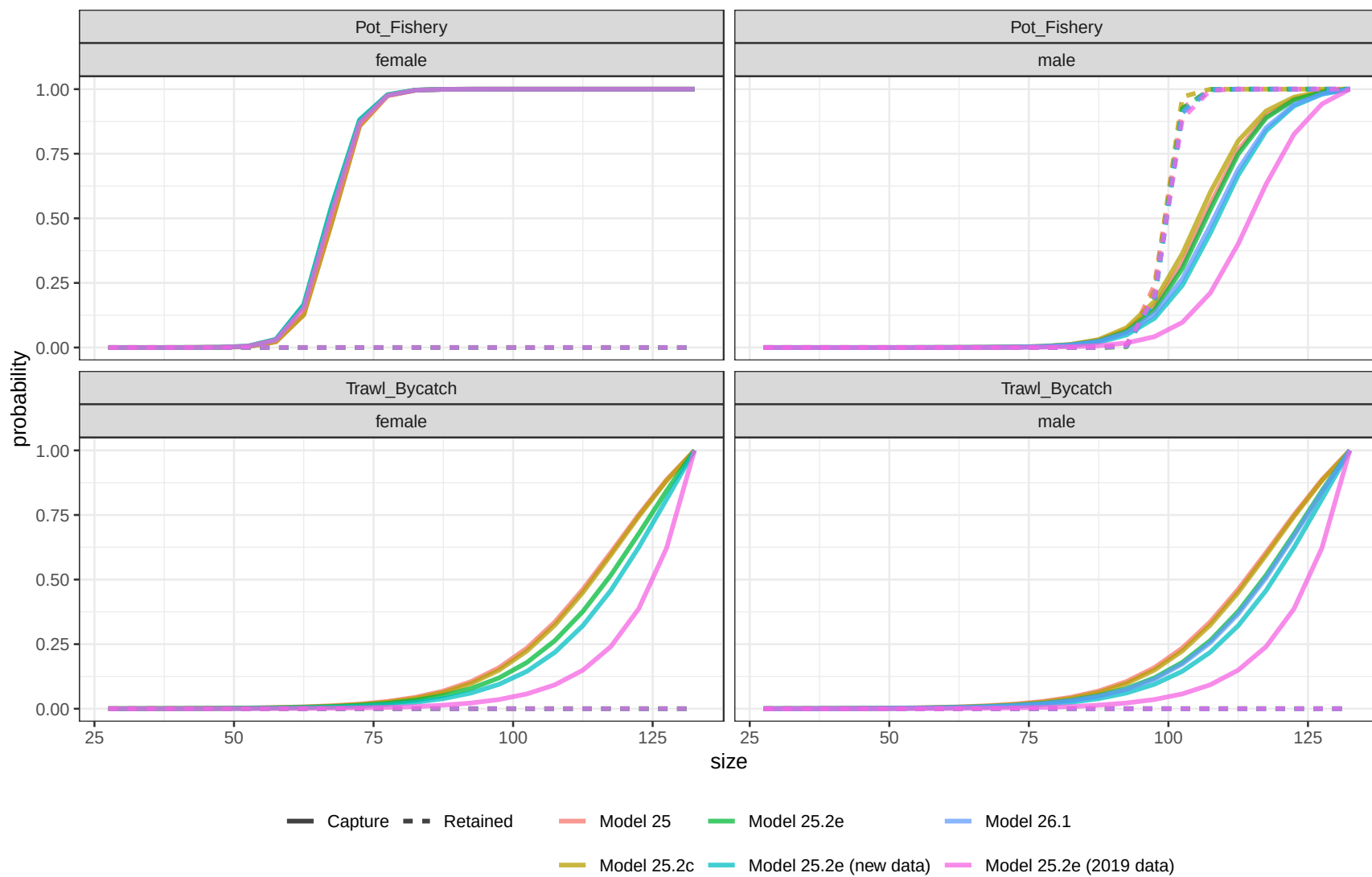


Figure 19: Estimated selectivities by fishing fleet and sex for capture and retained catches.

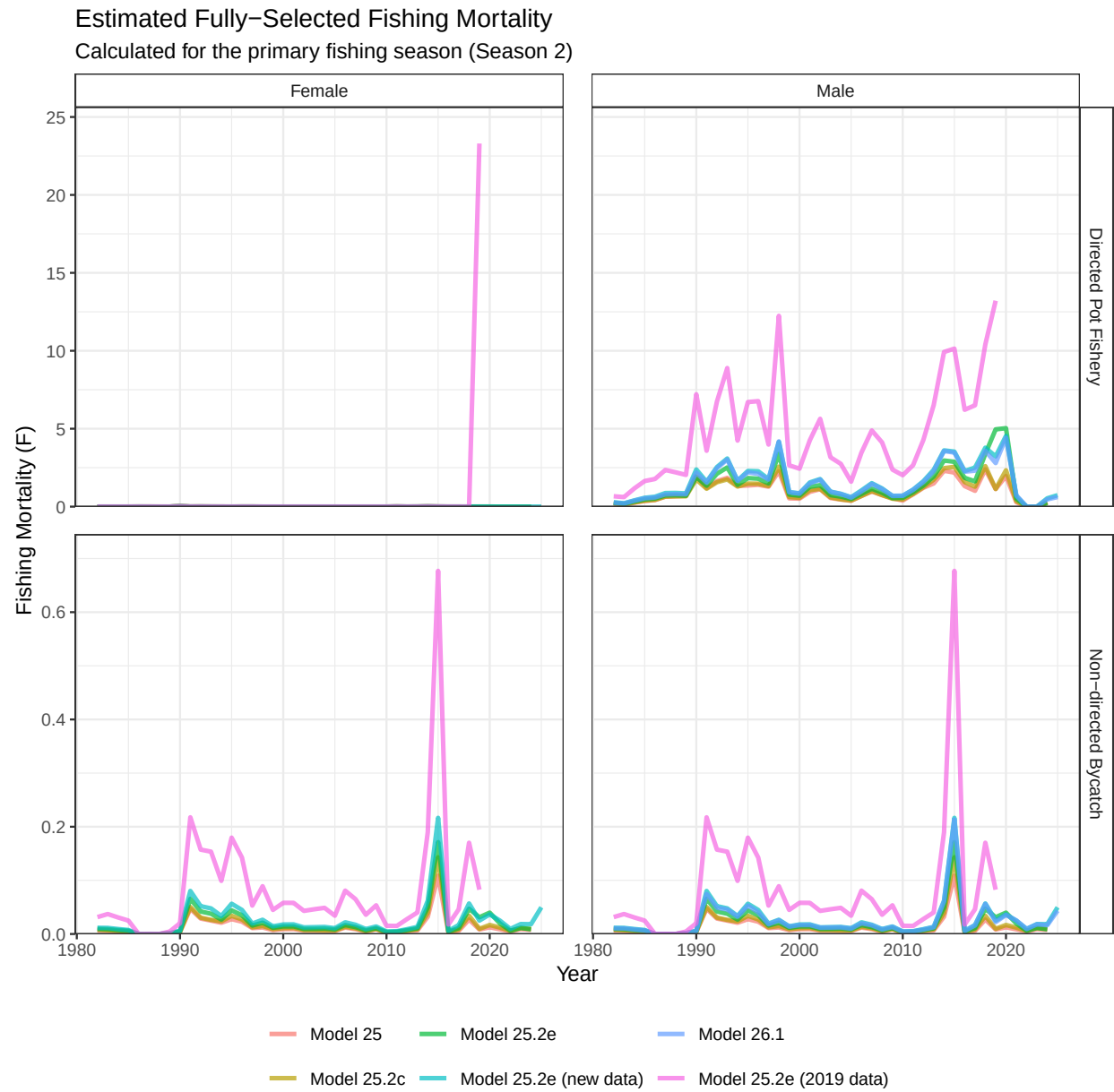


Figure 20: Estimated fishing mortalities (per year) for the directed and non-directed fisheries.

Probability of Maturing / Terminal Molt Separated by Maturity Data Specification

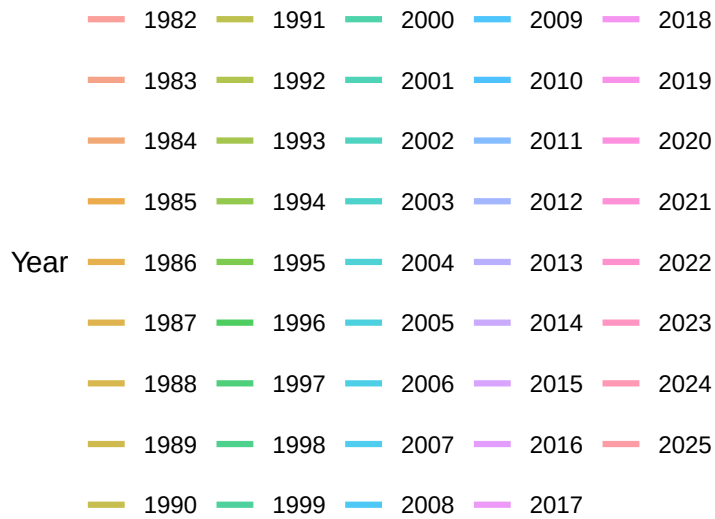
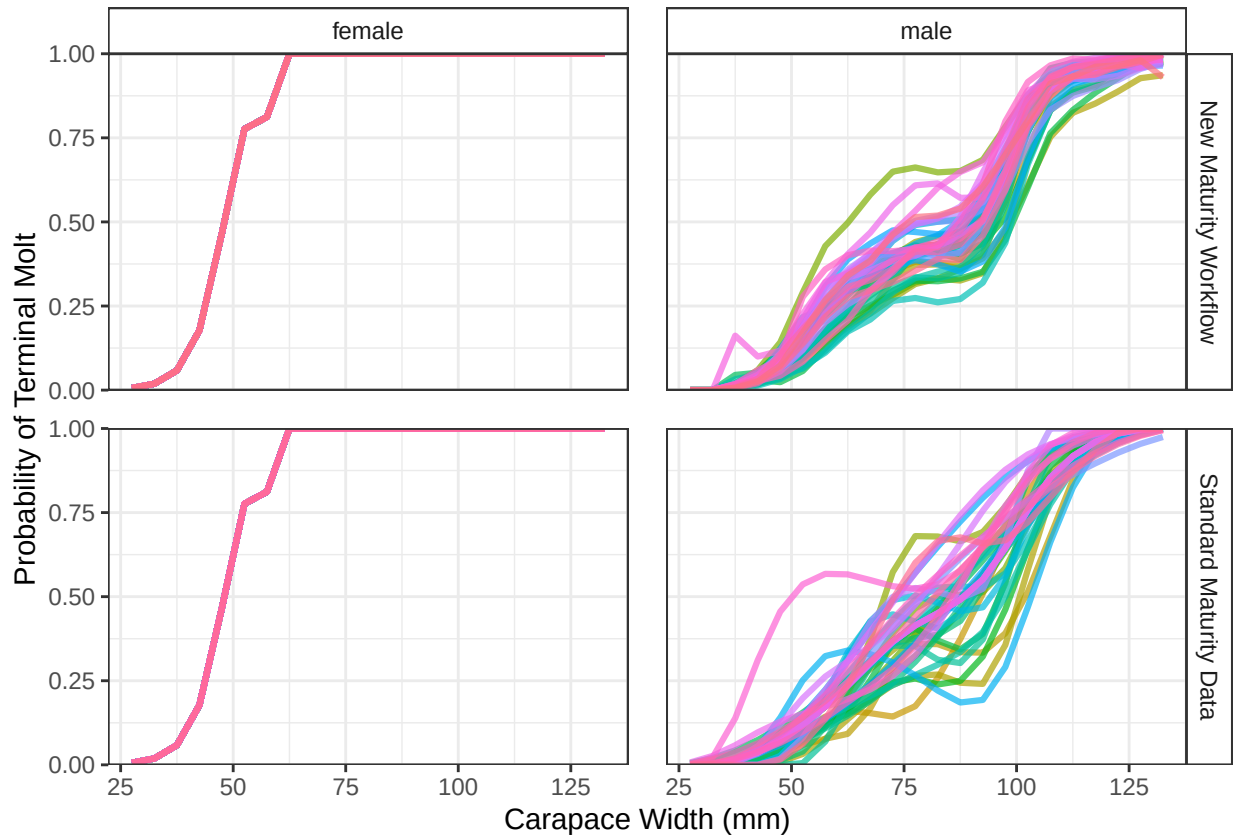


Figure 21: Estimated (black line) or specified (colored lines) probability(s) of maturing for male crab.

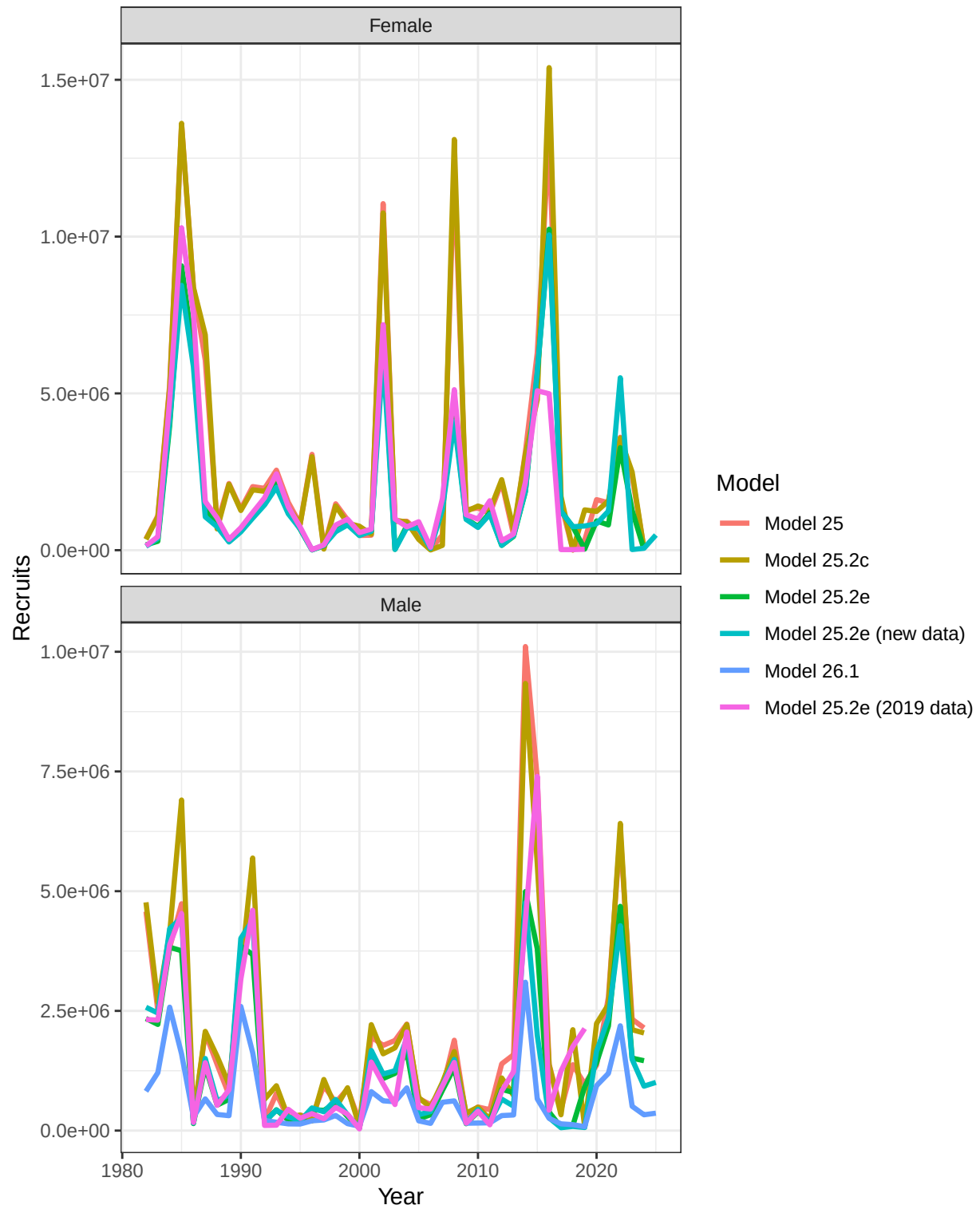


Figure 22: Estimated recruitment by sex (billions of crab), for all models presented.

Estimated Natural Mortality (M) Focusing on 2018–2019 Mortality Events

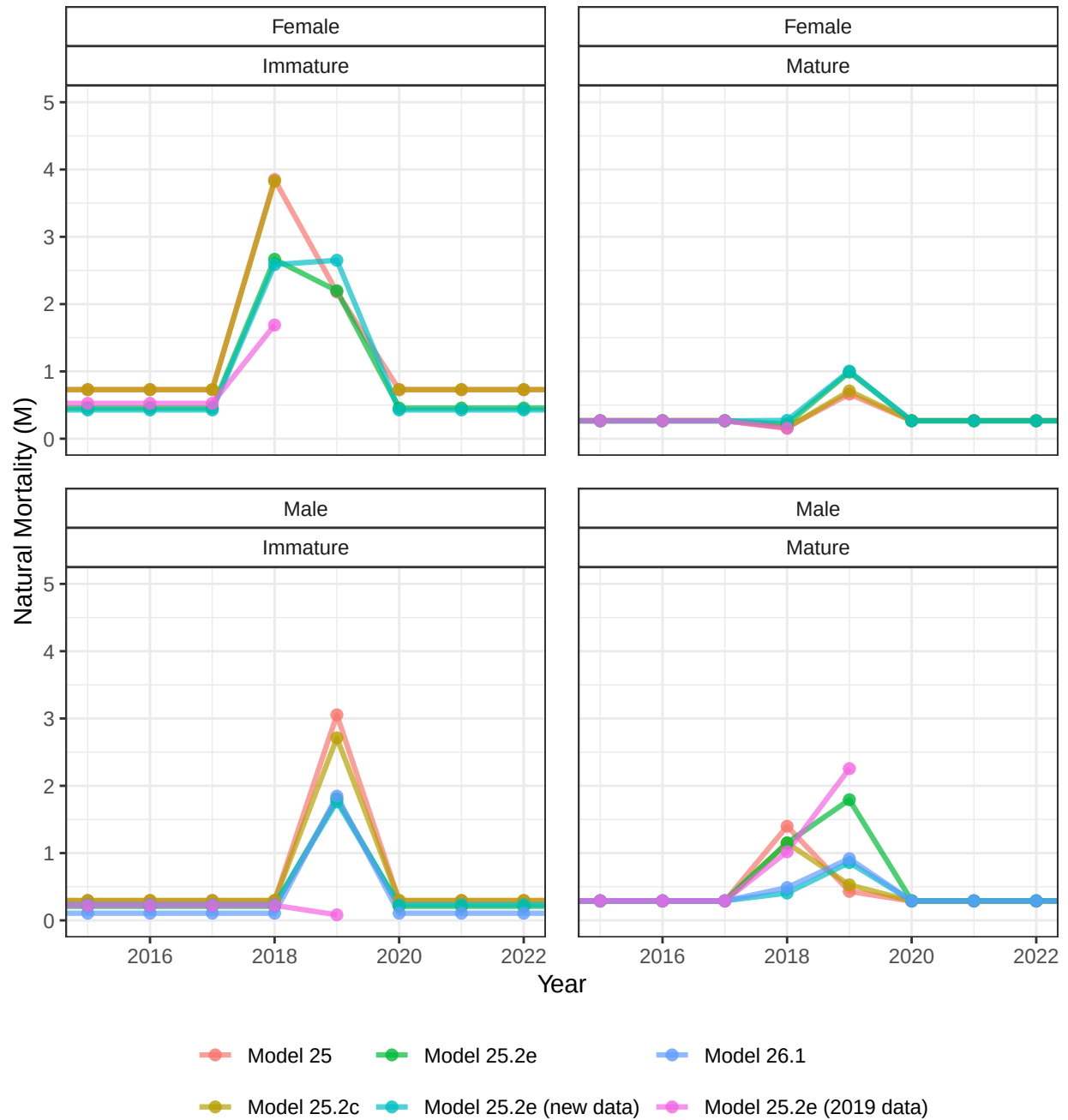


Figure 23: Estimated natural mortality (per year) by sex and maturity state. Natural mortality in all years previous to 2018 and after 2019 are equal to the estimated M in 2017. Note that the y-axis is capped at 5.

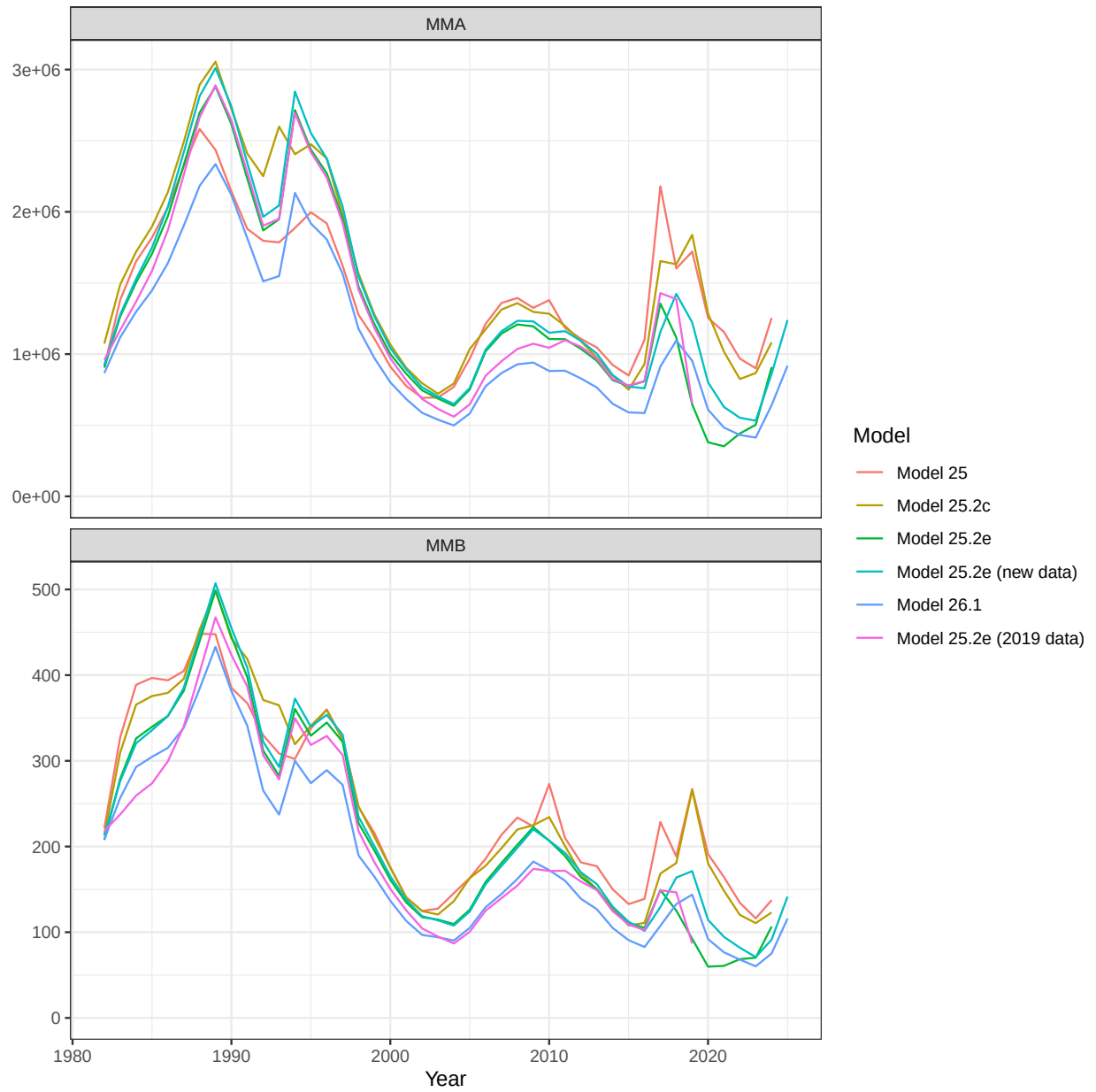


Figure 24: Model predicted mature male abundance (MMA; 1,000s) and biomass (MMB; 1,000 tonnes) and abundance at mating time.

Size composition fits

Fleet: Nmfs Trawl 1982

Female and Immature

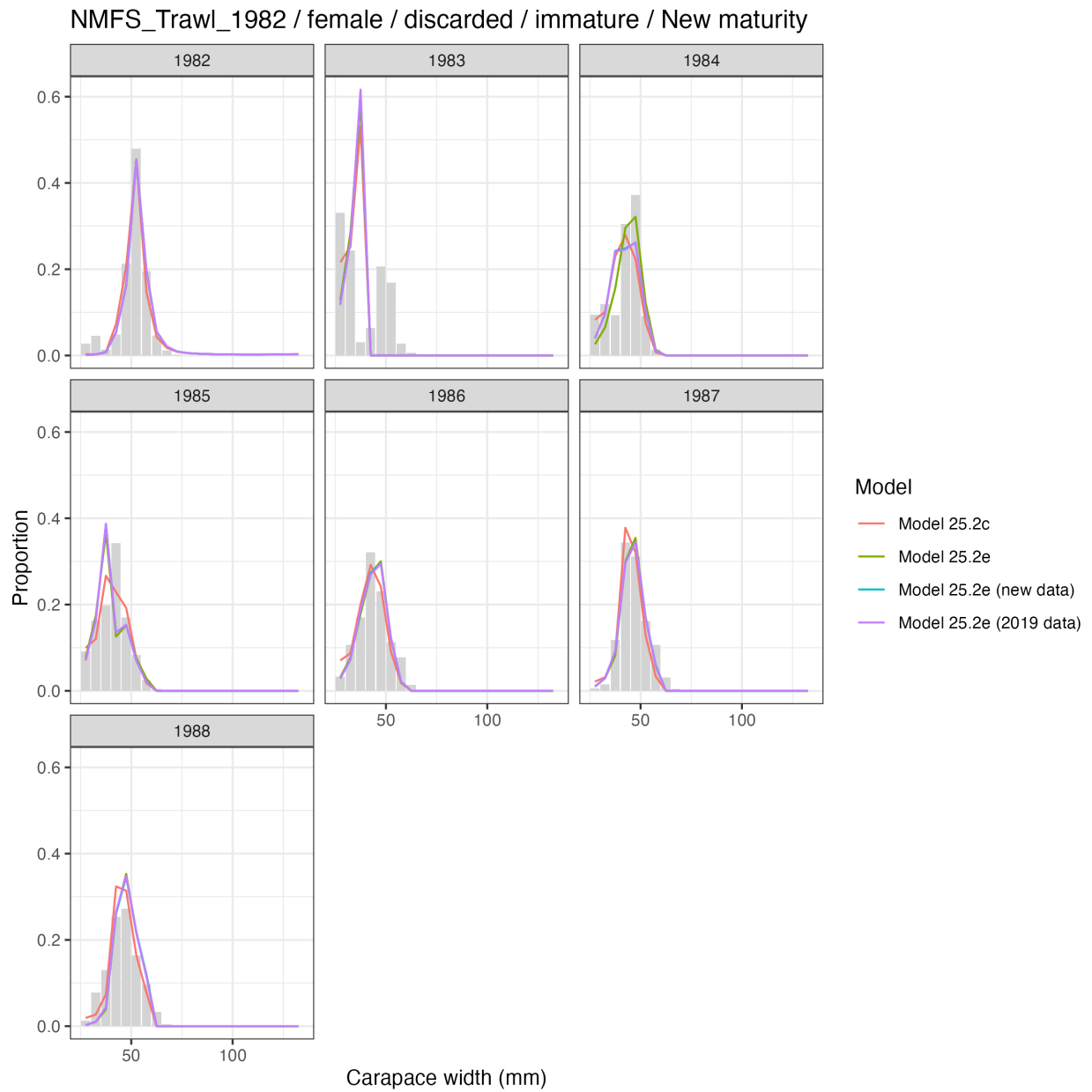


Figure 25: Model fits (lines) to size composition data (grey bars): NMFS Trawl 1982 female discarded immature New maturity.

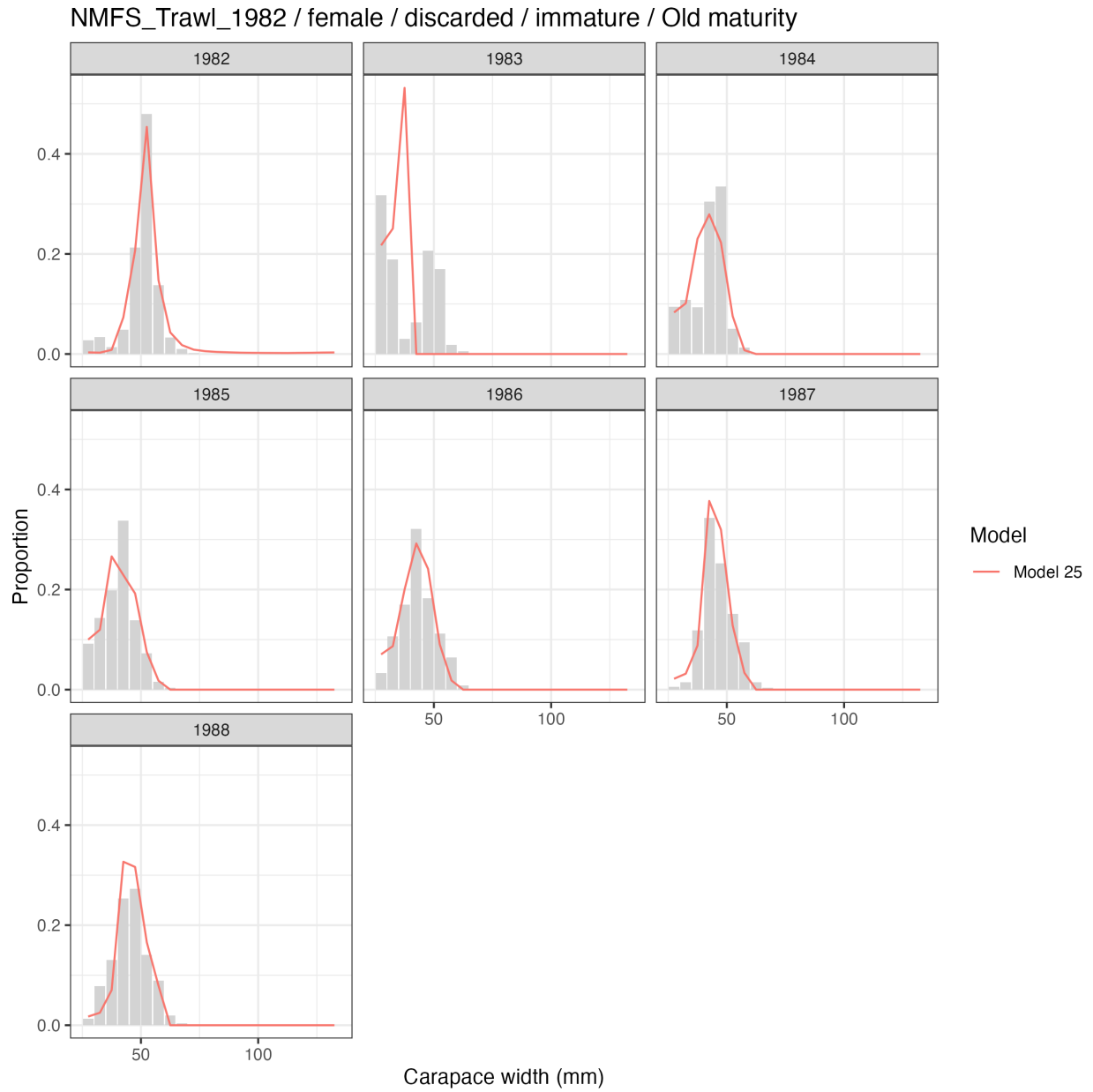


Figure 26: Model fits (lines) to size composition data (grey bars): NMFS Trawl 1982 female discarded immature Old maturity.

Female and Mature

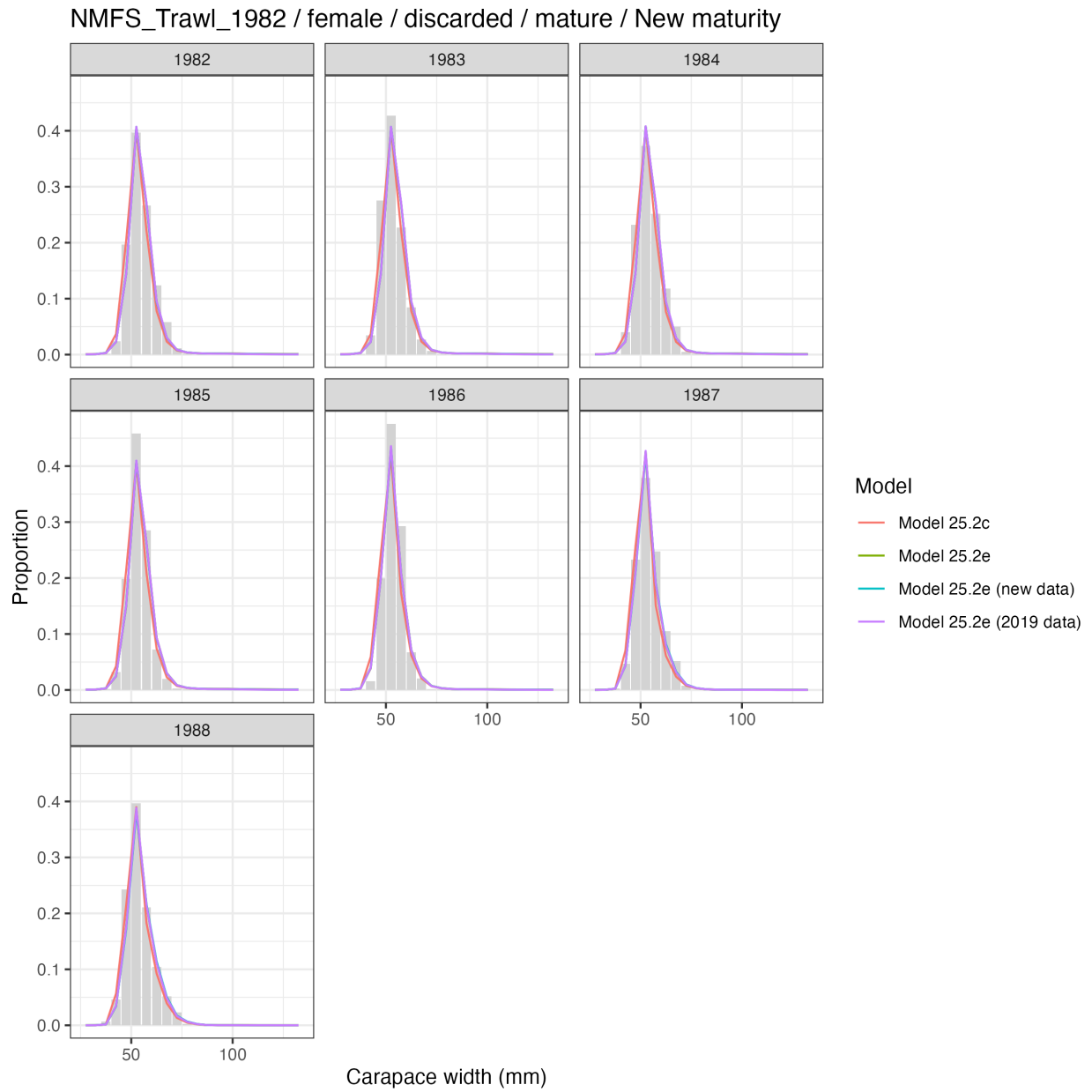


Figure 27: Model fits (lines) to size composition data (grey bars): NMFS Trawl 1982 female discarded mature New maturity.

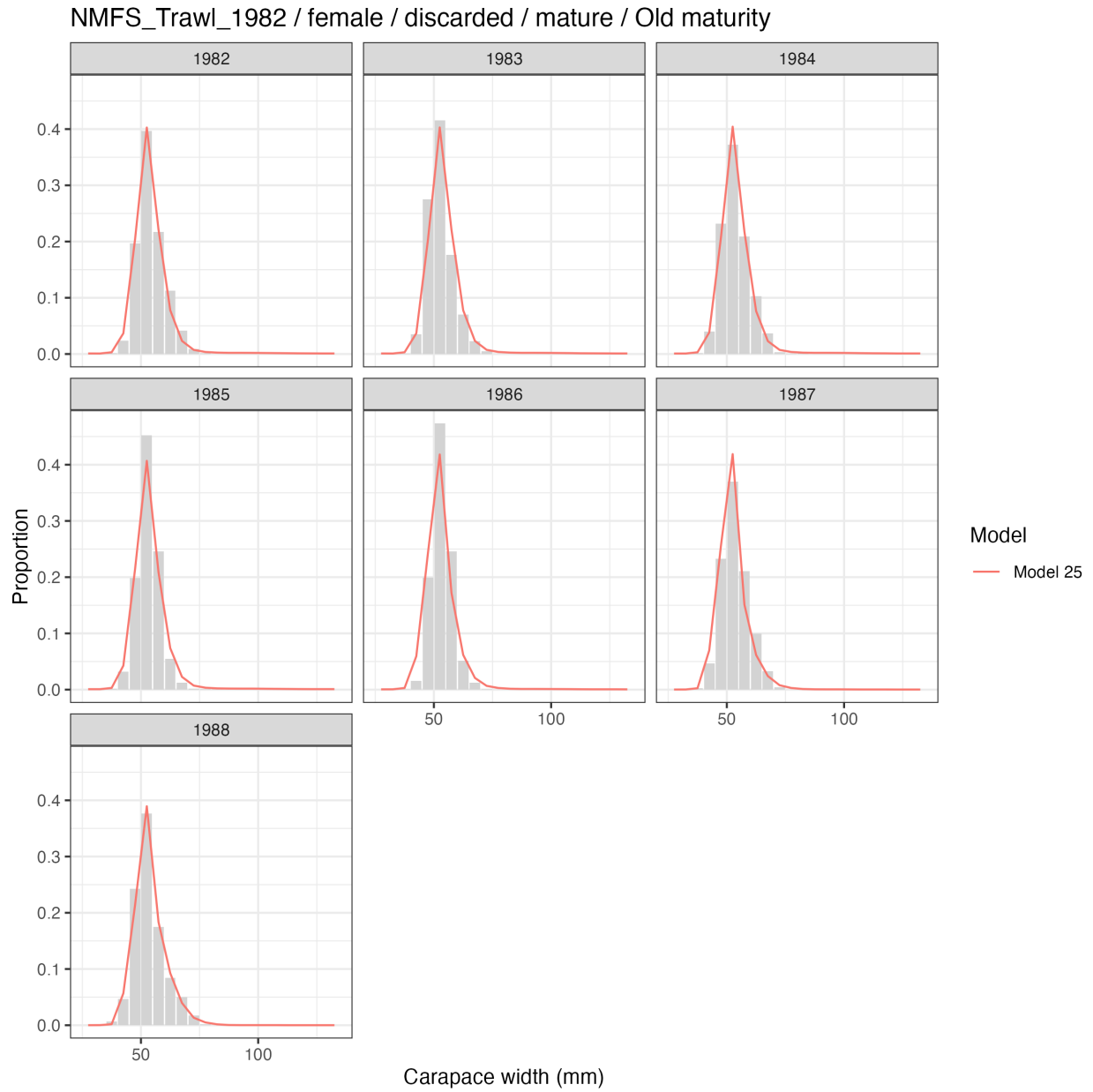


Figure 28: Model fits (lines) to size composition data (grey bars): NMFS Trawl 1982 female discarded mature Old maturity.

Male and Immature

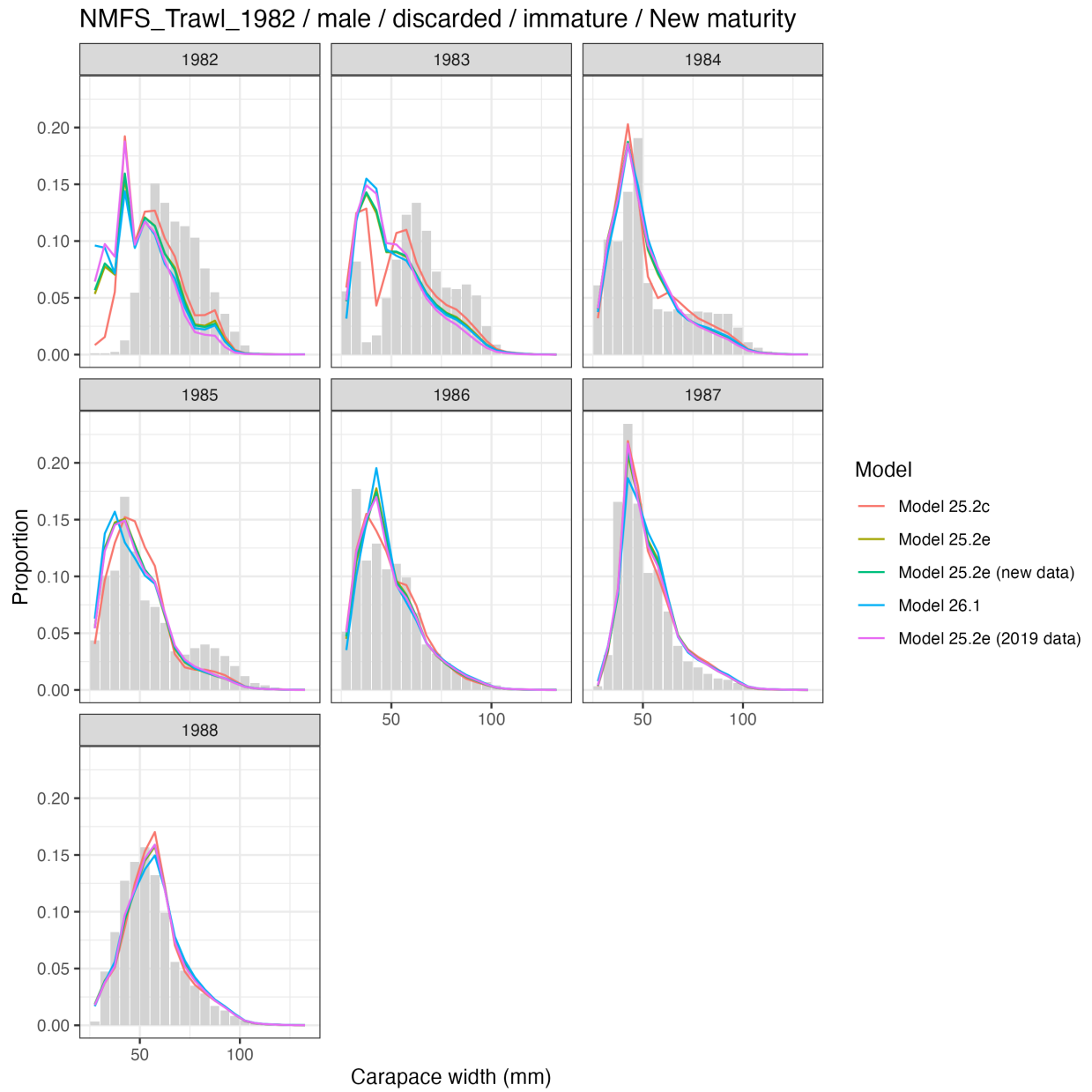


Figure 29: Model fits (lines) to size composition data (grey bars): NMFS Trawl 1982 male discarded immature New maturity.

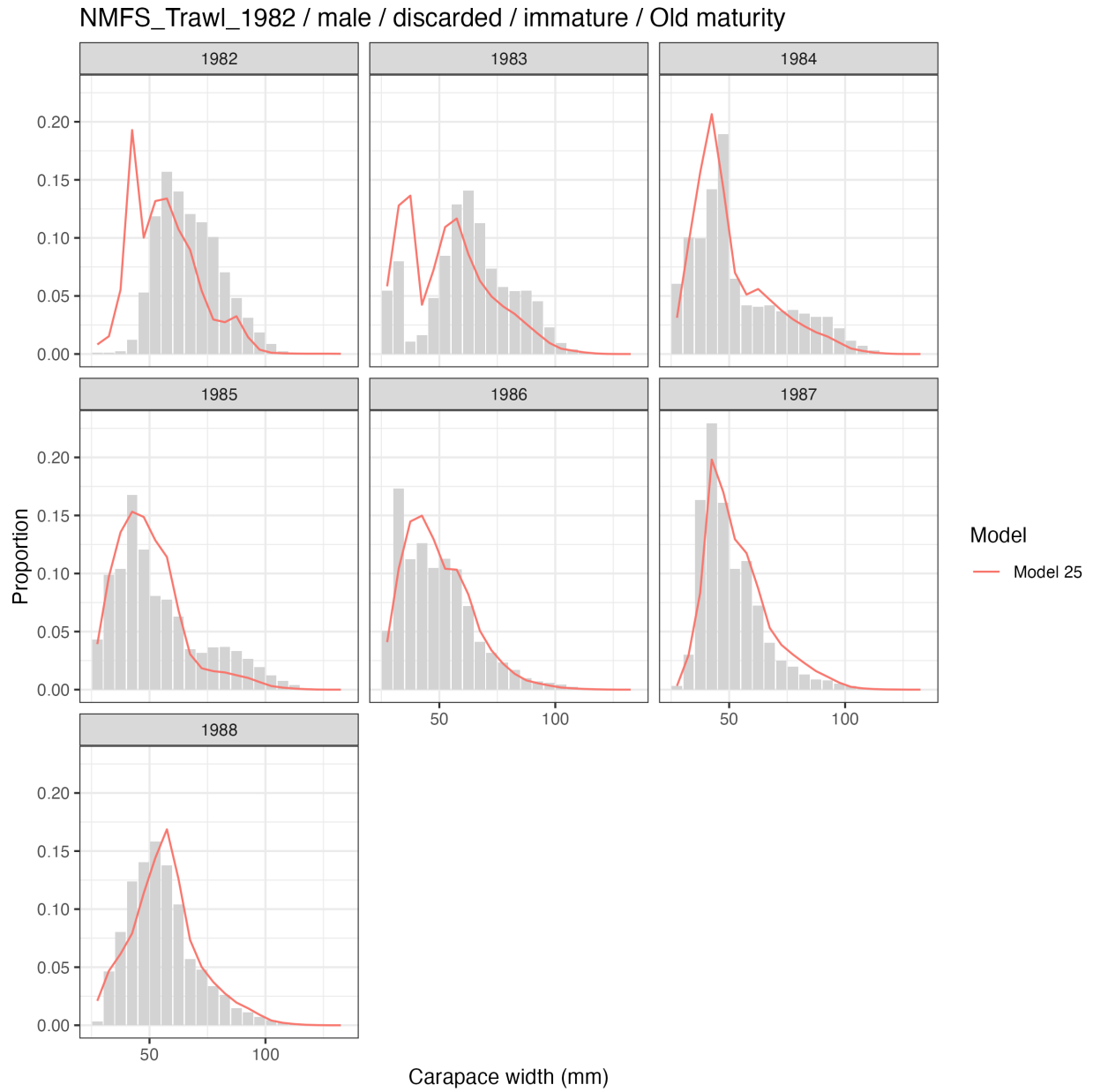


Figure 30: Model fits (lines) to size composition data (grey bars): NMFS Trawl 1982 male discarded immature Old maturity.

Male and Mature

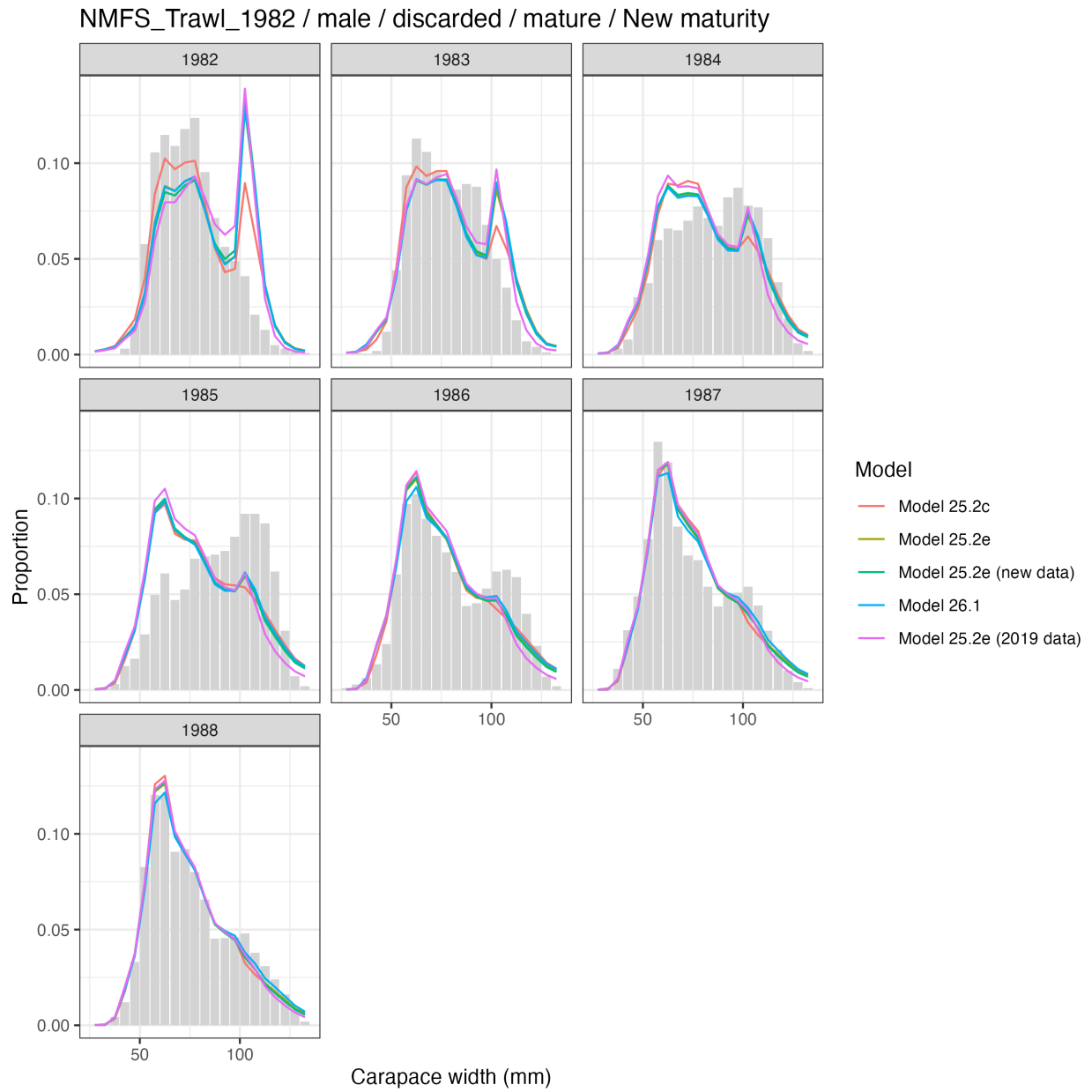


Figure 31: Model fits (lines) to size composition data (grey bars): NMFS Trawl 1982 male discarded mature New maturity.

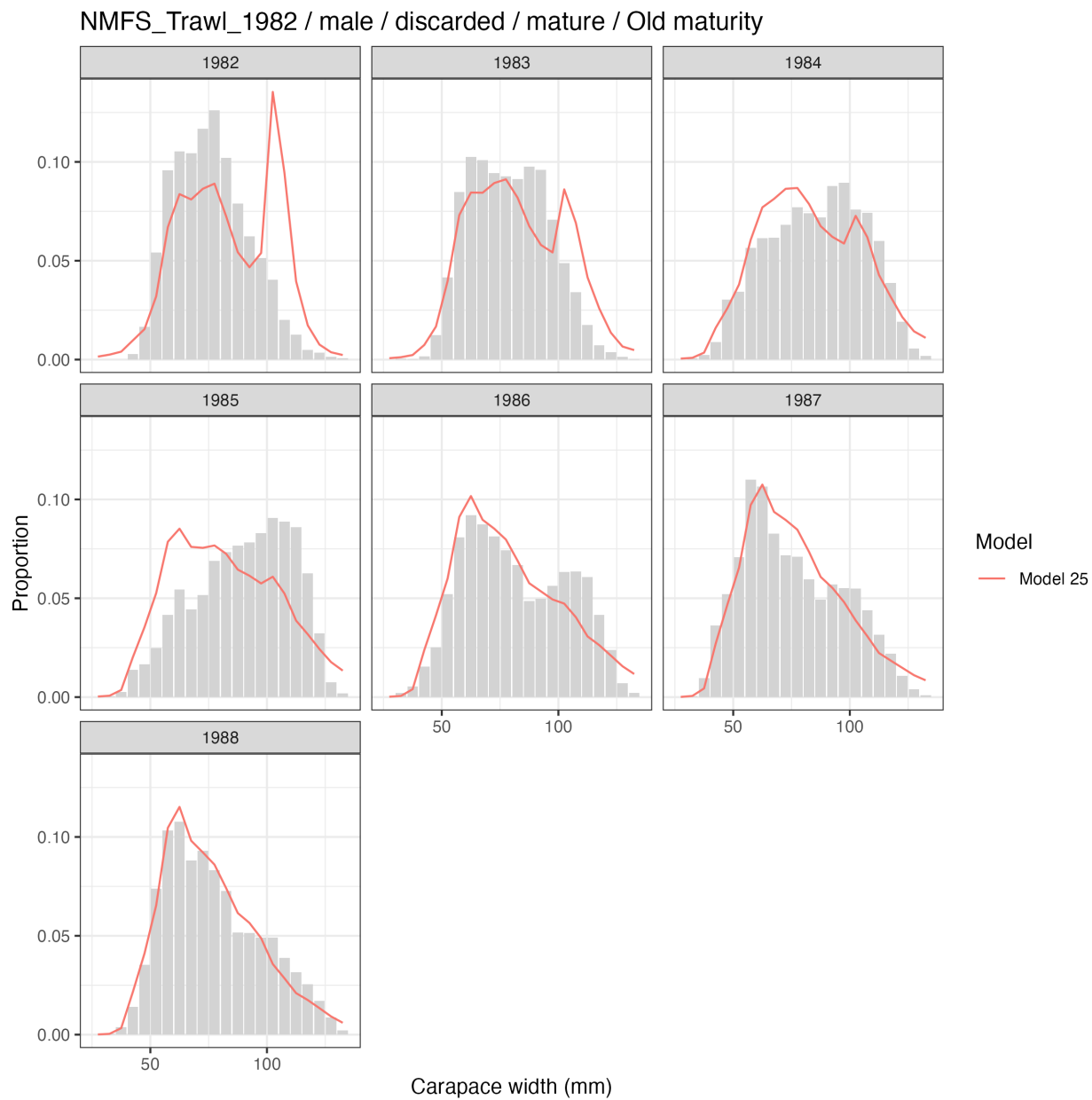


Figure 32: Model fits (lines) to size composition data (grey bars): NMFS Trawl 1982 male discarded mature Old maturity.

Fleet: Nmfs Trawl 1989

Female and Immature

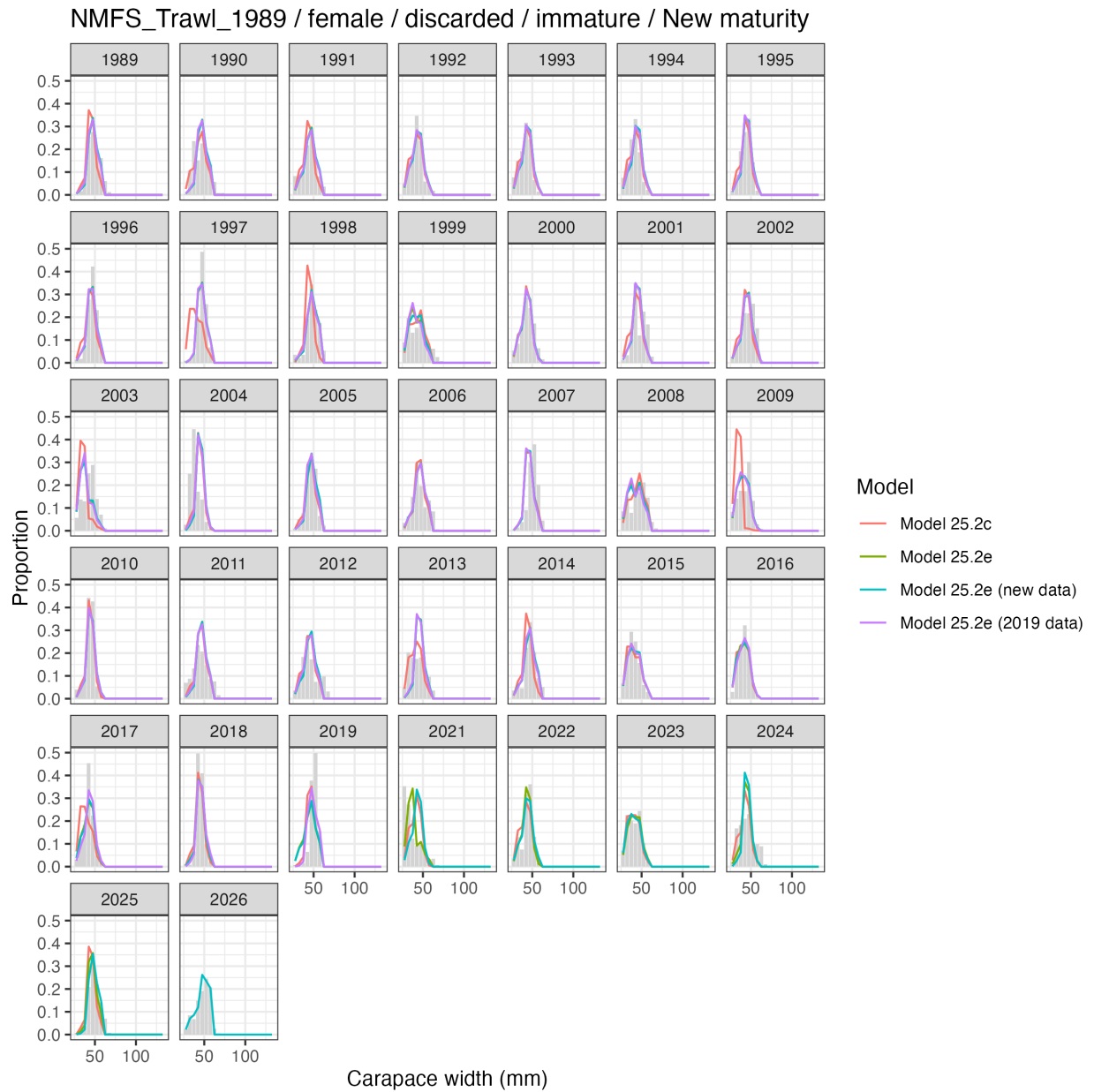


Figure 33: Model fits (lines) to size composition data (grey bars): NMFS Trawl 1989 female discarded immature New maturity.

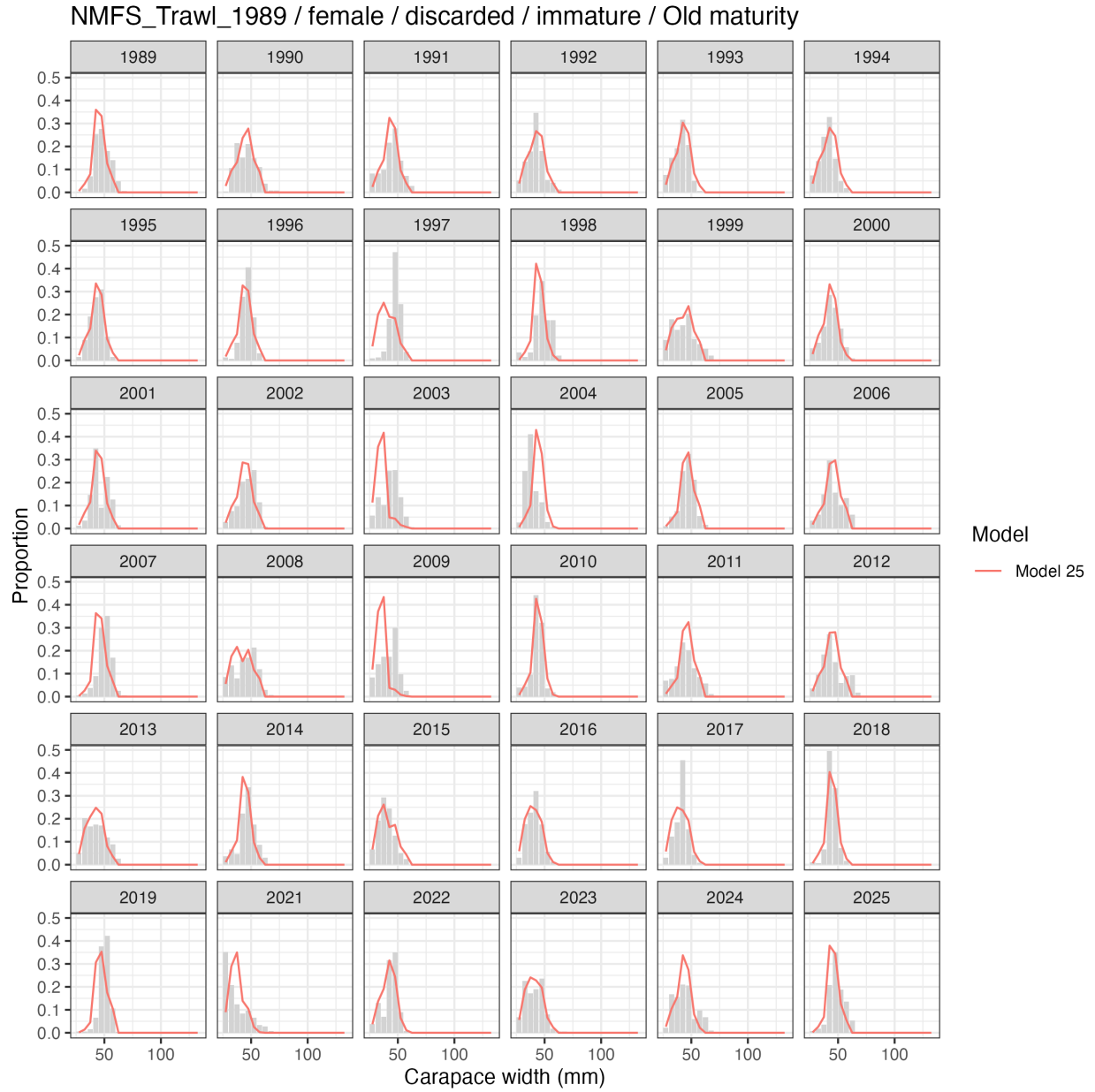


Figure 34: Model fits (lines) to size composition data (grey bars): NMFS Trawl 1989 female discarded immature Old maturity.

Female and Mature

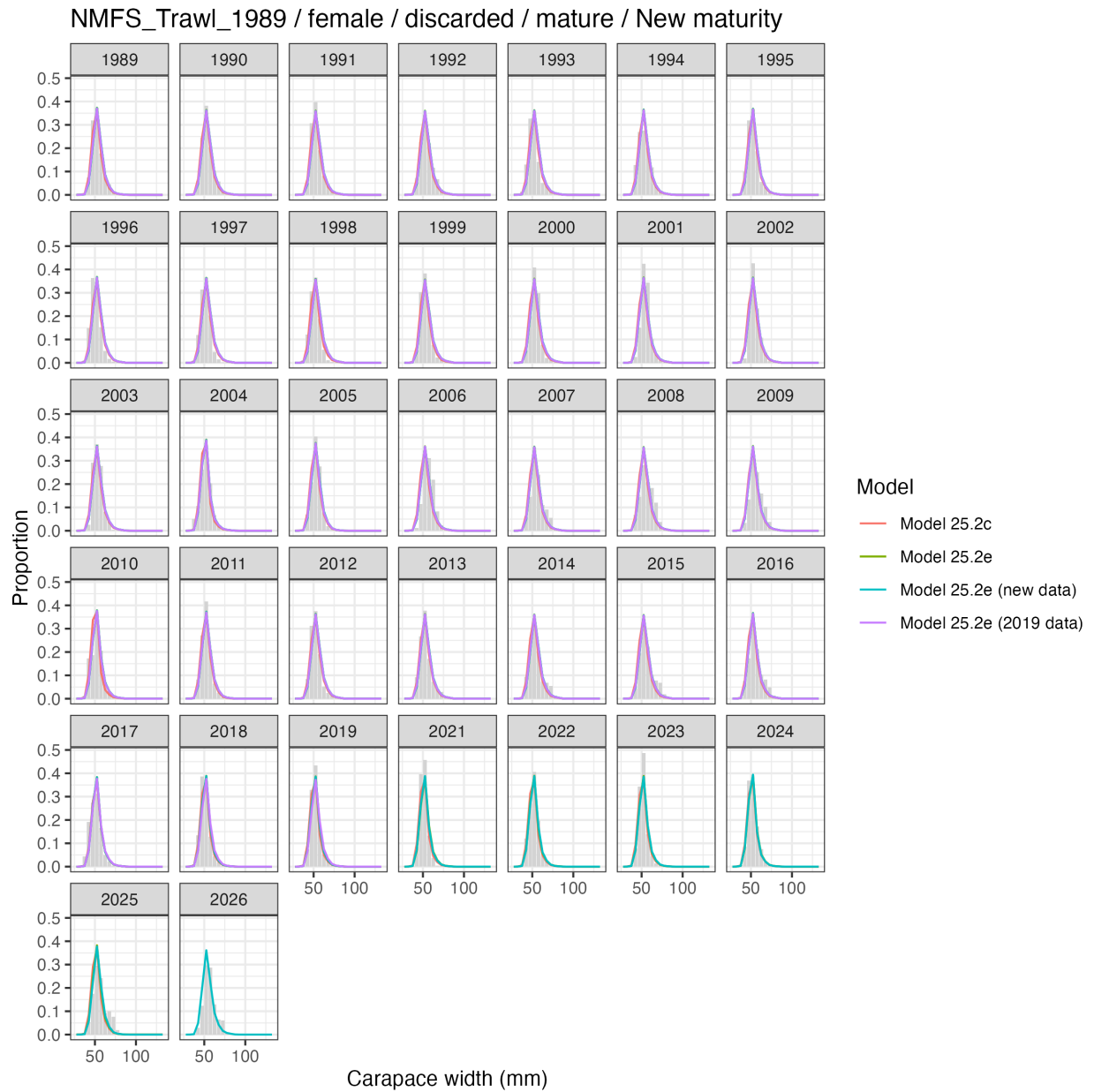


Figure 35: Model fits (lines) to size composition data (grey bars): NMFS Trawl 1989 female discarded mature New maturity.

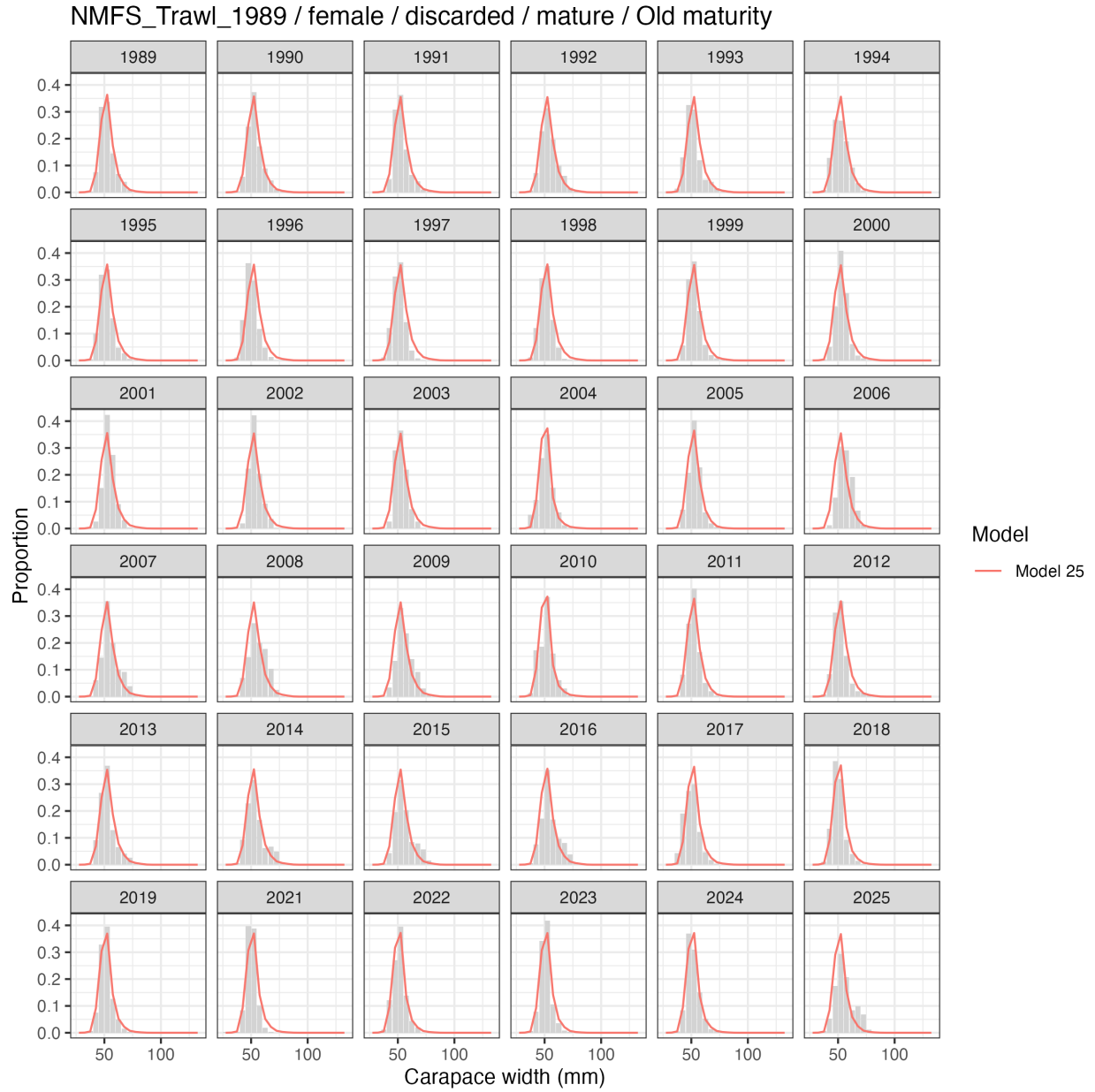


Figure 36: Model fits (lines) to size composition data (grey bars): NMFS Trawl 1989 female discarded mature Old maturity.

Male and Immature

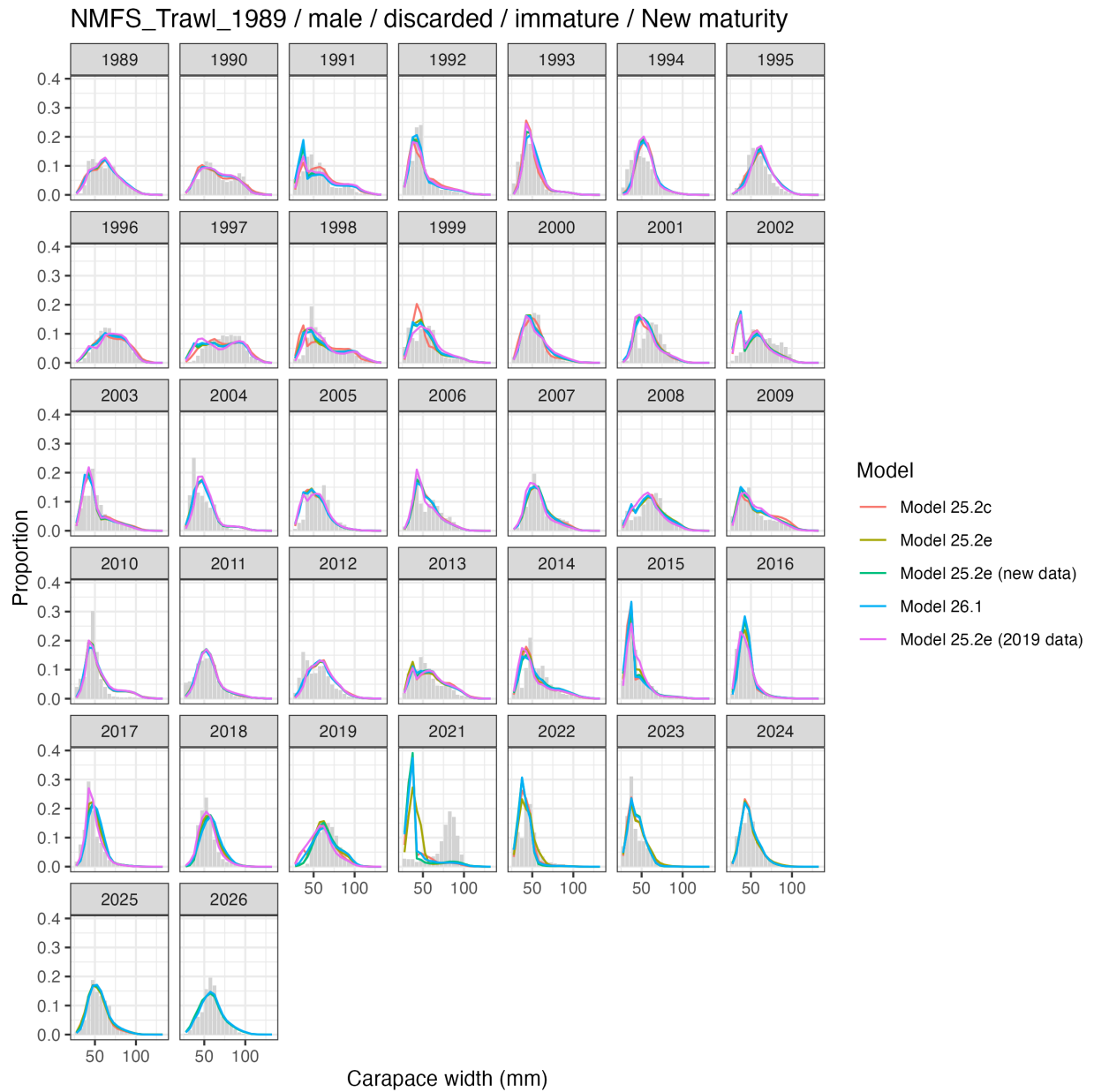


Figure 37: Model fits (lines) to size composition data (grey bars): NMFS Trawl 1989 male discarded immature New maturity.

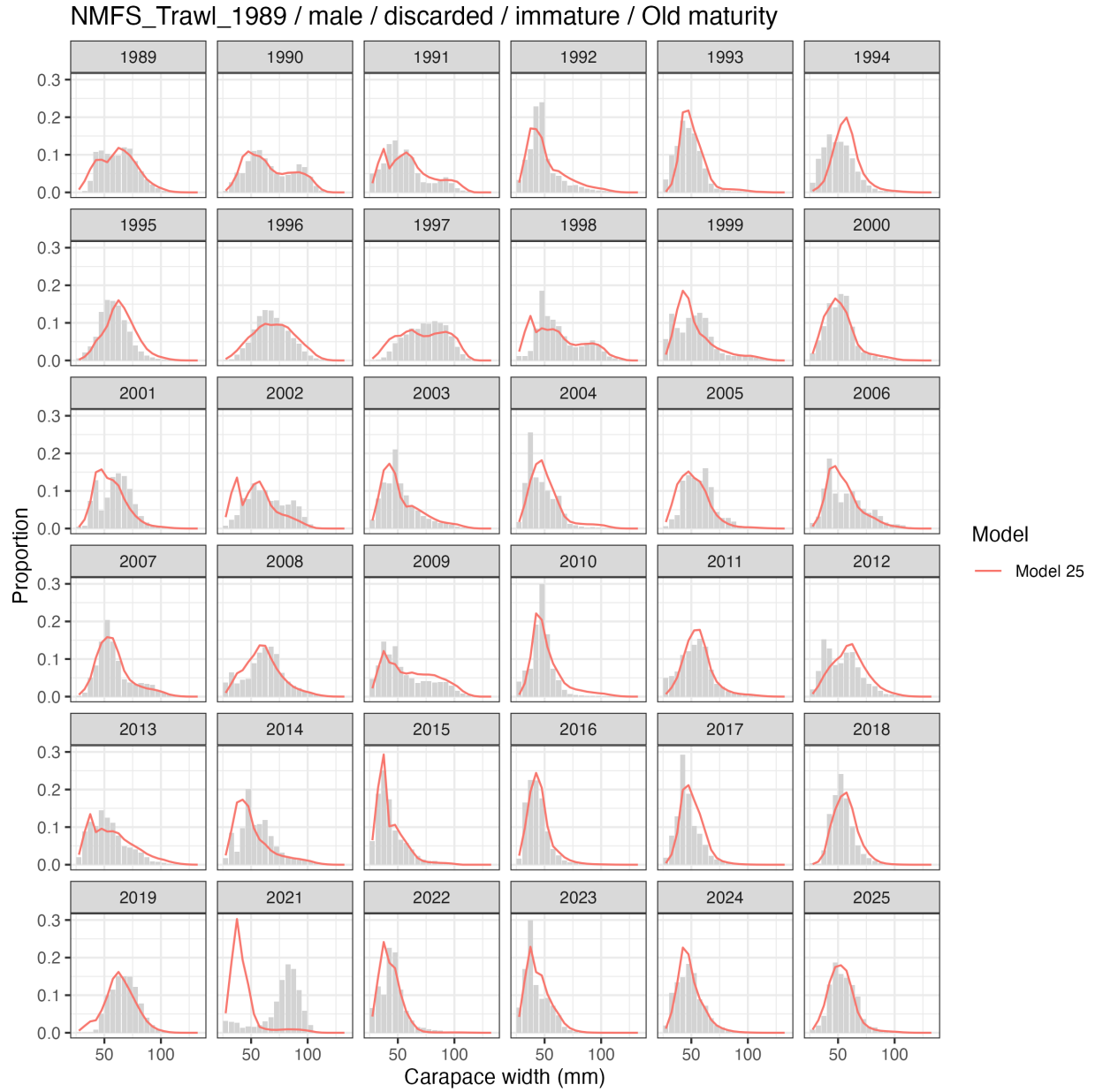


Figure 38: Model fits (lines) to size composition data (grey bars): NMFS Trawl 1989 male discarded immature Old maturity.

Male and Mature

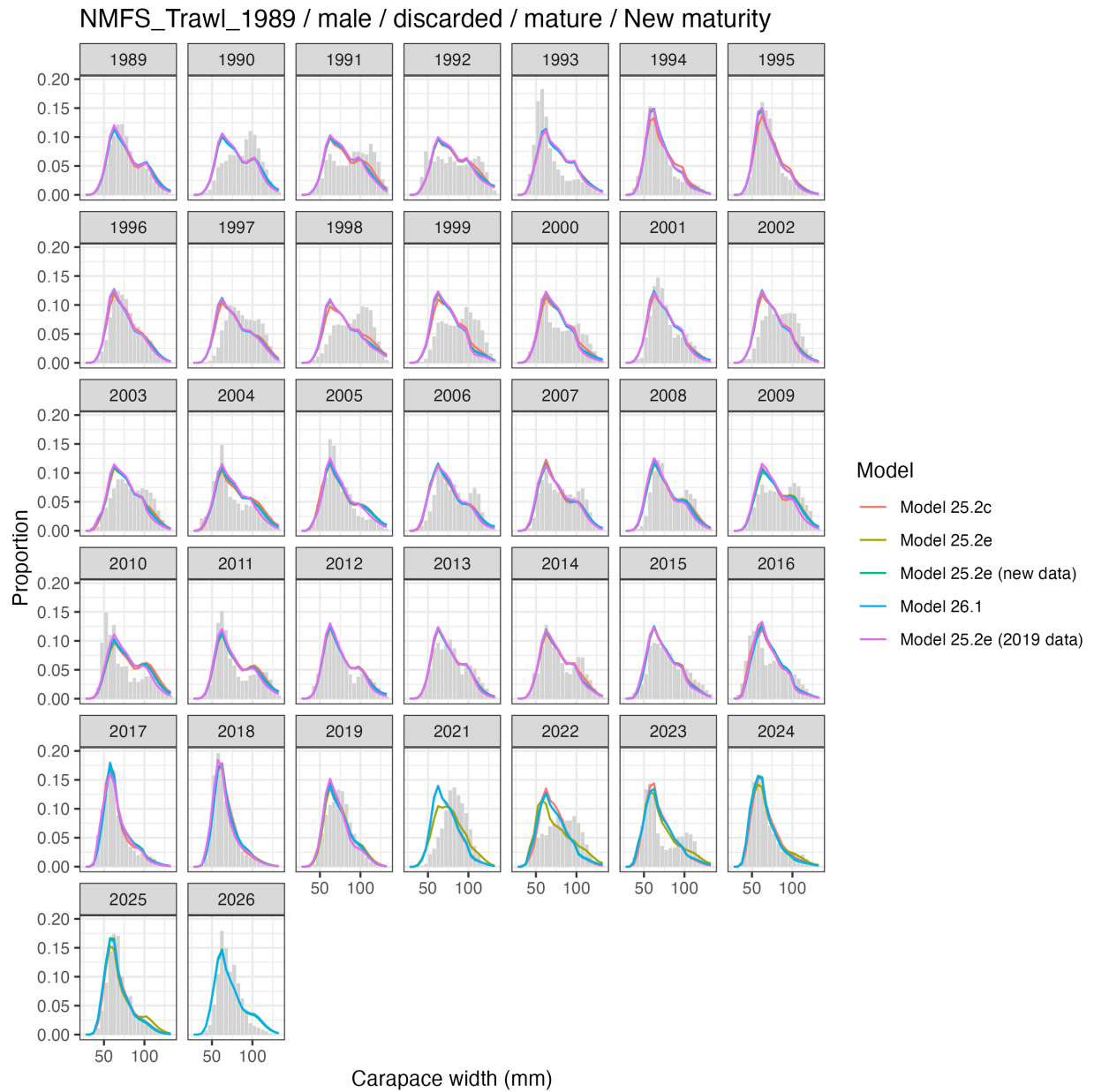


Figure 39: Model fits (lines) to size composition data (grey bars): NMFS Trawl 1989 male discarded mature New maturity.

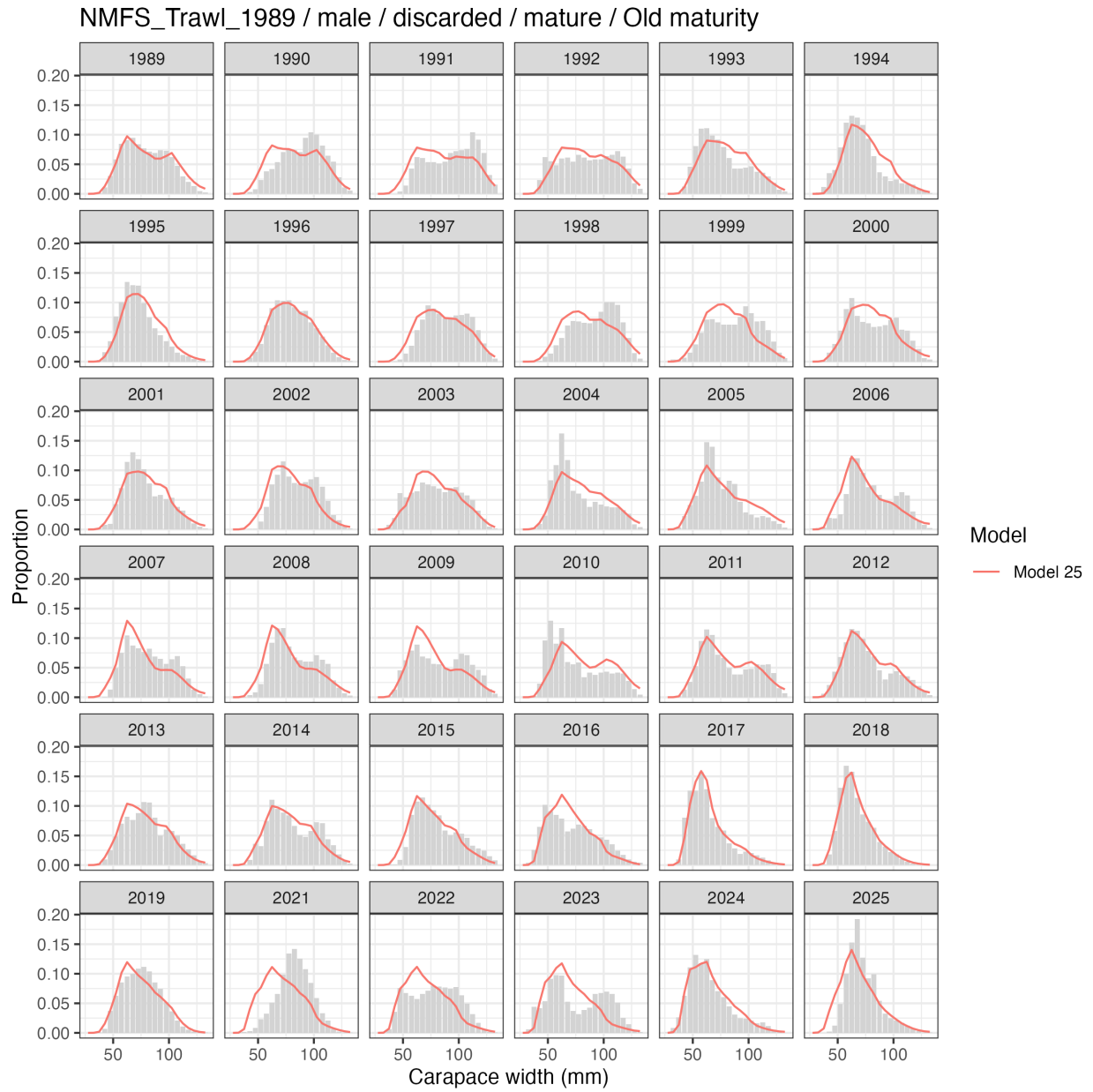


Figure 40: Model fits (lines) to size composition data (grey bars): NMFS Trawl 1989 male discarded mature Old maturity.

Fleet: Pot Fishery

Male and Undetermined

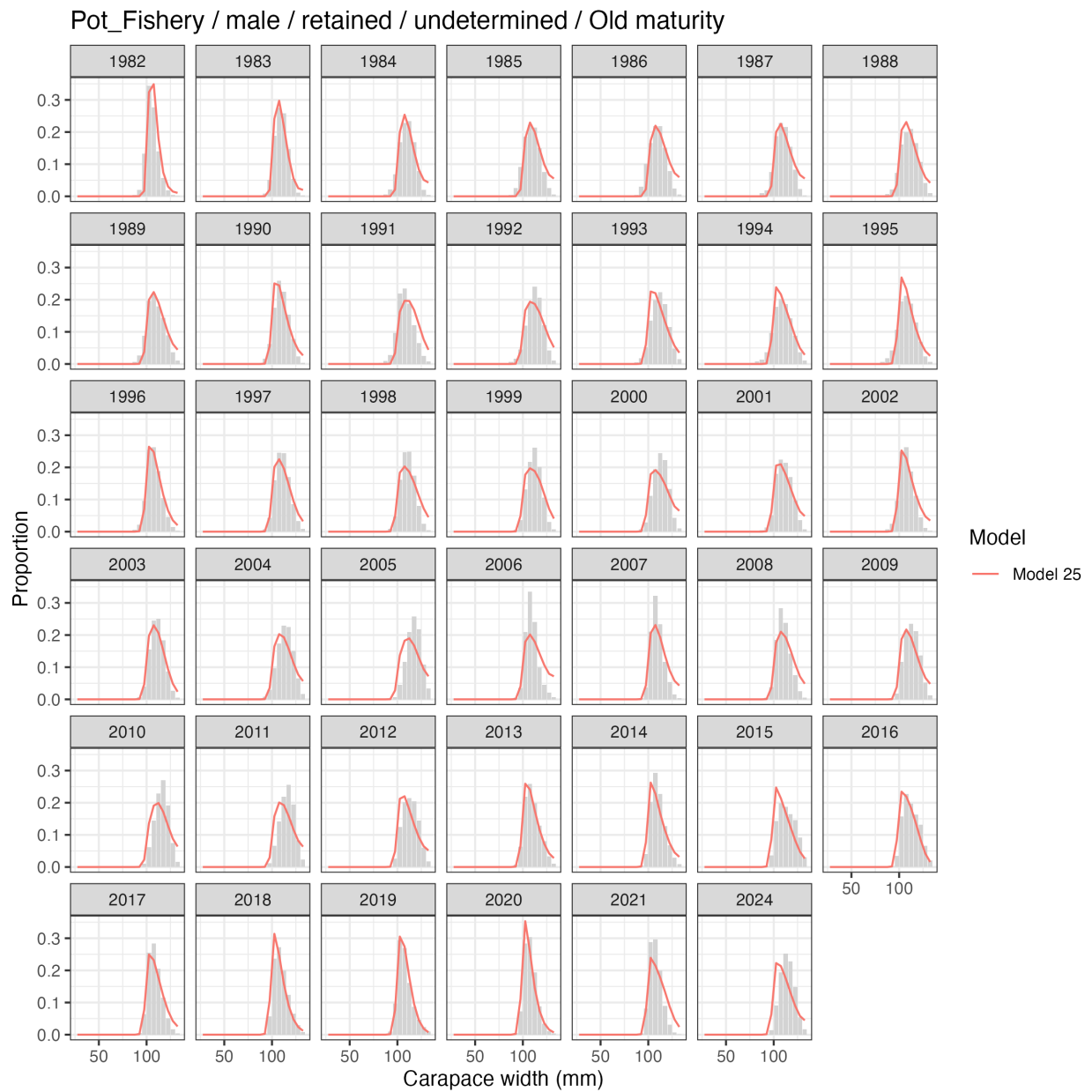


Figure 41: Model fits (lines) to size composition data (grey bars): Pot Fishery male retained undetermined Old maturity.

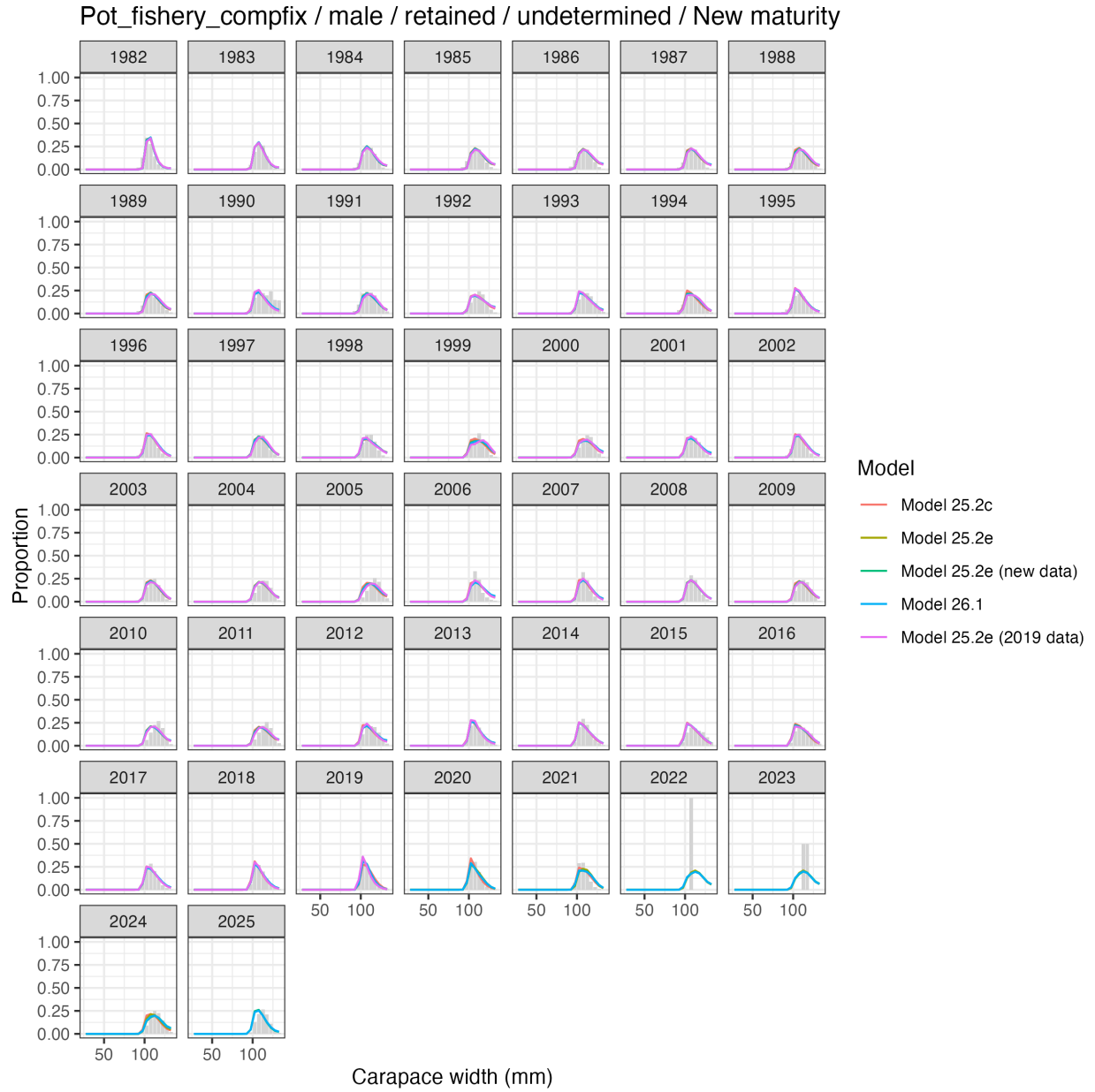


Figure 42: Model fits (lines) to size composition data (grey bars): Pot fishery compfix male retained undetermined New maturity.

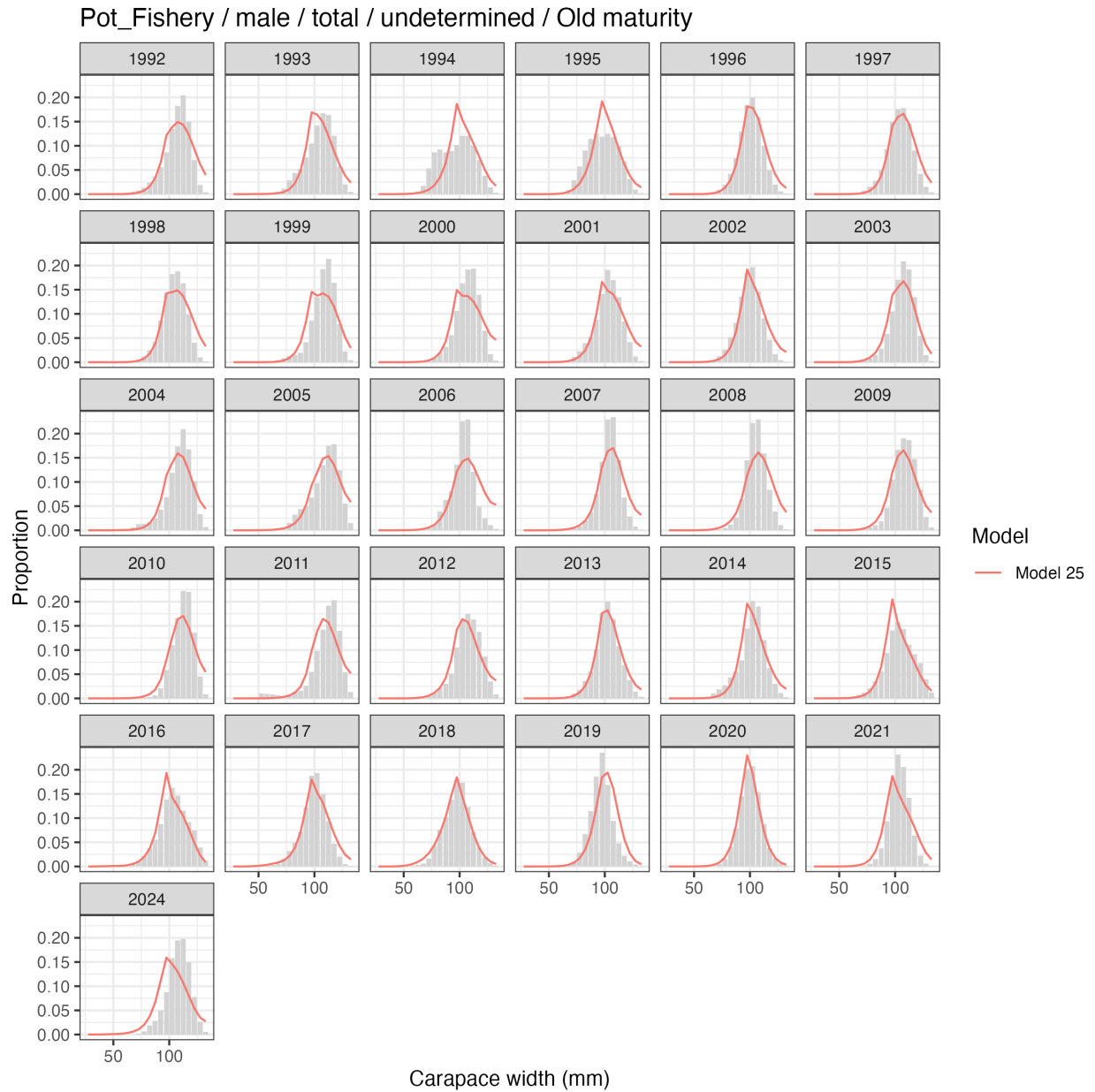


Figure 43: Model fits (lines) to size composition data (grey bars): Pot Fishery male total undetermined Old maturity.

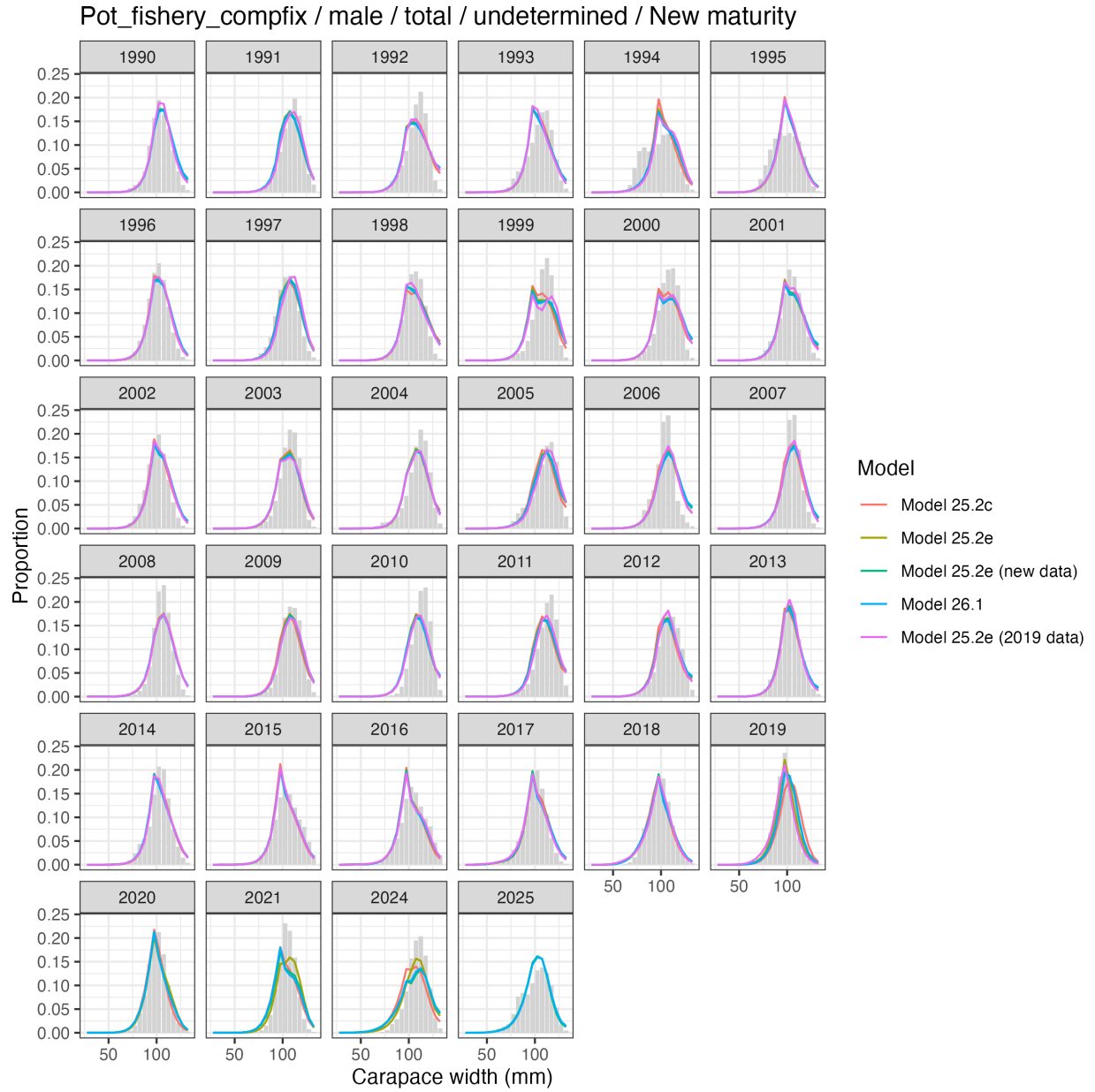


Figure 44: Model fits (lines) to size composition data (grey bars): Pot fishery compfix male total undetermined New maturity.

Female and Undetermined

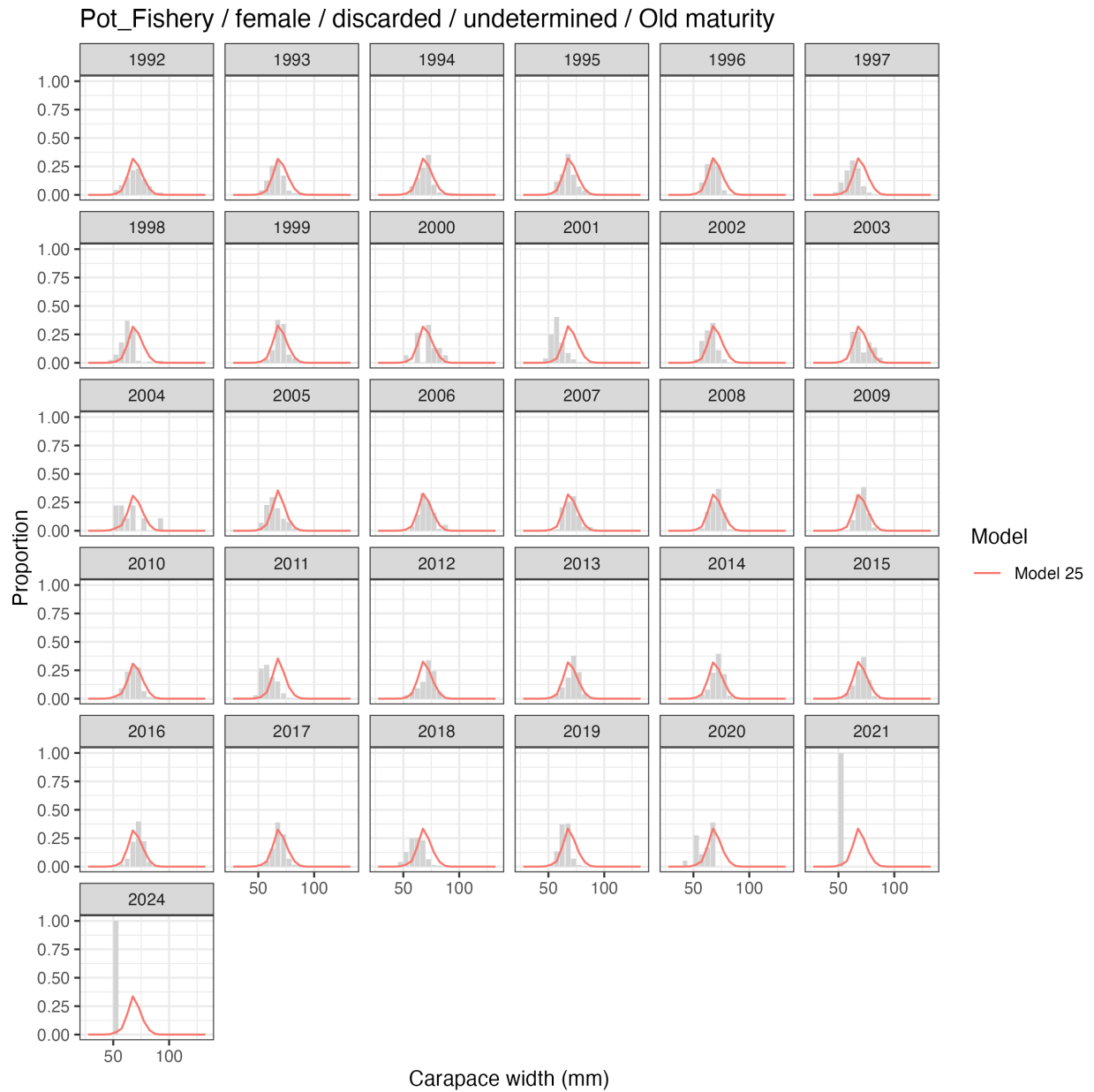


Figure 45: Model fits (lines) to size composition data (grey bars): Pot Fishery female discarded undetermined Old maturity.

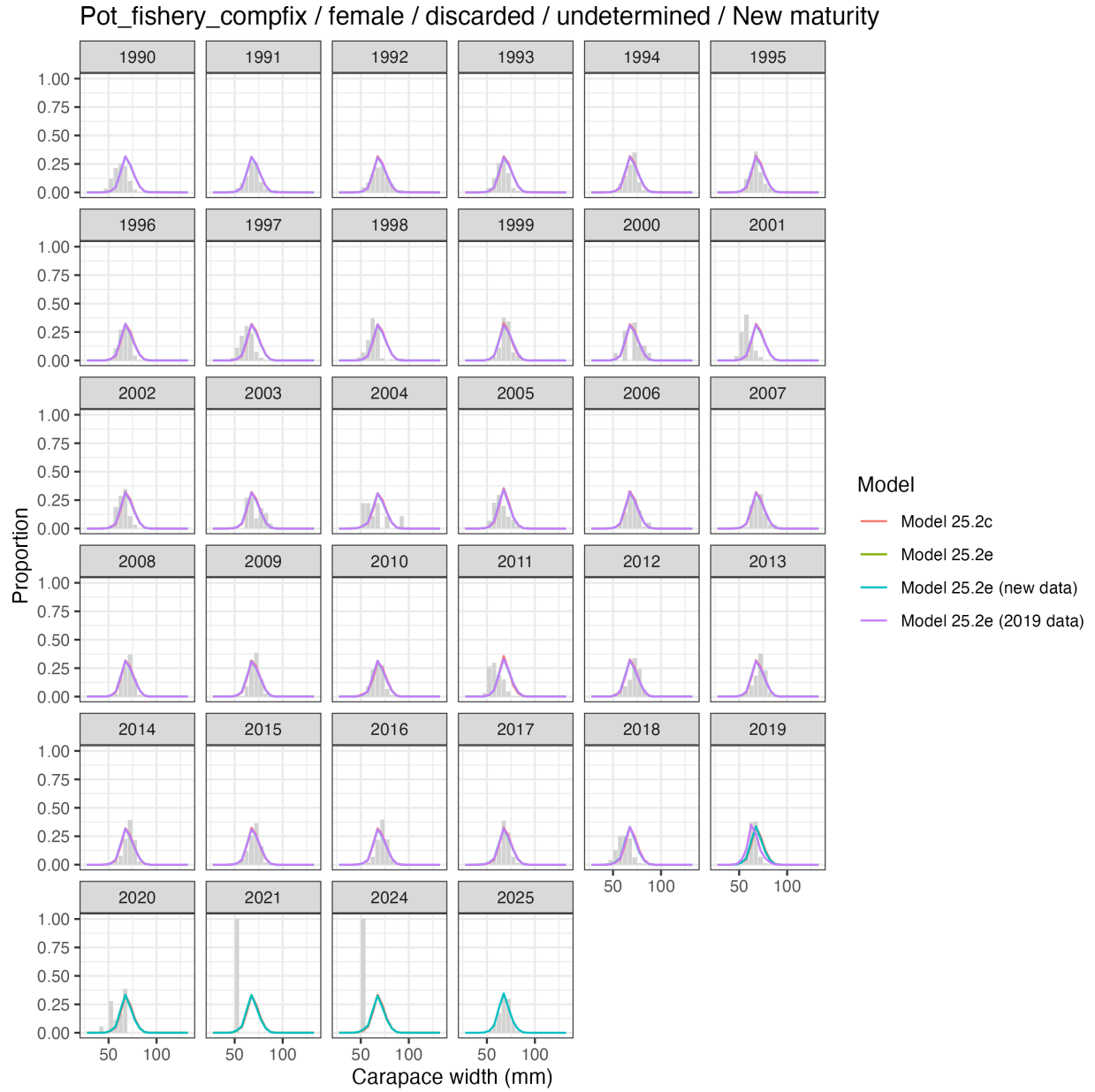


Figure 46: Model fits (lines) to size composition data (grey bars): Pot fishery compfix female discarded undetermined New maturity.

Fleet: Trawl Bycatch

Female and Undetermined

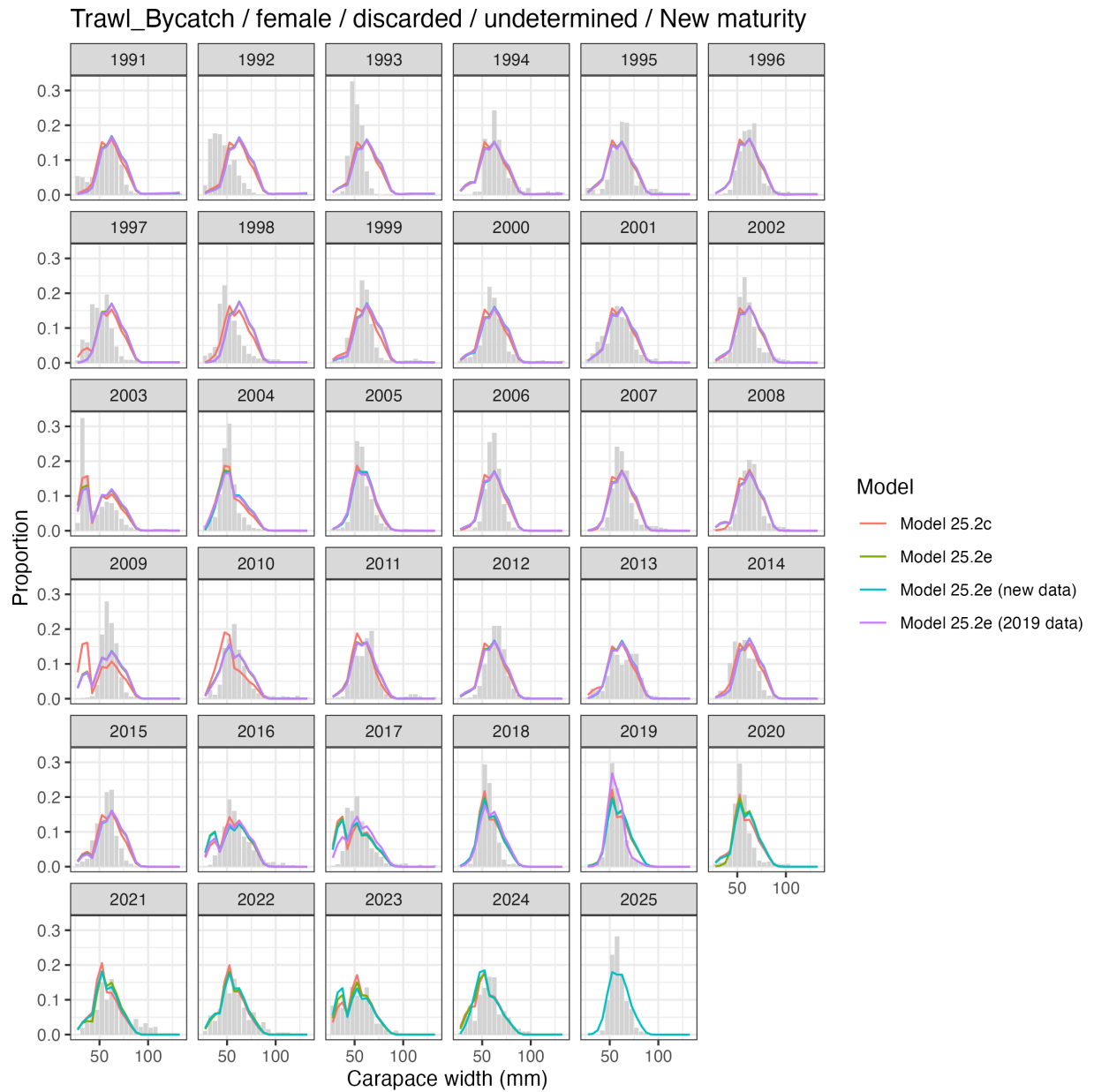


Figure 47: Model fits (lines) to size composition data (grey bars): Trawl Bycatch female discarded undetermined New maturity.

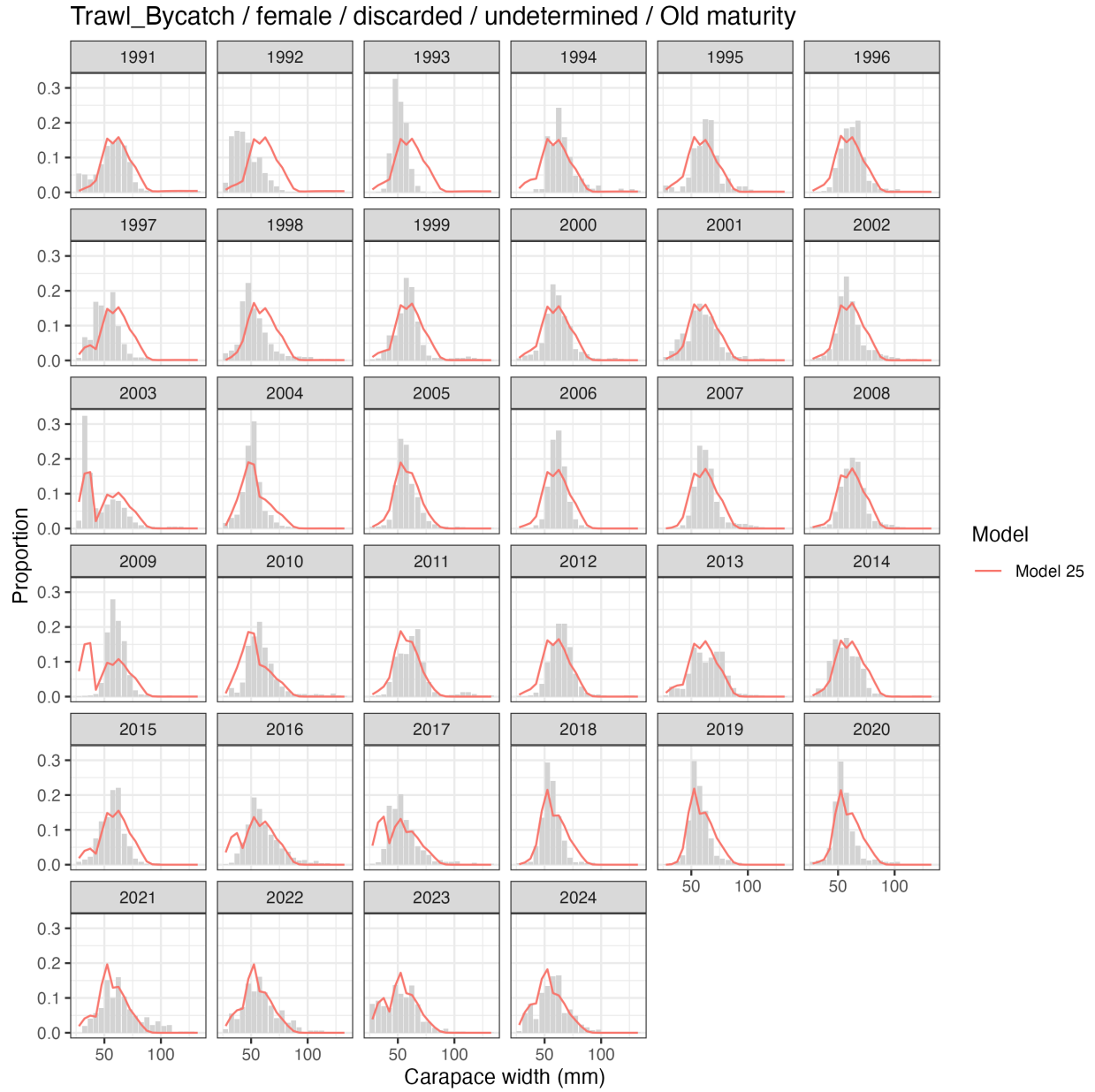


Figure 48: Model fits (lines) to size composition data (grey bars): Trawl Bycatch female discarded undetermined Old maturity.

Male and Undetermined

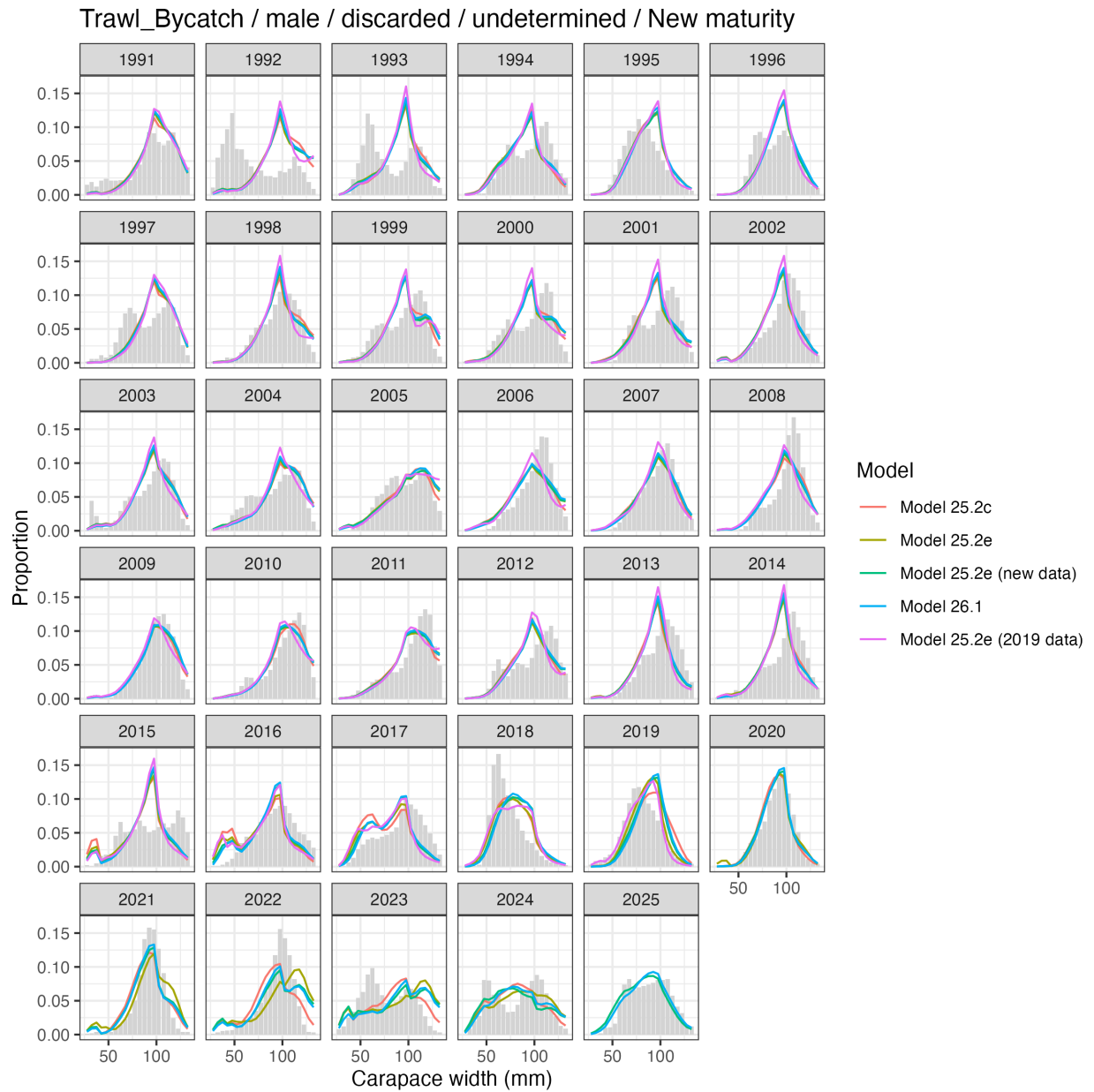


Figure 49: Model fits (lines) to size composition data (grey bars): Trawl Bycatch male discarded undetermined New maturity.

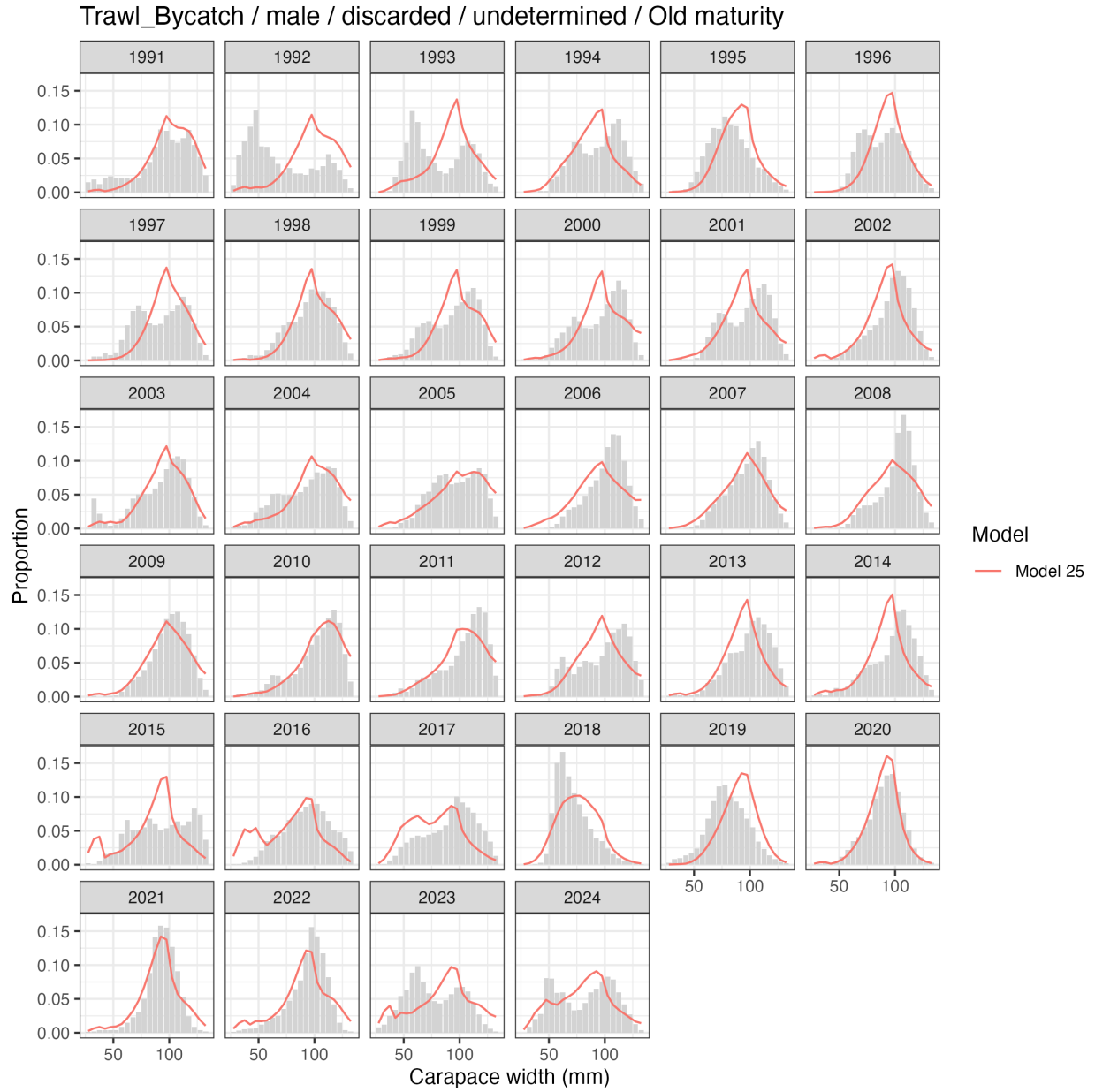


Figure 50: Model fits (lines) to size composition data (grey bars): Trawl Bycatch male discarded undetermined Old maturity.